

Yield-Curve Control Policy under Inelastic Financial Markets*

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Abstract

We develop a New Keynesian model that incorporates the term structure of financial markets, emphasizing the role of government and central bank balance sheet composition in monetary policy transmission. Our model accounts for microfounded market segmentation across asset classes and maturities based on finite and estimatable asset demand elasticities and households' endogenous portfolio reallocation. We show that unconventional monetary policy interventions, such as large-scale asset purchase programs and yield-curve control policies, are effective in stabilizing the economy during normal periods and at the zero lower bound.

Keywords: Term Structure, Yield-Curve Control, Market Segmentation

JEL codes: E32, E43, E44, E52, E58

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Note: The ordering of the authors' names is not indicative of relative contributions; it was chosen arbitrarily (or by mutual agreement) for practical reasons.

1 Introduction

Since the 2007–2008 Global Financial Crisis, unconventional monetary policies have become mainstream. Constrained by the zero lower bound (ZLB) on short-term rates, policymakers adopted strategies—such as expanding central bank balance sheets and increasing government debt issuance—to lower long-term rates, stimulate aggregate demand, and mitigate recessionary pressures. The COVID-19 pandemic further intensified this dynamic, prompting the Federal Reserve to implement another round of unconventional interventions as the policy rate once again reached the ZLB.¹

This paper develops a tractable New Keynesian model that incorporates an endogenous term structure of interest rates in bond markets, enabling an analysis of both conventional and unconventional monetary and fiscal policies. In contrast, standard log-linearized models typically include only a single policy rate, neglecting both the term structure of interest rates and heterogeneous asset returns. This limitation arises from the absence of frictions in asset substitution (e.g., assets demands are perfectly elastic), which leads to the equalization of expected returns across assets and maturities.² As a result, additional assets become redundant in monetary policy analysis.

To address these limitations, we develop a quantitative framework that underscores the role of market segmentation and inelastic demand across bonds of varying maturities in shaping the effectiveness of unconventional monetary policy. While a large existing literature incorporates preferred-habitat investors—who specialize in particular maturities—and arbitrageurs who absorb residual risk and help sustain an upward-sloping yield curve (Ray, 2019; Vayanos and Vila, 2021; Droste et al., 2021), we abstract from arbitrageurs entirely. Instead, our model features households that endogenously adjust their portfolio allocations across bond maturities and private loans, subject to finite own- and cross-price elasticities of demand, in the spirit of Jansen et al. (2024).³

Given these portfolio allocations, households face an effective return on their aggregate

¹In March 2020, the Federal Reserve reduced its policy rate to a range of 0%–0.25%. It also pledged to keep interest rates at these low levels until the economy reached full employment and sustained inflation at 2%. At the same time, the unprecedented CARES Act provided nearly \$500 billion in support of the Federal Reserve.

²This outcome also follows from the absence of a price of risk under log-linear approximations, resulting in the *expectations hypothesis*, which posits that long-term bond returns equal the average of expected future short-term rates.

³This asset demand system with finite own- and cross-elasticities, which generates market segmentation, builds on the recent literature on demand systems (Koijen and Yogo, 2019) and on inelastic markets (Gabaix and Koijen, 2022).

savings. This return enters directly into the Euler equation and serves as a sufficient statistic for the response of consumption to monetary policy. Balance-sheet policies affect the real economy through this channel, as households rebalance their portfolios both along the yield curve and across asset classes, adjusting the effective savings rate that governs consumption dynamics. This framework allows us to study how the size and composition of government and central bank balance sheets influence output, inflation, and welfare within a unified and tractable New Keynesian setting.

Under conventional monetary policy, the central bank directly sets the short-term interest rate—typically following a Taylor rule—while long-term yields are largely determined by household demand and the supply of government bonds. In this environment, substitution across maturities and asset classes dampens the impact of monetary policy rate changes on the broader economy. For instance, a contractionary policy shock prompts households to rebalance away from long-term bonds toward short-term ones, thereby muting the effect on the effective savings rate and the business cycle.⁴ If portfolio shares are sufficiently responsive to relative returns, the effective savings rate may even decline, potentially leading to an expansion.

Under such conditions, a reduction in government bond supply reduces long-term yields and encourages households to shift toward higher-yielding private loans, increasing the effective savings rate and triggering a downturn in aggregate demand, given other variables including the loan rates.⁵ This outcome—where a decline in bond supply leads to a recession—aligns with the safe-asset shortage literature (Caballero and Farhi, 2017; Caballero et al., 2021; Li and Merkel, 2025), though in our framework it arises from household portfolio rebalancing.

Yield-curve control (YCC) policies—in which the central bank directly shapes the entire yield curve—can be particularly effective. Recall that when short-term interest rates decline, households tend to rebalance their portfolios toward long-term bonds. This portfolio adjustment dampens the pass-through of changes in the policy rate to the effective return on household savings. By contrast, YCC policies influence both short- and long-term rates simultaneously, for example by guiding them through particular policy rules, thereby

⁴This occurs because short-term interest rates are generally lower than long-term rates. When households reallocate their portfolios toward short-term bonds, the overall composition shifts in a way that reduces the effective savings rate.

⁵As the economy enters a recession, in general equilibrium, the policy rate—the shortest-maturity interest rate—and loan rates decline by more than long-term yields. This reduces both the effective return on savings and the share of household portfolios allocated to loans. A lower loan portfolio share dampens firm activity and output, leading to a recession. See Section 5.2.1 for further details.

limiting substitution across maturities and reducing its dampening effect on the business cycle. Moreover, because the entire yield curve adjusts, YCC effectively strengthens the responsiveness of monetary policy to economic conditions. As a result, YCC plays a dual role: it curbs portfolio rebalancing while enhancing overall policy effectiveness, leading to improved stabilization of the business cycle. Overall, YCC improves household welfare compared with conventional policies focused solely on short-term interest rates.

However, YCC also entails potential side effects, including the possibility of prolonging ZLB episodes following certain shocks. By directly lowering long-term yields, YCC generates additional downward pressure on short-term interest rates through household portfolio rebalancing. As long-term bond returns decline, households shift savings toward: (i) short-maturity bonds, which further depress short-term yields; and (ii) private lending markets, reducing firms' borrowing costs and thereby lowering consumption prices through lower production costs.⁶

When the ZLB is binding, YCC operates primarily through its influence on long-term yields. While this policy boosts economic activity, it can also delay the economy's exit from the ZLB by keeping downward pressure on inflation and short-term rates. Nevertheless, YCC remains substantially more effective than conventional monetary policy in stabilizing the business cycle. As a result, overall, the economy encounters the ZLB less frequently under YCC, and the average duration of ZLB episodes is shorter despite the possibility of longer episodes in response to particular shocks.

Our asset demand system features *finite* own- and cross-elasticities across both asset classes and bond maturities. Thus, households adjust their portfolios in response to changes in relative returns and other fundamentals less frictionlessly than in standard log-linearized models. Compared with the literature (Ray, 2019; Droste et al., 2021; Jansen et al., 2024),⁷ our model captures market segmentation through a parsimonious household demand structure that governs substitution across maturities and between government bonds and private loans. A central advantage is that each asset-class demand elasticity serves as a sufficient statistic for its degree of market segmentation, which facilitates empirical testing and estimation of the segmented markets conjecture. We estimate bond demand elasticities, using

⁶A decline in the price index further intensifies downward pressure on the short-term policy rate under an inflation-targeting rule, potentially extending the duration of ZLB episodes.

⁷Although Jansen et al. (2024) estimate maturity-specific demands with investor heterogeneity and arbitrageurs, also featuring finite own- and cross-elasticities, we instead embed a tractable finite-elasticity household demand system within a quantitative New Keynesian model that explicitly incorporates government and central bank balance sheets.

the structure implied by the model. The resulting elasticities align with those documented in [Jansen et al. \(2024\)](#).

Finally, we develop a microfoundation for inelastic asset demand based on imperfect information about asset returns. We assume that each household consists of a continuum of families and members, each holding a different and imperfect information set about future asset returns, while consumption is fully insured within the household. Since there is no common signal, the household allocates its total savings evenly across members, who then invest their individual portfolios in the asset they perceive to be most profitable. This structure generates an inelastic portfolio demand function, where cross-sectional dispersion in expected returns governs the degree of market segmentation across asset classes.⁸

To facilitate aggregation of individual portfolio decisions, we model differences in expected returns as Fréchet-distributed shocks around rationally anticipated values. Borrowing this aggregation approach from the international trade literature ([Eaton and Kortum, 2002](#)), the framework naturally accommodates new asset types and varying degrees of segmentation across maturities and asset classes, while yielding tractable expressions for portfolio shares as functions of relative expected returns. Our specification nests the standard expectations hypothesis as a special case, while also allowing for deviations driven by imperfect information and behavioral frictions.

Related Literature We contribute to several strands of the macroeconomics and finance literature. Early work emphasizes the role of macroeconomic factors in explaining the term structure of interest rates ([Ang and Piazzesi, 2003](#); [Rudebusch and Wu, 2008](#); [Bekaert et al., 2010](#)).⁹ Part of this literature employs an affine term structure representation ([Duffie and Kan, 1996](#)). A related literature studies time-varying term premia in a New Keynesian model without market segmentation represented by a finite-elasticity portfolio demand system ([Rudebusch and Swanson, 2012](#); [Buncic and Lentner, 2016](#); [Kliem and Meyer-Gohde, 2022](#); [Mertens and Zhang, 2025](#)). As emphasized by [Gürkaynak and Wright \(2012\)](#), a central challenge for this literature is to maintain tractability while generating time variation

⁸Our microfoundation is related to a literature that derives smooth aggregate choice behavior from discrete individual decisions made under limited information. For example, [Matějka and McKay \(2015\)](#) show that rational inattention in a discrete-choice setting gives rise to generalized multinomial logit choice probabilities. Similarly, [Woodford \(2009, 2011, 2014a,b\)](#) demonstrate how aggregate choices can vary smoothly even when individual decisions are discrete, owing to information frictions and limitations in decision-making.

⁹By examining the joint dynamics of bond yields and macroeconomic variables in a VAR setting with a no-arbitrage restriction, [Ang and Piazzesi \(2003\)](#) show that models incorporating business cycle factors yield better forecasts than those relying solely on unobservable factors.

in term premia. We instead model the term structure of interest rates within a New Keynesian framework that includes multiple asset classes, such as government bonds and private loans, and a finite-elasticity portfolio demand system. By explicitly accounting for the balance sheets of the government and central bank, alongside endogenous portfolio decisions of households across the yield curve, we generate time-varying term premia that depend on macroeconomic shocks, household portfolio choices and the central bank and government balance sheets while keeping the model tractable and interpretable.

The preferred-habitat theory of the term structure has been developed in seminal works such as [Modigliani and Sutch \(1966\)](#), [Vayanos and Vila \(2021\)](#), and [Kekre et al. \(2023\)](#).¹⁰ Building on [Vayanos and Vila \(2021\)](#), [Ray \(2019\)](#) incorporates preferred-habitat investors and limits to arbitrage within a New Keynesian framework,¹¹ while [Droste et al. \(2021\)](#) analyze quantitative easing shocks that operate through these channels in a model grounded in [Ray \(2019\)](#).¹² In contrast, [Jansen et al. \(2024\)](#) focus on cross-substitution across Treasury maturities within a detailed market structure featuring heterogeneous investors and arbitrageurs.

Our approach differs from this literature both in the source of market segmentation and in the broader macroeconomic setting. Rather than modeling cross-substitution through investor clienteles and arbitrageurs, we use a representative-household demand system in which portfolio allocations across maturities and between government bonds and private loans respond endogenously to relative expected returns, subject to finite elasticities. Unlike [Ray \(2019\)](#) and [Droste et al. \(2021\)](#), we abstract from preferred-habitat investors and arbitrageurs, and capture segmentation directly through these finite elasticities. This formulation summarizes market segmentation with estimatable parameters, interprets the effective savings rate as an equilibrium portfolio return, and allows government and central bank balance-sheet policies to transmit jointly through the yield curve, output, and inflation.

In addition, our model generates positive term premia even in log-linearized form, both in and out of steady state. By contrast, the existing literature derives term premia from arbitrageurs' risk aversion, so term premia vanish once those models are log-linearized. The

¹⁰In international macroeconomics, [Gourinchas et al. \(2022\)](#) and [Greenwood et al. \(2022\)](#) examine how preferred-habitat mechanisms jointly shape exchange rates and the term structure of interest rates.

¹¹In [Ray \(2019\)](#), the arbitrageur's capacity to bear risk shapes term premia across the yield curve, which in turn determines the effective savings rate households face when making intertemporal consumption decisions.

¹²[Costa \(2019\)](#) shows by embedding a preferred-habitat theory of the term structure in a DSGE model that monetary stabilization is more difficult to achieve when the central bank considers time-varying term premia in the yield curve, which is consistent with our results. We also show that YCC policy is a powerful tool for stabilizing the business cycle in those cases.

log-linear approximation also allows us to study occasionally binding ZLB episodes. Finally, the endogenous portfolio rebalancing channel, also explored by [Alpanda and Kabaca \(2020\)](#) in the context of LSAP spillovers, gives rise to distinctive dynamic responses.

Within this demand system with finite elasticities, a demand shock to a particular bond maturity—such as quantitative easing targeting long-term bonds—spills over to other maturities with a positive but incomplete pass-through, a phenomenon referred to as the *localization effect*.¹³ In particular, it delivers a positive term premium response to contractionary monetary policy shocks: as households shift from long-term to short-term bonds in search of higher yields, long-term bond prices decrease and term premia increase.¹⁴ This contrasts with models based on [Vayanos and Vila \(2021\)](#), such as [Droste et al. \(2021\)](#), which predict the opposite pattern.¹⁵

A related literature studies how the maturity structure and composition of public liabilities matter when debt provides liquidity services, price is subject to issuance, or short-term asset markets are segmented. Work on public debt management emphasizes incomplete markets and maturity structure ([Bhandari et al., 2017](#); [Aparisi de Lannoy et al., 2022](#)).¹⁶ [d’Avernas and Vandeweyer \(2024\)](#) emphasize Treasury bill scarcity and segmented short-term asset markets from a large amount of excess reserves and a binding regulatory leverage ratio, and [Gómez Cram et al. \(2025\)](#) link Treasury risk premia to monetary-fiscal interactions and limited market participation. These papers complement our analysis by showing that the pricing and allocation of public debt depend on maturity structure, market segmentation, and the joint balance sheets of fiscal and monetary authorities.

Another strand of the literature ([Gertler and Karadi, 2011](#); [Cúrdia and Woodford, 2011](#); [Christensen and Krogstrup, 2018, 2019](#); [Karadi and Nakov, 2021](#); [Del Negro et al., 2017](#); [Iovino and Sergeyev, 2018](#); [Eggertsson et al., 2024](#)) studies balance-sheet policy transmission and liquidity frictions more directly.¹⁷ Most of this work emphasizes overall balance-

¹³This is shown by [Jansen et al. \(2024\)](#), who estimate maturity-specific asset demand functions in a model with heterogeneous investors and arbitrageurs, yielding finite own- and cross-elasticities across maturities.

¹⁴In this context, the mechanism resembles [Hanson and Stein \(2015\)](#), who emphasize the importance of yield-seeking investors in driving positive term premium responses to monetary policy tightening.

¹⁵Typically, monetary tightening depresses bond prices and increases preferred-habitat investors’ demand for their targeted maturity, forcing arbitrageurs to absorb less risk and thereby lowering term premia in equilibrium. This discrepancy has been addressed by incorporating arbitrageur wealth ([Kekre et al., 2023](#)). Under contractionary policy, a decline in arbitrageur wealth increases their effective risk aversion, leading to a higher required term premium.

¹⁶[Bigio et al. \(2023\)](#) show how liquidity costs shape optimal debt-maturity management.

¹⁷[Gertler and Karadi \(2011\)](#) note that compared with private market intermediaries, the central bank faces fewer balance sheet constraints, and as private intermediaries’ constraints tighten during crises, central bank intermediation becomes more beneficial; [Cúrdia and Woodford \(2011\)](#) find targeted asset purchases effective

sheet expansion, crisis-specific market disruptions, or the banking channel rather than substitution across bonds of different maturities within a unified term-structure environment. In contrast, our framework shows how the central bank can rebalance bond holdings across the yield curve to enhance stabilization. Our result that active management of long-term bonds improves welfare is consistent with [Sims and Wu \(2021\)](#).¹⁸

While our treatment of ZLB recessions builds on earlier work ([Swanson and Williams, 2014](#); [Dordal i Carreras et al., 2016](#); [Caballero and Farhi, 2017](#); [Caballero et al., 2021](#)), we place additional emphasis on how household portfolio reallocation—under finite elasticities—shapes outcomes during such episodes. We also highlight the advantages of jointly managing both the size and the maturity composition of the central bank’s balance sheet when the ZLB is binding. To our knowledge, this is among the first general equilibrium models to integrate the term structure of interest rates, a binding ZLB, multiple financial assets, yield-curve control policies, and the portfolio balance channel with demand functions featuring finite elasticities.

Layout Section 2 presents a simplified model that illustrates the key mechanisms of our framework. Section 3 introduces the full model. Section 4 examines the steady-state implications of various policies and calibration choices. Section 5 explores the short-run cyclical responses to different shocks under alternative monetary policy regimes and economic conditions, including the ZLB. Section 6 concludes. The Appendix contains additional figures and tables. Online Appendix A provides detailed derivations and proofs; Online Appendix B outlines our calibration and estimation strategies; and Online Appendix C derives the second-order approximation to welfare.¹⁹

2 Illustrative Model

This section presents a simplified framework designed to highlight the model’s core mechanisms, with a particular focus on how endogenous portfolio rebalancing and yield-curve control (YCC) shape the transmission of monetary policy. The analysis shows that substitution across maturities and asset classes weakens the effect of changes in the policy rate on the broader economy. For example, an expansionary shock leads households to shift their

under exogenous market disruptions.

¹⁸[Sims and Wu \(2021\)](#) assume that a wholesale firm and fiscal authorities issue perpetuities with declining coupon payments, whereas we consider bonds with varying maturities.

¹⁹Supplementary Material in Online Appendices D-G includes further figures and explanations.

portfolios away from short-term bonds toward higher-yielding long-term bonds. This substitution dampens the policy’s accommodative impact on the effective savings rate, which controls the intertemporal consumption decision, and, in turn, on the business cycle.

Within this setting, YCC policies can be especially powerful because they enable the central bank to influence both short- and long-term interest rates simultaneously. By doing so, YCC limits substitution across maturities—and across asset classes when multiple assets are available. It also enhances the overall effectiveness of monetary policy by allowing the central bank to move the entire yield curve in response to economic conditions.

For tractability, the simplified model is log-linearized and features two assets: short-term and long-term bonds, though they can be interpreted more broadly as different asset classes. It abstracts from inflation dynamics by assuming perfectly rigid prices.²⁰ The representative household allocates its portfolio based on the relative expected returns of these assets, with finite own- and cross-return elasticities. The central bank follows a Taylor-type rule to set short-term interest rates, while long-term rates depend on the household’s allocation to long-term bonds and government bond supply. In the full model presented later, conventional policy leads to *incomplete* pass-through to long-term rates—for instance, higher short-term rates can push long-term rates up as households rebalance toward short-term assets. In contrast, the simplified model abstracts from this mechanism by assuming that long-term rates remain fixed under conventional policy. YCC is introduced through a separate Taylor-type rule for long-term rates, allowing the central bank to directly manage the entire yield curve.

Setting The representative household faces the following Euler equation:

$$\hat{c}_t = \mathbb{E}_t [\hat{c}_{t+1} - \hat{r}_{t+1}^S], \quad (1)$$

where $\hat{c}_t$ and $\hat{r}_{t+1}^S$ are the log-deviation of consumption and the (gross) effective savings rate from the steady state, respectively. In a canonical New Keynesian model (Woodford, 2003), the effective savings rate $\hat{r}_{t+1}^S$ coincides with the policy rate set by the monetary authority. In our framework, however, this rate is a composite measure reflecting the returns on both short-term and long-term bonds, each weighted by its share in the aggregate portfolio. The

²⁰Our full model later relaxes this assumption and allows for inflation dynamics and multiple asset classes.

(gross) effective savings rate is given by:

$$R_{t+1}^S = \sum_{f=1}^2 \lambda_t^f R_{t+1}^f, \quad (2)$$

where λ_t^f is the household's portfolio share of the f -maturity bond in period t , and R_{t+1}^f is the f -maturity bond (one-period) return. As stated, the simplified model considers two assets: short-term and long-term bonds, so $f = 1, 2$. Log-linearizing (2) yields:

$$\hat{r}_{t+1}^S = \sum_{f=1}^2 \frac{\lambda^f R^f}{R^S} \left(\hat{\lambda}_t^f + \hat{r}_{t+1}^f \right), \quad (3)$$

where λ^f , R^f , and R^S are the steady state levels of the portfolio share, the bond return, and the effective savings rate, respectively. Note that the effective savings rate changes when either the portfolio share $\hat{\lambda}_t^f$ or the bond return $\hat{r}_{t+1}^f$ changes. For instance, a contractionary monetary policy shock, corresponding to a higher $\hat{r}_{t+1}^1$, will increase the effective savings rate given $\hat{\lambda}_t^f$. At the same time, $\hat{\lambda}_t^1$ increases as households shift their portfolios toward short-term bonds. This substitution mutes the increase in the effective savings rate.²¹

The household's portfolio share of the short-term bond is given by:

$$\hat{\lambda}_t^1 = \kappa_B \mathbb{E}_t \left[\hat{r}_{t+1}^1 - \hat{r}_{t+1}^2 \right], \quad (4)$$

where κ_B is the elasticity of the demand for the short-term bond with respect to the expected return spread between the short-term and long-term bonds. Although the full model later provides a microfoundation for this elasticity, we treat it as exogenous and finite here.²² $\kappa_B \rightarrow \infty$ corresponds to perfect substitution between the two bonds, rendering returns on the long-term bond irrelevant for the business cycle dynamics and leading to the expectations hypothesis.²³ For expositional simplicity, we assume that in that case the households weakly prefer to save into the short-maturity asset, i.e. $\lim_{\kappa_B \rightarrow \infty} \lambda^1 = 1$, which replicates the traditional single-asset model employed in the literature.

²¹We assume that $\lambda^1 R^1 < \lambda^2 R^2$, which implies that a larger share of short-term bonds lowers the effective savings rate given R_{t+1}^1 and R_{t+1}^2 . This compositional change dampens the increase in the effective savings rate.

²²In addition, we estimate this parameter based on U.S. data.

²³Note that as $\kappa_B \rightarrow \infty$, $R^1 = R^2 = R^S$ at the steady state.

Since $\lambda_t^1 + \lambda_t^2 = 1$, we obtain

$$\hat{\lambda}_t^2 = -\frac{\lambda^1}{\lambda^2} \hat{\lambda}_t^1, \quad (5)$$

where λ^1 and λ^2 denote levels of the portfolio share of the short-term and long-term bonds at the steady state, respectively.

Finally, because prices are rigid, the economy is demand-determined. We therefore abstract from the supply side and assume that output is pinned down by aggregate consumption demand, as characterized by the Euler equation (1).

Conventional Monetary Policy The central bank follows a Taylor rule to set the short-term interest rate:

$$\hat{r}_{t+1}^1 = \phi_C \hat{c}_t + \varepsilon_t \quad (6)$$

where $\phi_C > 0$ is the policy response coefficient to the output gap, and ε_t is a policy shock. Under conventional policy, we assume that the long-term interest rate is fixed at the steady state level, i.e., $\hat{r}_{t+1}^2 = 0$.²⁴ For simplicity, we further assume that ε_t is a i.i.d. random variable with $\mathbb{E}[\varepsilon_t] = 0$.

Yield-Curve Control Policy In addition to the short-term rate given by (6), the central bank follows a Taylor rule to set the expected long-term interest rate:

$$\mathbb{E}_t \hat{r}_{t+1}^2 = \phi_{C,2} \hat{c}_t \quad (7)$$

where $\phi_{C,2} > 0$ is the policy response coefficient to the output gap.²⁵

²⁴Our full model relaxes this assumption—the long-term interest rates are determined by the household's endogenous portfolio share of the long-term bonds and government bond issuance. In those cases, the long-term rates will be affected by the policy rate change and the portfolio rebalancing channel, with an incomplete pass-through.

²⁵Note that the long-term rate $\hat{r}_{t+1}^2$ is intrinsically uncertain, as future bond prices cannot be known with certainty from today's perspective. In the full model, rather than assuming the central bank directly selects $\mathbb{E}_t \hat{r}_{t+1}^2$, we posit that it chooses long-term bond yields using Taylor rules as current yields are observable.

2.1 Monetary Policy Transmission

We first consider the transmission of monetary policy under conventional policy and yield-curve control policy. The short-term interest rate is given by (6), and the long-term interest rate is fixed at the steady state level, i.e., $\hat{r}_{t+1}^2 = 0$, under conventional policy and given by (7), under yield-curve control policy. The effective savings rate is given by (3), and the portfolio share of the short-term bond is given by (4). For illustrative purposes, we assume that the only shock that drives the business cycle is the monetary policy shock ε_t .

Conventional Policy Merging (3) and (5) with $\hat{r}_{t+1}^2 = 0$, we obtain:

$$\hat{r}_{t+1}^S = \underbrace{\frac{\lambda^1 R^1}{R^S}}_{\equiv \eta_1 > 0} \hat{r}_{t+1}^1 - \underbrace{\frac{\lambda^1 (R^2 - R^1)}{R^S}}_{\equiv \eta_2 > 0} \hat{\lambda}_t^1 \quad (8)$$

where $\eta_1 < 1$ due to our assumption that $\lambda^1 R^1 < \lambda^2 R^2$, and $\eta_2 > 0$. This equation shows an important feature of our model: when $\hat{r}_{t+1}^1$ increases, its direct effect on the effective savings rate $\hat{r}_{t+1}^S$ is mitigated by a factor of $\eta_1 < 1$. In addition, households will substitute their portfolio towards the short-term bond, raising $\hat{\lambda}_t^1$. The endogenous change in portfolio shares further mitigates the increase in the effective savings rate $\hat{r}_{t+1}^S$ given $R^2 > R^1$ at the steady state.

The equilibrium is characterized by merging the Euler equation (1), the effective savings rate (8), the portfolio share of the short-term bond (4), and the short-term interest rate (6), which results in:²⁶

$$\hat{c}_t^{\text{CP}} = -\frac{1}{1 + (\eta_1 - \eta_2 \kappa_B) \phi_C} (\eta_1 - \eta_2 \kappa_B) \varepsilon_t \quad (9)$$

where the denominator is assumed to be positive, i.e., $1 + (\eta_1 - \eta_2 \kappa_B) \phi_C > 0$. We categorize different cases of the strength of substitution between the two bonds as follows:

Case 1 $\kappa_B \rightarrow \infty$: Perfect substitution between the two bonds, leading to the *expectations hypothesis*. In this case, $R^1 = R^2 = R^S$ at the steady state, so $\eta_2 = 0$; under the perfect-substitution normalization $\lim_{\kappa_B \rightarrow \infty} \lambda^1 = 1$ stated above, $\eta_1 = 1$. Thus,

²⁶This is the only rational expectation equilibrium that is consistent with the assumptions of the model. Because the policy shock is iid and the simplified model has no endogenous state variable, the equilibrium policy function depends only on the current shock, implying $\mathbb{E}_t[\hat{c}_{t+1}] = 0$.

the equilibrium consumption is given by:

$$\hat{c}_t^{\text{CP}} = -\frac{1}{1 + \phi_C} \varepsilon_t, \quad (10)$$

where stronger monetary policy response to the output gap ϕ_C leads to more stabilizing effect on the business cycle. This corresponds to a standard New Keynesian model (Woodford, 2003).

Case 2 $\kappa_B \rightarrow 0$: No substitution between the two bonds. In this case, $\hat{\lambda}_t^1 = 0$ and $\hat{r}_{t+1}^S = \eta_1 \hat{r}_{t+1}^1$. The equilibrium consumption is given by:

$$\hat{c}_t^{\text{CP}} = -\frac{1}{1 + \eta_1 \phi_C} \eta_1 \varepsilon_t. \quad (11)$$

Note that compared with (10), equation (11) shows that both the monetary policy shock ε_t and the monetary policy responsiveness $\phi_C > 0$ are reduced by a factor of $\eta_1 < 1$. This occurs because in this case, a shock to the policy rate $\hat{r}_{t+1}^1$ is not fully transmitted to the effective savings rate $\hat{r}_{t+1}^S$, with the degree of pass-through determined by the fixed weighted-return share $\eta_1 < 1$.

Case 3 $\kappa_B > 0$ and $\eta_2 \kappa_B < \eta_1$: Substitution between the two bonds, which further mitigates the pass-through from the policy rate to the effective savings rate $\hat{r}_{t+1}^S$ —both the monetary policy shock ε_t and the monetary policy responsiveness $\phi_C > 0$ are reduced by a smaller factor of $\eta_1 - \kappa_B \eta_2 < 1$. In this case, $\hat{r}_{t+1}^S$ still moves positively with the policy rate $\hat{r}_{t+1}^1$. Consumption $\hat{c}_t^{\text{CP}}$ decreases under contractionary monetary policy shocks $\varepsilon_t > 0$, but less than in the above two cases.²⁷

Case 4 $\kappa_B > 0$ and $\eta_2 \kappa_B > \eta_1$: Substitution between the two bonds is now strong enough that $\hat{r}_{t+1}^S$ moves inversely with the policy rate $\hat{r}_{t+1}^1$. Contractionary monetary policy shocks $\varepsilon_t > 0$ reduces the effective savings rate $\hat{r}_{t+1}^S$ and raises consumption $\hat{c}_t^{\text{CP}}$

²⁷This is because the first effect—where the monetary policy shock ε_t is dampened by the smaller factor $\eta_1 - \kappa_B \eta_2 < 1$ —outweighs the second effect, in which the policy responsiveness $\phi_C > 0$ is also mitigated by the same factor. Note that the coefficient

$$\frac{\eta_1 - \eta_2 \kappa_B}{1 + (\eta_1 - \eta_2 \kappa_B) \phi_C} > 0$$

is less than $\frac{\eta_1}{1 + \eta_1 \phi_C}$ for **Case 2** and $\frac{1}{1 + \phi_C}$ for **Case 1** when $\eta_2 \kappa_B < \eta_1$.

As a result, monetary policy becomes less effective in stabilizing the business cycle—remember, the effective monetary responsiveness ϕ_C is dampened by a factor of $\eta_1 - \eta_2\kappa_B$ —and, in certain cases, can even produce outcomes that run counter to those predicted by the standard New Keynesian model when the degree of substitution between the two bonds is high enough.²⁸

Yield-Curve Control Policy Under YCC policy, the equilibrium is characterized by the Euler equation (1), the general effective savings rate (3), the portfolio share of the short-term bond (4), the short-term interest rate (6), and the long-term interest rate (7). This results in:

$$\hat{c}_t^{\text{YCC}} = -\frac{1}{1 + \underbrace{(\eta_1\phi_C + (1 - \eta_1)\phi_{C,2})}_{\text{Policy Response}} - \eta_2\kappa_B \underbrace{(\phi_C - \phi_{C,2})}_{\text{Cross-Substitution}}} (\eta_1 - \eta_2\kappa_B) \varepsilon_t, \quad (12)$$

where in the policy-response term, the short-rate and long-rate responses enter the effective savings rate as a weighted average, with weights η_1 and $1 - \eta_1$, respectively.

Comparing equations (12) and (9) elucidates the key advantage of YCC policy: it allows the central bank to influence both short- and long-term interest rates simultaneously. This effectively limits substitution across the two bonds— $\eta_2\kappa_B\phi_C$ term in the denominator of (9) is replaced by $\eta_2\kappa_B(\phi_C - \phi_{C,2})$ in the denominator of (12), raising the effect of monetary policy in stabilizing the business cycle. In contrast, the policy shock ε_t is dampened by the same factor $\eta_1 - \eta_2\kappa_B$ as under conventional policy.

In addition, YCC policy effectively raises the overall responsiveness of monetary policy to the output gap— $(\eta_1\phi_C + (1 - \eta_1)\phi_{C,2})$ term in the denominator of (12) is greater than $\eta_1\phi_C$ in the denominator of (9). This is because the central bank can now shift the entire yield curve in response to economic conditions.

Overall, these two channels make YCC policy more effective in stabilizing the business cycle than conventional policy when the degree of substitution between the two bonds is finite— $|\hat{c}_t^{\text{YCC}}| < |\hat{c}_t^{\text{CP}}|$. While our simplified model is defined using a monetary policy shock, the stabilizing effect of YCC policy remains robust across various types of shocks. Under all shock scenarios, YCC proves more effective than conventional policy in stabilizing the business cycle. And this result continues to hold even when we consider a zero

²⁸A higher κ_B reduces $R^2 - R^1$ at the steady state, reducing η_2 . Under our calibration in the main text, $\eta_2\kappa_B$ still increases with κ_B .

lower bound (ZLB) on the short-term interest rate.

In the following sections, we provide a more formal model specification and empirical analysis to support this conclusion.

3 Model

As in Section 2, the model is built around a representative household framework. To provide a tractable foundation, we consider a large household with internal frictions, allowing us to isolate the effects of yield-curve interventions in a clean and controlled setting.

3.1 Representative Household

The representative household maximizes its expected discounted utility:

$$\max_{\{C_{t+j}, N_{t+j}\}} \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\log(C_{t+j}) - \frac{\eta}{\eta+1} \left(\frac{N_{t+j}}{\bar{N}_{t+j}} \right)^{1+\frac{1}{\eta}} \right], \quad (13)$$

where C_t denotes final good consumption and $N_t = \left(\int_0^1 N_t(\nu)^{\frac{\eta+1}{\eta}} d\nu \right)^{\frac{\eta}{\eta+1}}$ is the aggregate labor index with $N_t(\nu)$ representing labor supplied in intermediate industry ν . The parameter η is the Frisch elasticity of labor supply, and $\bar{N}_t$ is the balanced-growth population, growing at a constant gross rate GN .

Each period, the household invests in f -period zero-coupon government bonds (with $f = 1, \dots, F$) and provides loans to firms.²⁹ The budget constraint is:

$$C_t + \frac{L_t^H}{P_t} + \frac{\sum_{f=1}^F B_t^{H,f}}{P_t} = \frac{\sum_{f=0}^{F-1} R_t^f B_{t-1}^{H,f+1}}{P_t} + \frac{R_t^K L_{t-1}^H}{P_t} + \int_0^1 \frac{W_t(\nu) N_t(\nu)}{P_t} d\nu + \frac{\Lambda_t}{P_t}, \quad (14)$$

where L_t^H denotes one-period loans to firms provided by households, yielding a gross return R_t^K ; $W_t(\nu)$ is the wage in industry ν ; and Λ_t aggregates transfers from lump-sum taxation and profits from the central bank and firms.

$B_t^{H,f}$ denotes nominal investment in f -maturity government bonds that pay one dollar at time $t+f$, with Q_t^f denoting the bond price (by definition, $Q_t^0 = 1$ at maturity). Households cannot issue risk-free bonds. The one-period return on an f -period bond is given by $R_t^f =$

²⁹Alternatively, this may be interpreted as households purchasing one-period corporate bonds. Banks and financial intermediaries are abstracted from our framework; in the absence of intermediation frictions, both representations yield equivalent results.

$\frac{Q_t^f}{Q_{t-1}^{f+1}}$, capturing the one-period holding returns. The gross yield of a zero-coupon bond with f -maturity is defined as $YD_t^f := (Q_t^f)^{-1/f}$.³⁰

3.1.1 Inelastic Portfolio Demands

The representative household determines optimal consumption, employment, and savings S_t , which are allocated between government bonds $B_t^H = \sum_{f=1}^F B_t^{H,f}$ and firm loans L_t , so that $S_t = B_t^H + L_t^H$.

Following the literature (Koijen and Yogo, 2019; Gabaix and Koijen, 2022), we assume that households' portfolio shares are determined by downward-sloping demand functions with finite own- and cross-price elasticities. We microfound these portfolio shares through heterogeneous information within the household. In particular, after choosing total savings S_t , the representative household delegates portfolio decisions to a continuum of families $m \in [0, 1]$. Each family chooses between government bonds and private loans; if a family selects bonds, a continuum of members $n \in [0, 1]$ within that family chooses the bond maturity. Consumption is fully insured within the household as in Del Negro et al. (2017). Because there is no common signal about future asset returns, savings are allocated symmetrically across families and members before idiosyncratic information is realized, and each member then allocates assigned savings to the asset with the highest perceived expected discounted return.

Bond Family For bond-investing families, member n selects the maturity f that maximizes $\mathbb{E}_{m,n,t}[Q_{t,t+1}R_{t+1}^{f-1}]$. Under the rational expectations, all members would choose the same maturity, thereby equalizing expected discounted returns across maturities and reproducing the expectations hypothesis in the log-linearized model. To generate a downward-sloping demand curve, we instead assume that individual expectations satisfy

$$\mathbb{E}_{m,n,t} [Q_{t,t+1}R_{t+1}^{f-1}] = z_{n,t}^f \mathbb{E}_t [Q_{t,t+1}R_{t+1}^{f-1}], \quad \forall f = 1, \dots, F,$$

where $z_{n,t}^f$ follows an i.i.d. Fréchet distribution with scale parameter z_t^f and shape parameter κ_B . Aggregating across members implies that the household's portfolio share of the

³⁰It follows that bond returns can alternatively be expressed as $R_t^f = \frac{(YD_t^f)^{-f}}{(YD_{t-1}^{f+1})^{-(f+1)}}$.

f -maturity bond in total government bond holdings, $\lambda_t^{HB,f} \equiv \frac{B_t^{H,f}}{B_t^H}$, is given by:

$$\lambda_t^{HB,f} = \left(\frac{z_t^f \mathbb{E}_t[Q_{t,t+1} R_{t+1}^{f-1}]}{\Phi_t^B} \right)^{\kappa_B}, \quad (15)$$

where $\Phi_t^B \equiv \left[\sum_{j=1}^F (z_t^j \mathbb{E}_t[Q_{t,t+1} R_{t+1}^{j-1}])^{\kappa_B} \right]^{\frac{1}{\kappa_B}}$. Here, κ_B is the elasticity of portfolio demand for the f -maturity bond with respect to its expected discounted return.³¹ We assume $\kappa_B > 0$ throughout the paper.

Additionally, we introduce z_t^f as a shock to the portfolio preference for the f -maturity bond. For example, an increase in z_t^f raises the household's f -maturity share of government bond holdings, given returns and other macroeconomic states. In equilibrium, this shock will raise the price of the f -maturity bond Q_t^f and induce households to reallocate their portfolios toward other maturities and private loans, thus pushing down their returns.³² This shock, combined with the household's endogenous portfolio "rebalancing", is an important source of variation that generates an upward-sloping yield curve both at and away from the steady state.³³

Given equation (15), the aggregate return on the household's bond portfolio is:

$$R_{t+1}^{HB} = \sum_{f=0}^{F-1} \lambda_t^{HB,f+1} R_{t+1}^f. \quad (16)$$

Bond vs. Loans Family At the upper nest, family m compares the aggregate bond return R_{t+1}^{HB} with the loan return R_{t+1}^K . Under benchmark rational expectations, all families would make the same choice, equalizing $\mathbb{E}_t[Q_{t,t+1} R_{t+1}^{HB}]$ and $\mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$. We instead assume that family-specific expectations for loans satisfy

$$\mathbb{E}_{m,t} [Q_{t,t+1} R_{t+1}^K] = z_{m,t}^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K],$$

³¹In our framework, $\mathbb{E}_t[Q_{t,t+1} R_{t+1}^{f-1}]$ does not equal 1 due to the inelasticity of demand in (15), consistent with the literature.

³²Based on our calibration and the estimated value of κ_B , decreases in returns of other assets are smaller than the decrease in the original return R_{t+1}^f . This produces a phenomenon known as *localization* of demand shocks, as documented in Vayanos and Vila (2021) and Droste et al. (2021).

³³In the context of the preferred-habitat literature (Vayanos and Vila, 2021), this shock can be interpreted as a shock to the preferred-habitat demand for the f -maturity bond.

where $z_{m,t}^K$ is a Fréchet shock with scale parameter z_t^K and shape parameter κ_S . Aggregating across families yields the household's portfolio share of total savings allocated to loans:

$$\lambda_t^K = \left(\frac{z_t^K \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]}{\Phi_t^S} \right)^{\kappa_S}, \quad (17)$$

where $\Phi_t^S = [(\mathbb{E}_t[Q_{t,t+1} R_{t+1}^{HB}])^{\kappa_S} + (z_t^K \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K])^{\kappa_S}]^{\frac{1}{\kappa_S}}$. Here, κ_S is the elasticity of the portfolio demand for private loans with respect to their expected discounted return.³⁴

We assume $\kappa_S > 0$ throughout the paper. As the bond return R_{t+1}^{HB} depends on the bond portfolio $\{\lambda_t^{HB,f}\}_{f=1}^F$, given by (15), our portfolio demand follows a nested constant elasticity of substitution (CES) structure.

Additionally, we introduce z_t^K as a shock to the portfolio preference for private loans. Similar to the case for $\{z_t^f\}$, this shock induces households to rebalance their portfolios, which affects the equilibrium returns of bonds with different maturities and private loans.³⁵

Based on (15) and (17), the aggregate bond holdings by maturity f are:

$$B_t^{H,f} = (1 - \lambda_t^K) \lambda_t^{HB,f} S_t, \quad \forall f = 1, \dots, F,$$

and the aggregate return on household savings is:

$$R_t^S = (1 - \lambda_{t-1}^K) R_t^{HB} + \lambda_{t-1}^K R_t^K. \quad (18)$$

Therefore, R_t^S is a weighted average of returns from bonds (across maturities) and loans,³⁶ representing the effective savings rate of households. The household's budget constraint (14) can be rewritten as:

$$C_t + \frac{S_t}{P_t} = \frac{R_t^S S_{t-1}}{P_t} + \int_0^1 \frac{W_t(\nu) N_t(\nu)}{P_t} d\nu + \frac{\Lambda_t}{P_t}. \quad (19)$$

Despite the richer asset structure and market segmentation, the representative household's problem ultimately resembles that of a conventional New Keynesian model. This

³⁴Again, $\mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$ deviates from 1 in our framework, due to the assumed inelasticity of demand in (17), consistent with the literature.

³⁵Equation (17) indicates that the household allocates a larger share of their portfolio to loans when loan returns R_{t+1}^K rise, and reduce this share when the aggregate bond return R_{t+1}^{HB} increases.

³⁶While our analysis assumes a one-period duration for private loans, extending the model to incorporate a term structure in the loan market is feasible. However, given our focus on unconventional monetary policy via central bank purchases of government bonds, we restrict our discussion to the term structure for bonds to maintain model tractability.

structural representation offers two advantages. First, the estimated demand elasticities—see Online Appendix B—summarize market segmentation across maturities. Second, it nests the benchmark New Keynesian model as a special case.

Our microfoundation is related to a literature that derives smooth aggregate choice from discrete individual decisions under limited information. [Matějka and McKay \(2015\)](#) show that rational inattention in a discrete-choice environment yields generalized multinomial-logit choice probabilities. [Woodford \(2009, 2011, 2014a,b\)](#) develop closely related treatments in which information-processing constraints generate discrete adjustment, inattentive valuation, and stochastic choice. Our framework is similar in spirit, though not identical in formulation: each household member makes a discrete portfolio choice by selecting the asset or maturity with the highest perceived expected return, while aggregation across heterogeneously informed decision makers produces smooth portfolio shares and finite own- and cross-elasticities in a multi-asset term-structure environment.

Expectations Hypothesis Note that as $\kappa_B \rightarrow \infty$, $\kappa_S \rightarrow \infty$, and $z_t^f = z_t^K = 1$ for all f, t , our framework converges to a conventional New Keynesian model represented by:

$$\mathbb{E}_t[Q_{t,t+1}R_{t+1}^{f-1}] = \mathbb{E}_t[Q_{t,t+1}R_{t+1}^K] = 1, \quad \forall f. \quad (20)$$

In its log-linearized form, equation (20) becomes:

$$\mathbb{E}_t[\hat{r}_{t+1}^{f-1}] = \mathbb{E}_t[\hat{r}_{t+1}^K] = \hat{r}_{t+1}^0, \quad \forall f, \quad (21)$$

which renders all returns other than the policy rate $\hat{r}_{t+1}^0$ redundant. Equation (21) corresponds to the classic expectations hypothesis. Thus, our model specification with inelastic financial markets nests the conventional expectations hypothesis result as a special case.

3.1.2 Optimality Conditions

The solution to the household's problem (13), subject to the budget constraint (19), yields the following equilibrium conditions:

$$\left(\frac{N_t(\nu)}{\bar{N}_t}\right)^{\frac{1}{\eta}} = \left(\frac{C_t}{\bar{N}_t}\right)^{-1} \frac{W_t(\nu)}{P_t}, \quad (22)$$

$$1 = \beta \mathbb{E}_t \left[\frac{R_{t+1}^S C_t}{C_{t+1} \Pi_{t+1}} \right], \quad (23)$$

where $\Pi_{t+1} = P_{t+1}/P_t$ is the gross inflation rate. In (23), the effective savings rate R_{t+1}^S serves as a composite interest rate for intertemporal consumption decisions, reflecting the returns on bonds of various maturities and loans, each weighted by its share in the aggregate portfolio. As a result, the endogenous portfolio rebalancing channel becomes central to the business cycle dynamics under forward-looking unconventional monetary policies.

Remarks Our findings on imperfect substitutability across assets align with bond-in-the-utility models that feature finite elasticities (Alpanda and Kabaca, 2020) and with models incorporating portfolio adjustment costs (Costa, 2019). However, our approach retains the standard aggregate dynamics—see equations (22) and (23)—which improves tractability.³⁷

In contrast to the preferred-habitat literature, our framework does not depend on financial arbitrageurs to generate an upward-sloping yield curve. Instead, the financial market is segmented, with the extent of segmentation directly governed by the demand elasticities κ_B and κ_S . Household portfolios are also exposed to asset-specific and maturity-specific preference shocks $z_{t,f=1}^{fF}$ and z_t^K , which, together with central bank and government balance sheet policies, produce equilibrium variation in term premia across the yield curve.

3.2 Capital Producer

A representative capital firm produces capital K_t and rents it to intermediate goods producers at price P_t^K . Capital is accumulated by investing final goods, depreciates at rate δ , and new investment I_t is implemented with a one-period lag. Hence, capital evolves according to

$$K_t = (1 - \delta)K_{t-1} + I_{t-1}.$$

The capital firm's profits are given by

$$\Lambda_t^K = P_t^K K_t - P_t I_t, \quad (24)$$

where P_t denotes the final good price index. Profit maximization with respect to I_t yields the following standard first-order condition:

$$1 = \mathbb{E}_t \left[Q_{t,t+1} \Pi_{t+1} \left((1 - \delta) + \frac{P_{t+1}^K}{P_{t+1}} \right) \right], \quad (25)$$

³⁷Harrison (2024) likewise introduces a portfolio friction that generates a gap between short- and long-term bond returns, enabling the central bank's quantitative easing policy to be effective.

where $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ is the gross inflation rate.

3.3 Firms

There is a continuum $\nu \in [0, 1]$ of intermediate varieties, each produced by a monopolist using capital and labor according to

$$Y_t(\nu) = \left(\frac{K_t(\nu)}{\alpha} \right)^\alpha \left(\frac{A_t N_t(\nu)}{1 - \alpha} \right)^{1-\alpha}, \quad (26)$$

where $A_t = \exp(u_t^A)$ denotes aggregate labor-augmenting productivity with $u_t^A = \mu + u_{t-1}^A + \varepsilon_t^A$, $\varepsilon_t^A \sim N(0, \sigma_A^2)$, and the gross growth rate is $GA_t \equiv A_t/A_{t-1} = \exp(\mu + \varepsilon_t^A)$.

A representative competitive firm aggregates intermediate varieties into a final good using a Dixit–Stiglitz aggregator:

$$Y_t = \left[\int_0^1 Y_t(\nu)^{\frac{\epsilon-1}{\epsilon}} d\nu \right]^{\frac{\epsilon}{\epsilon-1}},$$

with $\epsilon > 1$ as the elasticity of substitution. The demand for intermediate good ν is

$$Y_t(\nu) = \left(\frac{P_t(\nu)}{P_t} \right)^{-\epsilon} Y_t, \quad (27)$$

and the aggregate price index is defined as

$$P_t = \left[\int_0^1 P_t(\nu)^{1-\epsilon} d\nu \right]^{\frac{1}{1-\epsilon}}. \quad (28)$$

Intermediate producers face sticky prices à la **Calvo (1983)** and reset their prices each quarter with probability $1-\theta$. Firms that adjust set the optimal price P_t^* , and the equilibrium aggregate price index satisfies $P_t^{1-\epsilon} = (1-\theta)(P_t^*)^{1-\epsilon} + \theta P_{t-1}^{1-\epsilon}$.

Working-Capital Constraint and Loan Market Each intermediate firm rents capital from the capital producer at price P_t^K . To finance production, firm ν must borrow a fraction γ_L of its subsidized revenue, $(1+\zeta^F)P_t(\nu)Y_t(\nu)$, where ζ^F denotes the production subsidy. A firm borrowing $L_t(\nu)$ in period t repays $R_{t+1}^K L_t(\nu)$ in period $t+1$, where the gross loan rate R_{t+1}^K is contracted at time t . This requirement introduces a financial friction in the form of a working-capital constraint.

Loan supply has two sources, households' loans L_t^H and central bank credit L_t^{CB} ; the latter is a stylized representation of the central bank's purchases of corporate bonds under quantitative easing.³⁸ In equilibrium the loan market clears, so aggregate firm borrowing equals total loan supply, $L_t \equiv \int_0^1 L_t(\nu) d\nu = L_t^H + L_t^{CB}$.

Firm ν maximizes its discounted profit stream:

$$\max \sum_{j=0}^{\infty} \mathbb{E}_t \left[Q_{t,t+j} \left((1 + \zeta^F) P_{t+j}(\nu) Y_{t+j}(\nu) - W_{t+j}(\nu) N_{t+j}(\nu) - P_{t+j}^K K_{t+j}(\nu) - R_{t+j}^K L_{t+j-1}(\nu) + L_{t+j}(\nu) \right) \right], \quad (29)$$

where $Q_{t,t+j} = \beta^j \left(\frac{P_{t+j}}{P_t} \frac{C_{t+j}}{C_t} \right)^{-1}$ is the stochastic discount factor between periods t and $t+j$. At period $t+j$, firm ν repays $R_{t+j}^K L_{t+j-1}(\nu)$ on the loan taken in period $t+j-1$.

Minimizing production costs with respect to labor and capital yields the input demand functions:

$$N_t(\nu) = (1 - \alpha) \frac{Y_t(\nu)}{A_t} \left(\frac{\frac{P_t^K}{P_t}}{\frac{W_t(\nu)}{P_t A_t}} \right)^{\alpha}, \quad \frac{K_t(\nu)}{A_t} = \alpha \frac{Y_t(\nu)}{A_t} \left(\frac{\frac{P_t^K}{P_t}}{\frac{W_t(\nu)}{P_t A_t}} \right)^{-(1-\alpha)}. \quad (30)$$

3.4 Bond Market

The bond market equilibrium condition is given by

$$B_t^{H,f} + B_t^{G,f} + B_t^{CB,f} = 0, \quad \forall f = 1, \dots, F, \quad (31)$$

where $B_t^{H,f}$ represents households' holdings of bonds with maturity f bonds, while $B_t^{G,f}$ and $B_t^{CB,f}$ denote the bond positions of the government³⁹ and the central bank, respectively. We assume that only the government and the central bank are able to issue risk-free claims and therefore may take on negative bond positions.⁴⁰

³⁸Modeling the central bank as an intermediary that channels credit to firms follows [Gertler and Karadi \(2011\)](#). In practice, several central banks have purchased corporate bonds directly as part of their quantitative easing operations: the European Central Bank began acquiring investment-grade euro-denominated corporate bonds under its Corporate Sector Purchase Programme in June 2016 ([Zaghini, 2019](#); [Grosse-Rueschkamp et al., 2019](#)), and the Federal Reserve made its first direct purchases of corporate bonds through the Primary and Secondary Market Corporate Credit Facilities in 2020 ([Boyarchenko et al., 2022](#)).

³⁹By assumption, the government issues bonds across maturities, so $B_t^{G,f} \leq 0$ for all f .

⁴⁰In practice, a negative position of the central bank can be interpreted as issuing interest-bearing excess reserves—for this, see [Frost \(1971\)](#), [Güntner \(2015\)](#), [Mattingly and Abou-Zaid \(2015\)](#), [Primus \(2017\)](#), and

Allowing the government to issue bonds—and maintain negative positions—is crucial. Our assumptions regarding technology growth GA_t and population growth GN ensure that, in steady state, government debt remains positive yet stable, avoiding explosive dynamics. As a result, the government continues to supply risk-free debt even under cyclical fluctuations.

Defining $\lambda_t^{G,f}$ and $\lambda_t^{CB,f}$ as portfolio shares of f -maturity bonds held by the government and the central bank, respectively, we can rewrite (31) as

$$\lambda_t^{HB,f} B_t^H + \lambda_t^{G,f} B_t^G + \lambda_t^{CB,f} B_t^{CB} = 0, \quad \forall f = 1, \dots, F. \quad (32)$$

3.5 Government

The government's budget constraint is given by

$$G_t + \zeta^F Y_t + \frac{B_t^G}{P_t} = T_t + \frac{R_t^G B_{t-1}^G}{P_t}, \quad \text{with } B_t^G = \sum_{f=1}^F B_t^{G,f}, \quad R_t^G = \sum_{f=0}^{F-1} \lambda_{t-1}^{G,f+1} R_t^f. \quad (33)$$

B_t^G denotes the government's nominal bond position, G_t represents real government spending, T_t captures tax revenues, and R_t^G is the overall return on the government's bond portfolio. Here, $\lambda_t^{G,f}$ indicates the share of government debt allocated to bonds of maturity f . By assumption, the government's holdings satisfy $B_t^{G,f} = \lambda_t^{G,f} B_t^G$ for all $f = 1, \dots, F$, with both $\lambda_t^{G,f}$ and B_t^G treated as exogenous.⁴¹

Rearranging (33), the budget constraint can be written as

$$\frac{B_t^G}{P_t} = \frac{R_t^G B_{t-1}^G}{P_t} - [\zeta_t^G + \zeta^F - \zeta_t^T] Y_t, \quad (34)$$

where $\zeta_t^G = \frac{G_t}{Y_t}$ and $\zeta_t^T = \frac{T_t}{Y_t}$ denote government spending and taxation as shares of GDP, respectively, and are exogenous in our framework.

Ennis (2018).

⁴¹We abstract from the government's optimal maturity structure problem and assume its gross bond positions and portfolios across maturities are exogenous, focusing on the central bank's monetary policy. For this issue, see e.g., Bhandari et al. (2017); Aparisi de Lannoy et al. (2022).

3.6 Central Bank

The profits earned on the corporate loans and government bonds held by the central bank are given by

$$\begin{aligned} \Lambda_t^{CB} &= R_t^K L_{t-1}^{CB} + R_t^{CB} B_{t-1}^{CB} - (L_t^{CB} + B_t^{CB}), \\ \text{with } B_t^{CB} &= \sum_{f=1}^F B_t^{CB,f}, \quad R_t^{CB} = \sum_{f=0}^{F-1} \lambda_{t-1}^{CB,f+1} R_t^f, \end{aligned} \quad (35)$$

where B_t^{CB} denotes the central bank's total nominal bond holdings, and R_t^{CB} is the aggregate bond return on its portfolio $\{B_{t-1}^{G,f}\}_{f=1}^F$. The share of bonds allocated to maturity f is given by $\lambda_t^{CB,f}$. Formally,

$$B_t^{CB,f} = \lambda_t^{CB,f} \cdot B_t^{CB}, \quad \forall f = 1, \dots, F. \quad (36)$$

Both B_t^{CB} and $\lambda_t^{CB,f}$ depend on the monetary policy rules, which will be described shortly. The profits earned on central bank balance sheets Λ_t^{CB} are transferred as a lump-sum payment to the household, forming part of the total transfer Λ_t in (14).

3.7 Monetary Policy

The above equation (36) introduces F additional conditions, so that the central bank's monetary policy now has F degrees of freedom that must be specified to achieve a determinate nominal equilibrium.⁴² The monetary authority implements policy by one of the following three approaches:

Policy 1 Setting a rule for each f -maturity bond's holdings $B_t^{CB,f}$, allowing bond prices (and yields) to adjust accordingly.

Policy 2 Setting a rule for each f -maturity bond's yield YD_t^f —or equivalently, its price Q_t^f , and then adjusting the purchase amounts $B_t^{CB,f}$ to achieve the target yield.

Policy 3 Employing a combination of the two approaches across different maturities.

In **Policy 1**, the central bank directly controls the expansion of its bond holdings, resem-

⁴²The central bank selects its bond portfolio across maturities and its total debt position. In conventional New Keynesian models, this portfolio problem is typically abstracted away since the explicit term structure is absent, as discussed in Section 3.1.1.

bling traditional rules on money supply growth.⁴³ **Policy 2** exemplifies yield-curve control (YCC) policies, as implemented by the Bank of Japan in 2016.⁴⁴ **Policy 3** combines elements of both and includes the traditional short-term rate target of conventional monetary policy.

While unconventional policies could be implemented according to any of the three cases above, we assume that their defining characteristic is an intent to influence returns along the entire yield curve (in contrast to conventional policies, which focus on short-term rates). Consequently, we choose a YCC policy rule as the representative unconventional policy within our framework. In what follows, we formally characterize the equations describing both conventional and YCC policy rules.

Conventional Policy Conventional monetary policy targets the short-term interest rate in **Policy 3**: the central bank sets a Taylor rule for the one-period bond yield YD_t^1 , while normalized long-term bond positions follow the exogenous portfolio-holding process below. We assume the central bank's normalized long-term positions are adjusted for technology, population, and price growth as follows:

$$R_{t+1}^0 \equiv YD_t^1 = \max \left\{ YD_t^{1*}, 1 \right\}, \quad (37a)$$

$$\frac{YD_t^{1*}}{\bar{YD}^1} = \left(\frac{YD_{t-1}^{1*}}{\bar{YD}^1} \right)^{\rho_1} \left(\frac{YD_{t-2}^{1*}}{\bar{YD}^1} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\gamma_\pi} \left(\frac{Y_t}{\bar{Y}_t} \right)^{\gamma_y} \exp \left(\tilde{\varepsilon}_t^{YD^1} \right) \right]^{1-(\rho_1+\rho_2)}, \quad (37b)$$

$$\frac{B_t^{CB,f}}{A_t \bar{N}_t P_t} = \frac{\bar{B}^{CB,f}}{\bar{A} \bar{N} \bar{P}} \exp \left(\tilde{u}_t^{CB,f} \right), \quad \forall f = 2, \dots, F, \quad (37c)$$

where YD_t^{1*} follows a standard Taylor rule that responds to deviations in inflation and output from the steady state, and $\tilde{\varepsilon}_t^{YD^1}$ denotes a standard monetary policy shock. The term $\tilde{u}_t^{CB,f}$ denotes a shock to the central bank's bond holdings, following the exogenous process described in Appendix I. When YD_t^{1*} falls below 1, the ZLB binds and $R_{t+1}^0 \equiv YD_t^1 = 1$, as specified in (37a).

⁴³This approach reflects the idea that bond purchases increase money supply. It also specifies the portfolio composition across maturities, even though we do not explicitly incorporate money as a separate variable.

⁴⁴On September 21, 2016, the Bank of Japan set a short-term rate target of -0.1% and capped its 10-year government bond yield near zero. For the U.S. case, see e.g., [Humpage \(2016\)](#) regarding the Fed's yield-curve control during World War II.

Yield-Curve Control Policy Yield-curve control targets the entire yield curve by applying a Taylor rule to each bond maturity, including the short-term rate rule in (37). For each maturity $f \geq 2$, the YCC target $YD_t^{YCC,f}$, is given by the equation below. For the short maturity, the same Taylor/ZLB rule applies

$$\frac{YD_t^{YCC,f}}{\overline{YD}^f} = \left(\frac{YD_{t-1}^{YCC,f}}{\overline{YD}^f} \right)^{\rho_1} \left(\frac{YD_{t-2}^{YCC,f}}{\overline{YD}^f} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\overline{\Pi}} \right)^{\gamma_\pi^f} \left(\frac{Y_t}{\overline{Y}_t} \right)^{\gamma_y^f} \exp \left(\tilde{\varepsilon}_t^{YD^f} \right) \right]^{1-(\rho_1+\rho_2)}, \quad (38)$$

where γ_π^f and γ_y^f capture the responsiveness to inflation and output deviations for maturities $f = 1, \dots, F$, and $\tilde{\varepsilon}_t^{YD^f}$ is a monetary policy shock affecting the yield on an f -maturity bond. The shock processes, including the factor representation of $\{\tilde{\varepsilon}_t^{YD^f}\}_{f=1}^F$, are specified in Appendix I.

More broadly, one can consider a more general yield policy $\{YD_t^{GP,f}\}_{f=1}^F$ that combines the yields implied by conventional policy and YCC, nesting both as special cases:

$$YD_t^{GP,f} = \overline{YD}^f \left(\frac{YD_t^{CP,f}}{\overline{YD}^f} \right)^{\gamma_{CP}^f} \left[\frac{YD_t^{YCC,f}}{\overline{YD}^f} \right]^{1-\gamma_{CP}^f}, \quad (39)$$

where $\gamma_{CP}^f \in [0, 1]$ governs the weight on conventional policy relative to yield-curve control. When $\gamma_{CP}^f = 1$, the maturity- f yield follows conventional policy; when $\gamma_{CP}^f = 0$, it follows yield-curve control.

Corporate Loan Policy For simplicity, we assume the central bank sets a target for the corporate loans L_t^{CB} it holds on its balance sheet and lets the loan rate R_t^K adjust endogenously to clear the loan market. Formally, the target is given by:

$$\frac{L_t^{CB}}{A_t \bar{N}_t P_t} = \overline{\left(\frac{L^{CB}}{ANP} \right)} \cdot \exp(u_t^L), \quad (40)$$

where the balanced-growth-path target $\overline{\left(\frac{L^{CB}}{ANP} \right)}$ is a fraction γ_{CB} of balanced-growth-path real output, $\overline{\left(\frac{L^{CB}}{ANP} \right)} = \overline{\left(\frac{Y}{AN} \right)} \gamma_{CB}$, and u_t^L is an AR(1) process with shock ε_t^L , specified in Appendix I. We therefore treat expansions of the central bank's corporate loan portfolio as an exogenous shock, though the rule could be extended to respond to inflation and output deviations, as in (37b).

3.8 Market Clearing

Using the bond market equilibrium condition (31), we obtain the total transfers to households—comprising profits from firms, the central bank, and capital producers, net of government taxes as

$$\Lambda_t = P_t Y_t - P_t G_t - P_t I_t - \int_0^1 W_t(\nu) N_t(\nu) d\nu - R_t^K L_{t-1}^H + L_t^H + B_t^H - R_t^{HB} B_{t-1}^H. \quad (41)$$

Combining (41) with the household's budget constraint (14) yields the standard aggregate market-clearing condition:

$$C_t + G_t + I_t = Y_t. \quad (42)$$

3.9 Aggregation

Aggregating the labor demand—equation (30)—across intermediate firms yields:

$$\frac{N_t}{\bar{N}_t} = (1 - \alpha)^{\left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{C_t}{A_t \bar{N}_t}\right)^{-\alpha \left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{Y_t}{A_t \bar{N}_t}\right)^{\left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{P_t^K}{P_t}\right)^{\alpha \left(\frac{\eta}{\eta+\alpha}\right)} \Delta_t^{\frac{\eta}{\eta+1}}, \quad (43)$$

where Δ_t is a measure of price dispersion, recursively defined as:

$$\Delta_t = (1 - \theta) \left(\frac{P_t^*}{P_t}\right)^{-\epsilon \left(\frac{\eta+1}{\eta+\alpha}\right)} + \theta \Pi_t^{\epsilon \left(\frac{\eta+1}{\eta+\alpha}\right)} \Delta_{t-1}. \quad (44)$$

Note from equation (43) that, for a given (normalized) level of consumption and output, the required labor N_t increases with price dispersion Δ_t —a proxy for the inefficiency induced by nominal rigidities. Moreover, a higher $\frac{P_t^K}{P_t}$ raises the rental cost of capital, prompting firms to substitute capital with labor, thereby raising N_t . This substitution channel is also captured in the following aggregate capital equilibrium condition.⁴⁵

$$\frac{K_t}{A_{t-1} \bar{N}_{t-1}} = \alpha (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot G A_t \cdot G N \cdot \left(\frac{C_t}{A_t \bar{N}_t}\right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_t}{A_t \bar{N}_t}\right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t}\right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha}\right)} \Delta_t. \quad (45)$$

⁴⁵We normalize K_t by $A_{t-1} \bar{N}_{t-1}$, as K_t is determined in period $t-1$, while N_t , C_t , and Y_t are normalized by $A_t \bar{N}_t$, and P_t^K is normalized by P_t .

Consequently, aggregate capital K_t increases when consumption, output, or price dispersion increases and/or when the rental price of capital decreases. These two equations highlight the role of firms' substitution between capital and labor during production.

The household satisfies the Euler equation (23), with effective savings rate R_{t+1}^S determined by (18), λ_t^K given by (17), and $\lambda_t^{HB,f}$ specified in (15). The equilibrium condition for the household's allocation between loans and bonds is:

$$\frac{L_t^H}{B_t^H} = \frac{\gamma_L(1 + \zeta^F)P_t Y_t - L_t^{CB}}{B_t^H} = \frac{\lambda_t^K}{1 - \lambda_t^K}, \quad (46)$$

where B_t^H and L_t^H represent the aggregate household bond and loan holdings, respectively.

3.9.1 Conventional Policy

Under conventional monetary policy (37), the central bank sets the short-term rate and lets normalized long-term bond positions follow the exogenous process in (37c), so the long-maturity yields $\{YD_t^f\}_{f \geq 2}$ adjust endogenously to clear the bond market. In this setting, the household's bond portfolio share, $\lambda_t^{HB,f}$, must satisfy:

$$\lambda_t^{HB,f} = -\frac{\frac{B_t^{G,f}}{A_t N_t P_t} + \frac{B_t^{CB,f}}{A_t N_t P_t}}{\frac{B_t^H}{A_t N_t P_t}}, \quad \forall f > 1. \quad (47)$$

3.9.2 Yield-Curve Control Policy

In the yield-curve control case—equation (38)—monetary policy affects households' bond portfolio across maturities $\{\lambda_t^{HB,f}\}_{f=1}^F$ and the effective bond return R_t^{HB} via the following relations:

$$\lambda_t^{HB,f} = \left(\frac{z_t^f \mathbb{E}_t [Q_{t,t+1} R_{t+1}^{f-1}]}{\Phi_t^B} \right)^{\kappa_B}, \quad R_{t+1}^{f-1} = \frac{(YD_{t+1}^{f-1})^{-(f-1)}}{(YD_t^f)^{-f}}.$$

Changes in the household's bond portfolio composition affect the effective bond rate R_t^{HB} from (16), the loan rate R_t^K from (17), and the effective savings rate R_t^S from (18). These adjustments, in turn, influence consumption via the Euler equation (23) and other aggregate outcomes as captured by equations (42), (43), (45), and (46).

Figure III.2 in Appendix III provides a graphical illustration of the model. Appendix

I provides shock processes for portfolio preference shocks $\{z_t^f\}_{f=1}^F$ and z_t^K , government spending ratio ζ_t^G and the revenue ratio ζ_t^T , government bond shares $\{\lambda_t^{G,f}\}_{f=1}^F$, and monetary policy shocks $\{\tilde{\varepsilon}_t^{YDf}\}_{f=1}^F$.

We present all other equilibrium conditions and derivations in Online Appendix A.

4 Steady-State (Long-Run) Analysis

4.1 Steady-State Relations

We assume that the central bank's total bond holdings, B^{CB} , equal a constant fraction ζ^{CB} of the total government bond issuance, B^G ; that is, $B^{CB} = \zeta^{CB} B^G$.⁴⁶ Given the portfolio shares $\{\lambda^{CB,f}\}_{f=1}^F$, the steady-state relation in the bond market—equation (32)—becomes:

$$\lambda^{HB,f} = \frac{\lambda^{G,f} + \lambda^{CB,f} \zeta^{CB}}{1 + \zeta^{CB}}. \quad (48)$$

Thus, the household's steady-state bond portfolio shares $\{\lambda^{HB,f}\}_{f=1}^F$ are determined by the exogenous parameters $\{\lambda^{G,f}, \lambda^{CB,f}\}_{f=1}^F$ and ζ^{CB} .

At the steady state, the government's budget constraint can be expressed as:

$$\frac{B^G}{ANP} = - \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right)^{-1} [\zeta^G + \zeta^F - \zeta^T] \frac{Y}{AN}. \quad (49)$$

Given the normalized output level $\frac{Y}{AN}$ and a positive primary deficit ratio $\zeta^G + \zeta^F - \zeta^T > 0$, an increase in the interest rate on government debt, R^G , results in a larger volume of bond issuance $|B^G|$ (recall that $B^G < 0$) and a higher debt-to-output ratio, as the government must pay more in interest on its debt.⁴⁷

The remaining steady-state relationships and the procedures for characterizing these conditions are detailed in Online Appendix A.1.2.

⁴⁶Since $B^G < 0$ and $B^{CB} > 0$ at the steady state, it follows that $\zeta^{CB} < 0$.

⁴⁷To ensure a positive primary deficit through non-explosive bond issuance at the steady state, we require that $R^G < \Pi \cdot GA \cdot GN$, a condition that holds under our calibration. This condition is consistent with recent literature on debt sustainability, such as Blanchard (2019).

4.2 Results

4.2.1 Calibration

We use publicly available data on (i) Treasury yields, (ii) the Federal Reserve’s holdings of Treasury bonds (December 2002–June 2007), and (iii) the U.S. Treasury’s outstanding bonds (January 1990–January 2007) to calibrate the model’s term structure parameters. We set $F = 120$ to represent maturities up to 30 years (i.e., 120 quarters).⁴⁸ Parameter values are summarized in Table II.2 in Appendix II, with standard parameters calibrated to widely accepted values in the literature.

Household demand for maturity- f bonds, $\lambda_t^{HB,f}$, is driven by the maturity-specific preference shock z_t^f and the demand elasticity κ_B ; similarly, loan demand λ_t^K is driven by z_t^K and κ_S . We calibrate the portfolio preference parameters $\{z^f\}_{f=1}^F$ based on the slope of the yield curve, and z^K based on the level of the yield curve, with fixed values for the demand elasticities κ_B and κ_S . Specifically, we set $\kappa_B = 10$, estimated directly from macro data using our bond portfolio equation (15) in Online Appendix B,⁴⁹ and $\kappa_S = 6$, based on [Kekre and Lenel \(2023\)](#). With these values, $\{z^f\}$ is calibrated to match the relative yields across maturities, whereas z^K is calibrated to reproduce the steady-state return on the household bond portfolio, R^{HB} . The detailed calibration procedure is provided in Online Appendix B.1. We set $\gamma_L = 3$, which constitutes an upper bound for private debt-to-GDP ratios in advanced economies. We set $\gamma_{CB} = 0.01$ to reflect the fact that the Federal Reserve’s Primary and Secondary Market Corporate Credit Facilities in 2021 amounted to roughly 1% of nominal GDP.

Under our calibration, the untargeted steady-state value of R^K is 8.12%, close to the average Moody’s Seasoned BAA Corporate Bond Yield of 7.88% during 1990–2007.

Figure 1 illustrates the distribution of bond holdings across maturities for households, the government, and the central bank, as well as the implied steady-state yield curve. The calibrated values of z^K and $\{z^f\}_{f=1}^F$ are reported in Table II.1 and Figure II.1 in Appendix II. Notably, $z^1 = 1$ is relatively large compared to z^f for $f \geq 2$, reflecting the fact that short-term yields have historically been lower than longer-term yields. This gap likely reflects a safety and/or liquidity premium on short-term bonds, a well-documented feature in the literature—see, for example, [Krishnamurthy and Vissing-Jorgensen \(2012\)](#) and [Caballero](#)

⁴⁸When yield data are missing, we interpolate to generate a smooth yield curve. Data source: <https://fiscaldata.treasury.gov/datasets/monthly-statement-public-debt>

⁴⁹Note that our κ_B estimate is in the range of the elasticity estimates in [Jansen et al. \(2024\)](#), who find that elasticity estimates range roughly from 3.8 to 436.6 across different granular investors.

and Farhi (2017).

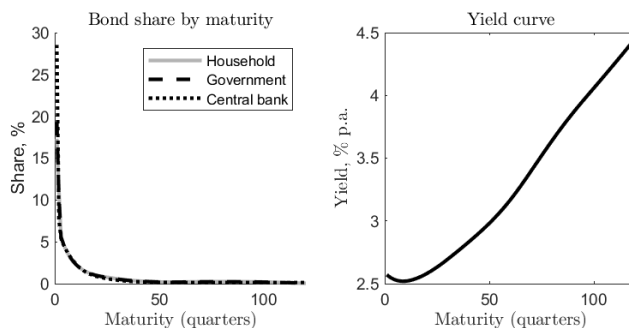


Figure 1: Steady-state bond shares of different entities, with the yield curve

Lastly, our calibration of Taylor coefficients, γ_π , $\gamma_\pi^{f \geq 2}$, γ_y , $\gamma_y^{f \geq 2}$, satisfies model determinacy both in conventional policy and yield-curve control policy regimes, where increasing monetary responsiveness to output and inflation generically guarantees that our model yields unique equilibrium. We characterize the determinacy conditions in Online Appendix G.

4.2.2 Supply and Demand for Bonds

We now examine how changes in the government's debt structure—captured by its Treasury issuance shares $\{\lambda^{G,f}\}_{f=1}^F$ —affect the yield curve at the steady state.⁵⁰ Figure 2 shows alternative debt issuance arrangements (left panel) and the corresponding yield curve shifts (right panel); dashed and dotted lines indicate higher long-term issuance compared to the benchmark (solid line).⁵¹ The model generates a positive correlation between yields and the relative supply $\lambda^{G,f}$; higher issuance at a given maturity increases its yield. This effect not only targets that maturity but also influences equilibrium returns in both Treasury and pri-

⁵⁰For the composition experiment in Sections 4.2.2 and 4.2.3, we add a small uniform maturity-weight vector to the benchmark government portfolio, using increments of 0.01 and 0.03, and then renormalize the portfolio shares so that they sum to one.

⁵¹For illustration purposes, the shifts in portfolio shares are arbitrary.

vate loan markets via household portfolio rebalancing, thereby affecting the government's overall bond issuance in general equilibrium.

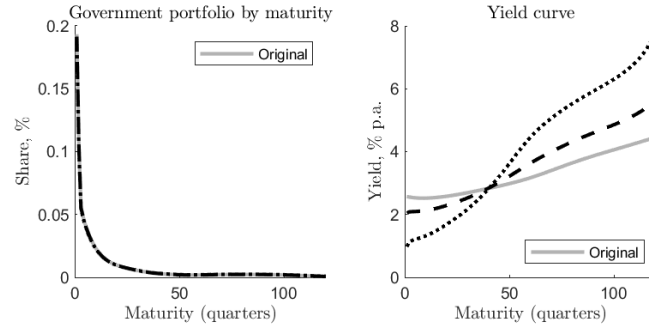


Figure 2: Government's bond issuance portfolio and yield curve

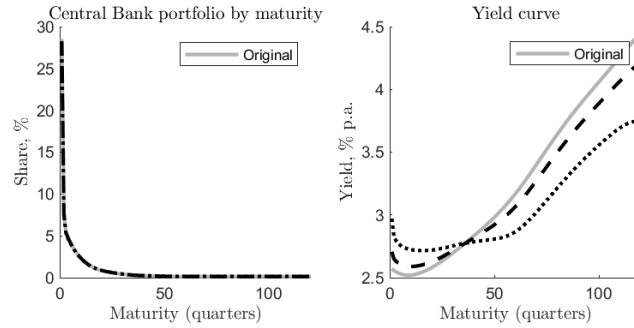


Figure 3: Variations in central bank's bond portfolio across maturities

Figure 3 illustrates an alternative scenario where the central bank adjusts its bond portfolio composition. Here, the dashed and dotted lines represent higher long-term bond purchases relative to the benchmark (solid line). Note that an increased relative purchase of a

given maturity is associated with a lower yield, which is consistent with evidence that central bank bond purchases act as an additional demand shock in segmented markets under general equilibrium.⁵²

4.2.3 Comparative Statics with Deficit Ratio

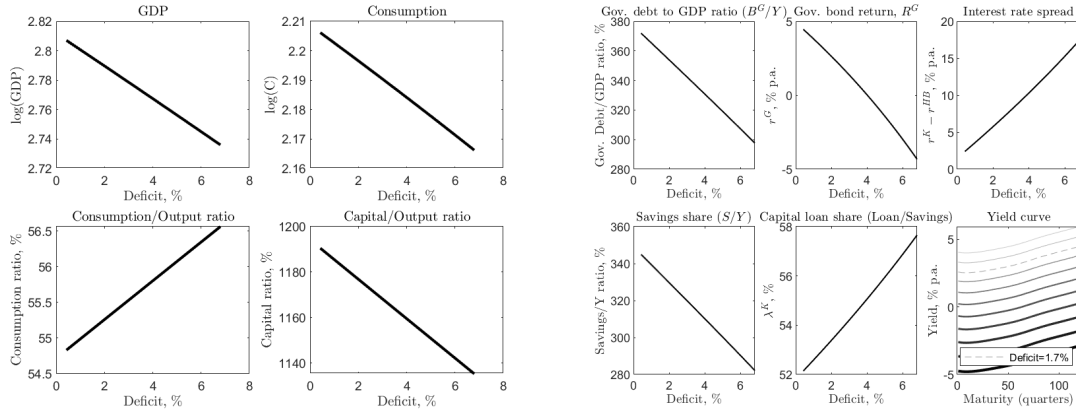


Figure 4: Variations in deficit ratio $\zeta^F + \zeta^G - \zeta^T$

Figure 4 shows comparative statics for changes in the deficit ratio $\zeta^F + \zeta^G - \zeta^T$. A higher deficit ratio can be sustained only if (i) the government issues more bonds, (ii) the effective bond rate R^G decreases, or (iii) output declines, thereby lowering nominal deficit expenditure.

Examining the first case, if the government issues more debt for a given output level, the supply effect discussed in Section 4.2.2 raises R^G , which in turn forces even more debt issuance to cover higher interest costs—a process that spirals indefinitely. In contrast, the second and third cases operate jointly: a higher deficit ratio reduces output, consumption, and capital, which lowers the nominal deficit and government bond issuance, thereby depressing R^G . Meanwhile, the loan rate R^K remains relatively stable, leading to an increased credit spread $r^K - r^{HB}$. Notably, our finding that the debt-to-GDP ratio $\frac{B^G}{Y}$ declines and the entire yield curve shifts downward in response to an increased deficit ratio aligns with

⁵²Krishnamurthy and Vissing-Jorgensen (2011) show that QE2, which concentrated on Treasury markets, had a larger effect on Treasuries and Agency bonds than on mortgage-backed securities and corporate bonds. D'Amico and King (2013) similarly find evidence of both stock and flow effects of QEs on Treasury yields, consistent with imperfect substitutability in the Treasury market.

previous literature.⁵³ Households in turn shift their portfolios towards loans, increasing the capital loan share λ^K .

Additional comparative statics results are provided in Online Appendix E.

5 Short-Run Analysis

5.1 Log-linearization

We now present the log-linearized solution of the model. Lower-case letters denote normalized variables, e.g., $k_t \equiv \frac{K_t}{A_{t-1}N_{t-1}}$, $y_t \equiv \frac{Y_t}{A_tN_t}$, $c_t \equiv \frac{C_t}{A_tN_t}$, $n_t \equiv \frac{N_t}{N_t}$, $p_t^K \equiv \frac{P_t^K}{P_t}$, while hats denote log-deviations from the steady state. Given our system's complexity, we highlight a few key equilibrium equations here; complete derivations are provided in Online Appendix A.

Linearizing the Euler equation (23) yields the dynamic IS equation featuring the effective savings rate $\hat{r}_{t+1}^S$.⁵⁴

$$\hat{c}_t = \mathbb{E}_t [\hat{c}_{t+1} - (\hat{r}_{t+1}^S - \hat{\pi}_{t+1})], \quad (50)$$

where $\hat{r}_t^S$ is derived from (18) as

$$\hat{r}_t^S = \frac{\lambda^K (R^K - R^{HB})}{R^S} \hat{\lambda}_{t-1}^K + \frac{(1 - \lambda^K) R^{HB}}{R^S} \hat{r}_t^{HB} + \frac{\lambda^K R^K}{R^S} \hat{r}_t^K. \quad (51)$$

Note that $\hat{r}_t^S$ depends on the effective bond rate $\hat{r}_t^{HB}$, the loan rate $\hat{r}_t^K$, and the loan share $\hat{\lambda}_{t-1}^K$, capturing the portfolio rebalancing between bonds and loans. The effective bond rate $\hat{r}_t^{HB}$ is obtained by linearizing (16):

$$\hat{r}_t^{HB} = \sum_{f=1}^F \frac{\lambda^{HB,f} (Y D^{f-1})^{-(f-1)}}{R^{HB} (Y D^f)^{-f}} \left[\hat{\lambda}_{t-1}^{HB,f} - (f-1) \cdot \hat{y} d_t^{f-1} + f \cdot \hat{y} d_{t-1}^f \right]. \quad (52)$$

It depends on past yields $\{\hat{y} d_{t-1}^f\}_{f=1}^F$, current yields $\{\hat{y} d_t^{f-1}\}_{f=1}^F$, and household bond portfolio shares $\{\hat{\lambda}_{t-1}^{HB,f}\}_{f=1}^F$, capturing the impact of holding-period returns and portfolio composition on overall bond returns.

To study portfolio reallocation, we log-linearize the household's optimal bond portfolio

⁵³Laubach (2009) estimates that a 1% point increase in the projected debt-to-GDP ratio raises long-term interest rates by approximately 3–4 basis points.

⁵⁴Note the similarity between this equation and the one in the simple model—see equation (1).

equation (15) and express the bond shares as functions of past and current yields:

$$\hat{\lambda}_{t-1}^{HB,f} = \kappa_B \mathbb{E}_{t-1} \left[\hat{z}_{t-1}^f - \hat{\pi}_t + \hat{c}_{t-1} - \hat{c}_t - (f-1) \cdot \hat{y}d_t^{f-1} + f \cdot \hat{y}d_{t-1}^f - \hat{\phi}_{t-1}^B \right], \quad (53)$$

where $\hat{\phi}_t^B$ depends on $\{\hat{y}d_t^{f-1}, \hat{y}d_{t-1}^f\}_{f=1}^F$ and other aggregate variables. Substituting equation (53) into (52) represents the household's effective bond rate as a function of the entire yield curve.

Finally, linearizing the household's optimal portfolio choice between bonds and loans (17) yields:

$$\hat{\lambda}_t^K = \kappa_S (1 - \lambda^K) (\hat{z}_t^K + \mathbb{E}_t [\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]). \quad (54)$$

Here, increases in $\hat{z}_t^K$ and in the expected spread $\mathbb{E}_t[\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]$ raise the loan share $\hat{\lambda}_t^K$, which affects capital accumulation and output via supply-side channels. As $\hat{r}_{t+1}^K$ directly affects the effective savings rate $\hat{r}_{t+1}^S$ in equation (50), it affects consumption demand via the intertemporal substitution channel.

5.2 Results

5.2.1 Impulse-Response without the ZLB

We first analyze impulse responses to various shocks in an economy that remains above the ZLB. The shocks considered are z_t^1 (household preference shock for short-term bonds), z_t^K (household preference shock for loans), and ε_t^T (tax shock). Graphs for additional shocks appear in Online Appendix F.

Short-term Bond Preference Shock, z_t^1 : Figure 5a indicates that a positive shock in z_t^1 increases households' demand for shortest-maturity bonds, which carry the lowest returns along the steady-state yield curve. Under conventional policy (dashed lines), the shock is expansionary: households rebalance their portfolios toward lower-yield short-term bonds, reducing the aggregate bond return R_{t+1}^{HB} . This induces households to allocate more savings toward loans, which boosts capital accumulation of firms and output through supply-side channels, and raises the effective savings rate. Therefore, the central bank raises the short-term rate to mitigate the expansionary effects of the shock.⁵⁵ Note that, although the shock

⁵⁵This increase in the short-term policy rate, in equilibrium, induces households to reallocate more savings toward shorter-term bonds, thereby amplifying the portfolio reallocation toward the shortest-maturity bonds.

is expansionary, the effective term premium for the government, $\mathbb{E}_t[R_{t+1}^G - R_{t+1}^0]$,⁵⁶ declines, which is consistent with the literature (Muir, 2017; Kekre et al., 2023).

While conventional policy would typically induce an expansion, YCC achieves near-perfect stabilization as it mitigates portfolio reallocation along the yield curve.

Loan Preference Shock, z_t^K : Figure 5b illustrates that a positive z_t^K shock causes households to allocate more savings toward loans. This portfolio reallocation increases aggregate capital and output via supply-side channels, which in turn boosts inflation as aggregate investment demand increases as well. As a result, the central bank raises the policy rate.⁵⁷ As in the previous case, YCC dampens these fluctuations more effectively than conventional policy.⁵⁸

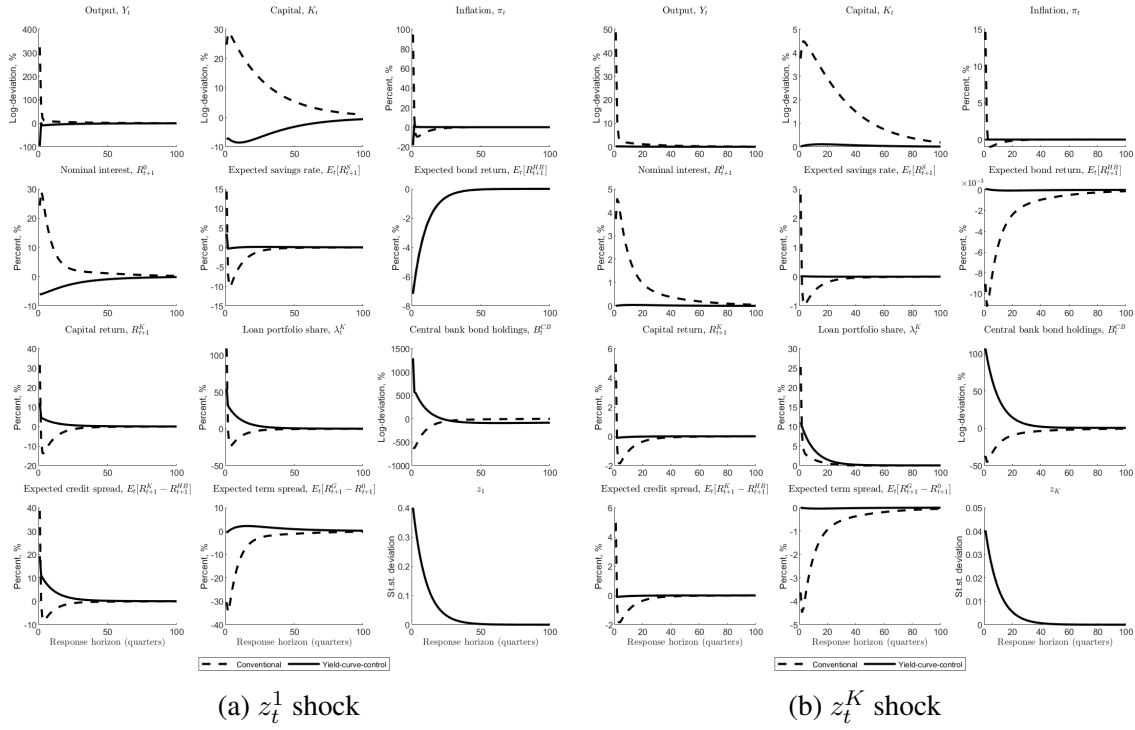


Figure 5: Impulse response to z_t^1 and z_t^K without ZLB

⁵⁶This represents the effective average term premium across the entire yield curve as perceived by the government.

⁵⁷Again, that while the shock is expansionary, the effective term premium, i.e., $\mathbb{E}_t[R_{t+1}^G - R_{t+1}^0]$, falls,

⁵⁸In this case, YCC policy shifts the entire yield curve upward, diluting the impact of the shock on aggregate capital and output

Tax Shock, ε_t^T : Figure 6 presents the impulse responses to a positive tax shock, ε_t^T . Under the conventional policy regime, higher tax revenues reduce the government’s issuance of bonds. In equilibrium, this generates a recession characterized by lower capital demand, output, inflation, and bond returns.

In response to the downturn, the central bank lowers the policy rate R_{t+1}^0 . The decline in short-term interest rates encourages households to shift a larger share of their savings toward longer-term bonds, increasing the effective household bond return, R_{t+1}^{HB} . The higher return on bonds further incentivizes households to allocate savings away from capital investment and toward bond holdings, reducing the loan share λ_t^K . As a result, fewer loans are available to firms, leading to lower investment, output, and capital demand, reinforcing the recessionary effects of the shock.⁵⁹ Note again the effective term premium for households, $\mathbb{E}_t[R_{t+1}^{HB} - R_{t+1}^0]$, rises as the economy undergoes a recession.⁶⁰

More broadly, Figure 6 highlights *non-Ricardian* features of our model, as tax shocks directly influence the business cycle via their effect on government bond issuance and the household’s portfolio reallocation.

In contrast, the YCC policy achieves near perfect stabilization because the entire yield curve shifts downward in response to the shock.

These results highlight that while conventional policy may induce significant business cycle fluctuations following shocks, a yield-curve control regime can more effectively stabilize the economy by actively managing the entire term structure.⁶¹

5.2.2 Impulse-Response at the ZLB

This section presents impulse responses to various shocks when the ZLB is binding. To illustrate independent ZLB episodes, we calibrate shocks to extremely large—though highly unlikely—magnitudes.⁶²

⁵⁹This result is consistent with Li and Merkel (2025), who find that a reduction in the supply of safe assets can lead to a recession even outside the ZLB.

⁶⁰This safe-asset shortage issue can be understood as follows. If financial markets were perfectly elastic and assets were perfect substitutes, a reduction in the supply of safe assets would lead investors to rebalance smoothly into other assets, causing returns across markets to adjust together in equilibrium. The macroeconomic effects in that case would be relatively limited. In our model, however, assets are imperfect substitutes. When the supply of safe bonds falls, their prices rise and their yields decline, but households cannot fully substitute into other assets. As a result, they continue to hold bonds despite their becoming relatively overpriced. This reduces aggregate demand, generating contractionary effects on the economy.

⁶¹See Online Appendix H for robustness of our impulse responses to alternative calibrations.

⁶²Note that the model is log-linearized around the steady state, and the ZLB introduces a piecewise-linear structure: outside the ZLB, the policy rate follows the Taylor-type rule; inside the ZLB, the short rate is fixed

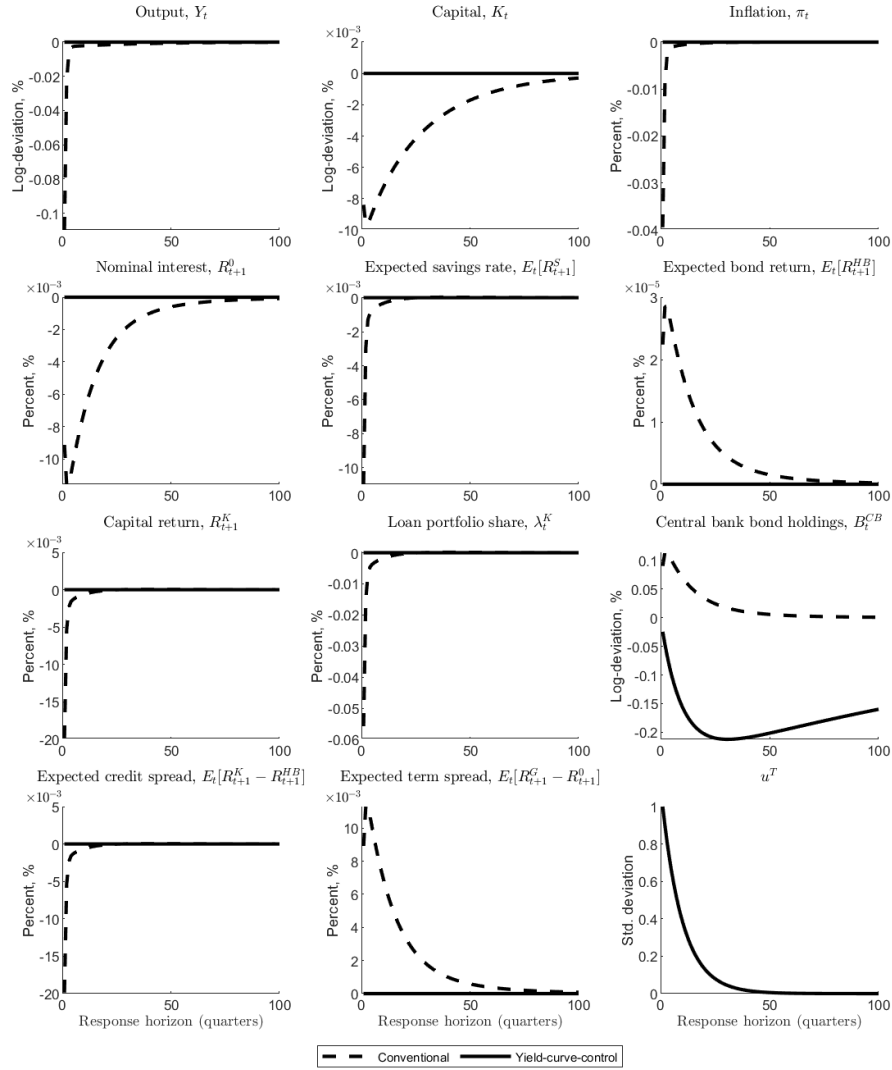


Figure 6: Impulse-response to ε_t^T shock without ZLB

Short-term Bond Preference Shock, z_t^1 : Figure 7a presents that a negative z_t^1 shock—decreasing household demand for shortest-term bonds—lowers the short-term rate under conventional policy (dashed lines) and generates a recession with deflation and output drop by 3–4% per standard deviation. In contrast, YCC (solid lines) significantly mitigates the recession.

However, this intervention also extends the ZLB episode, as the downward pressure on long-term yields feeds back into the household’s portfolio reallocation. Households further

at its lower bound and the corresponding policy equation is replaced by the binding constraint. For solution methods, see Coibion et al. (2012); Dordal i Carreras et al. (2016).

tilt their portfolios toward loans, tightening the zero lower bound on the short-term rate as the loan rate declines. This, in turn, puts downward pressure on inflation, as firms substitute labor with relatively cheaper capital, further lengthening the ZLB episode.

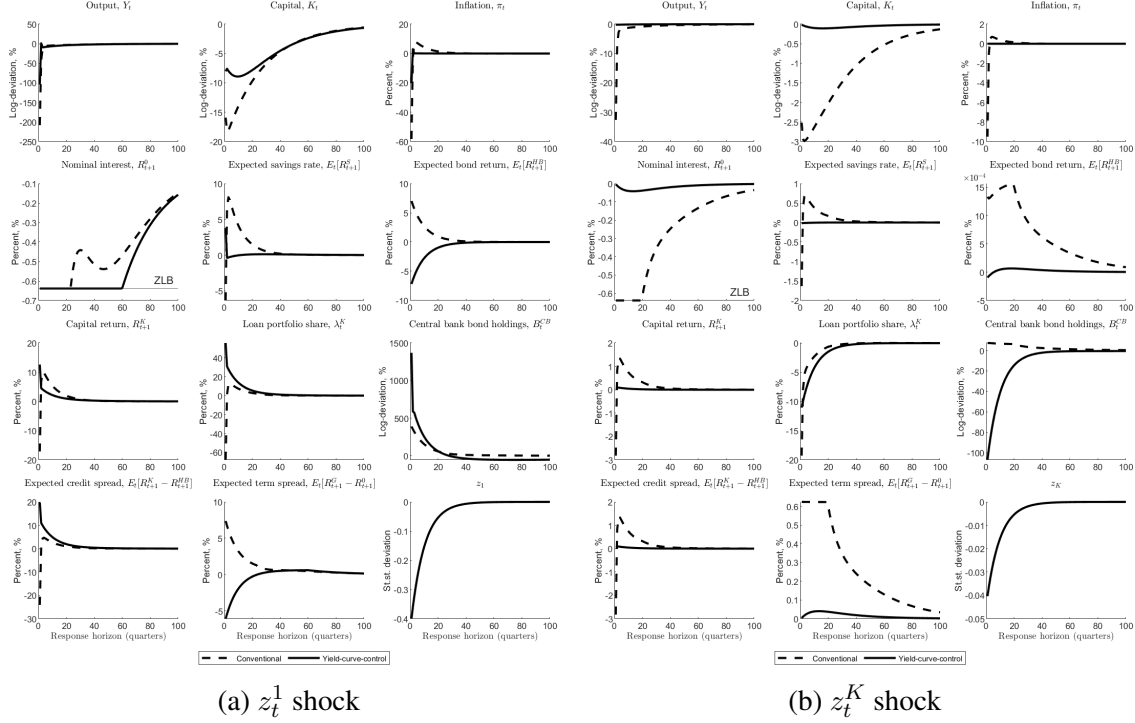


Figure 7: Impulse response to z_t^1 and z_t^K with ZLB

Loan Preference Shock, z_t^K : Figure 7b depicts that a sizable negative z_t^K shock induces households to reduce loan investments and reallocate their portfolios toward bonds. Consequently, bond rates decline and the policy rate remains at the ZLB, while output, capital, inflation, and loan rates all fall. YCC achieves superior stabilization so the economy almost never hits the ZLB.

Tax Shock, ε_t^T : Figure 8 illustrates that, under conventional policy, a positive tax shock leads to reduced bond issuance. Similar to Figure 6, the resulting declines in bond issuance, along with weaker investment demand, interact with endogenous portfolio adjustments by households and firms' input substitution to intensify the downturn—lowering output, capital, inflation, and returns to capital. Under YCC, the central bank compresses the entire yield curve, thereby achieving almost perfect stabilization. The economy almost never hits the ZLB under YCC.

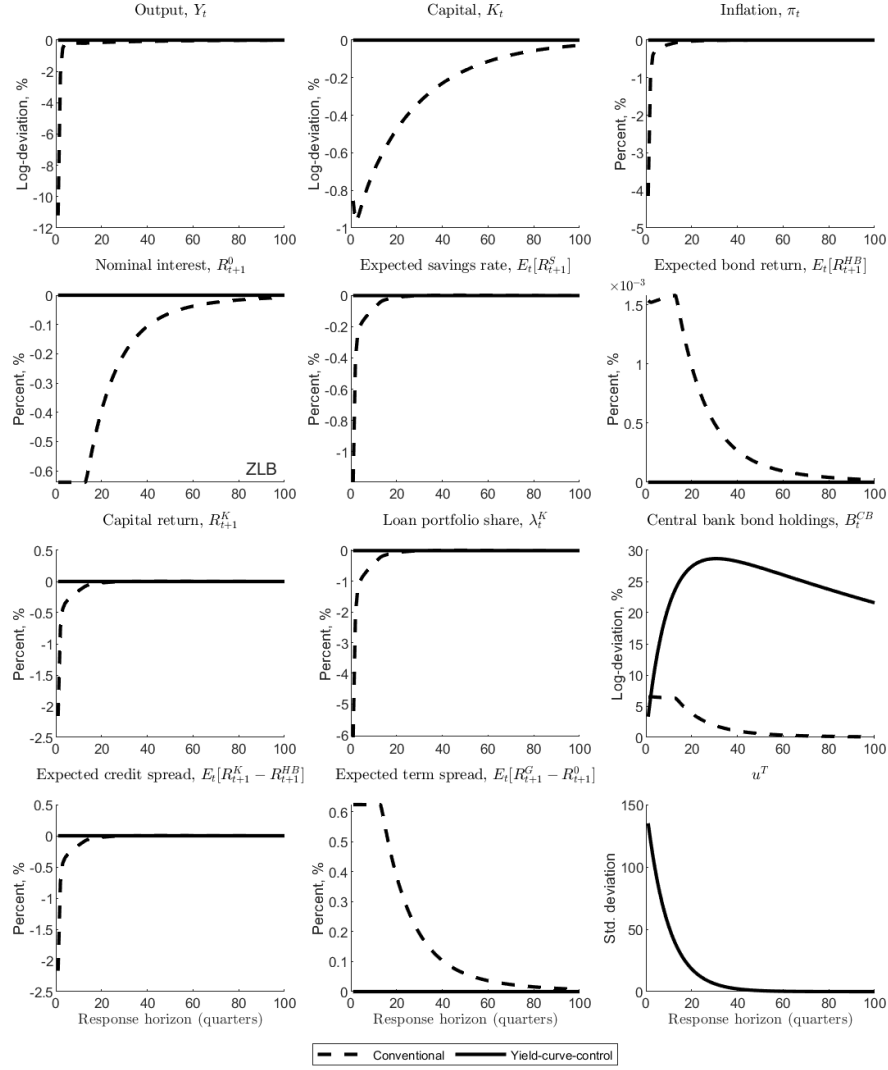


Figure 8: Impulse-response to ε_t^T shock with ZLB

In summary, while yield-curve control policies effectively stabilize the economy by offsetting shocks across the term structure, they tend to extend the duration of ZLB episodes compared with conventional policy, highlighting a trade-off in the stabilization benefits of unconventional monetary measures.

5.2.3 Quantity Shocks

We now examine the impulse responses to quantity shocks of the central bank, namely, a shock to the central bank's holdings of loans, u_t^L , and a shock to its holdings of long-term government bonds, $u_t^{CB,f}$. For illustration, we set $f = 20$ for the long-term bond shock.

As expected, an increase in the central bank's loan holdings, u_t^L , is expansionary. By increasing the supply of credit to firms, it stimulates capital accumulation and output through supply-side channels while also raising the effective return on household savings. Quantitatively, a 1% increase in central bank loan holdings raises output by approximately 0.6%. In response to the resulting expansion, the central bank raises the policy rate R_{t+1}^0 . As in the previous exercises, YCC is highly effective at stabilizing these fluctuations. Therefore, we can see that the Federal Reserve's liquidity injections via various programs such as the Main Street Lending Program and the Municipal Liquidity Facility are expansionary while the policy rate is at the ZLB.

Interestingly, a positive shock in the central bank's holdings of long-term bonds, $u_t^{CB,f}$, $f = 20$, is contractionary in our model. The mechanism is similar to the tax-shock case in Figure 6: by purchasing long-term government bonds, the central bank reduces the supply of safe assets available to households. In general equilibrium, this safe-asset shortage generates recessionary forces, lowering output, capital accumulation, inflation, and returns to capital, which is consistent with the literature on safe-asset shortages (Caballero and Farhi, 2017; Li and Merkel, 2025).^{63,64}

5.2.4 Policy Comparison

For policy comparison, we follow the literature (Woodford, 2003; Coibion et al., 2012) in calculating a second-order approximation of household utility.

Proposition 1 *A second-order approximation of the expected per-period household welfare is given by*

$$\mathbb{E}U_t - \bar{U}^F = \Omega_n \text{Var}(\hat{n}_t) + \Omega_\pi \text{Var}(\hat{\pi}_t) + t.i.p. + h.o.t.,$$

where Ω_n and Ω_π are coefficients that capture the disutility arising from fluctuations in labor and inflation, respectively; $\bar{U}^F$ denotes the efficient (i.e., flexible-price) steady-state⁶⁵ utility around which the approximation is centered; and the acronyms *t.i.p.* and *h.o.t.* refer to terms independent of policy and to higher-order terms, respectively.

⁶³YCC is automatically perfectly stabilizing as the quantity shock is absent under the YCC regime.

⁶⁴In Online Appendix I, we show that operation twist shocks, where the central bank changes the composition of its portfolio while leaving the balance sheet size fixed, are expansionary, while pure quantity shocks, where it changes the total balance sheet size while leaving the composition unaffected, can be contractionary.

⁶⁵Due to assumed trend inflation at the steady state ($\Pi > 1$), our steady state is not efficient; see Coibion et al. (2012).

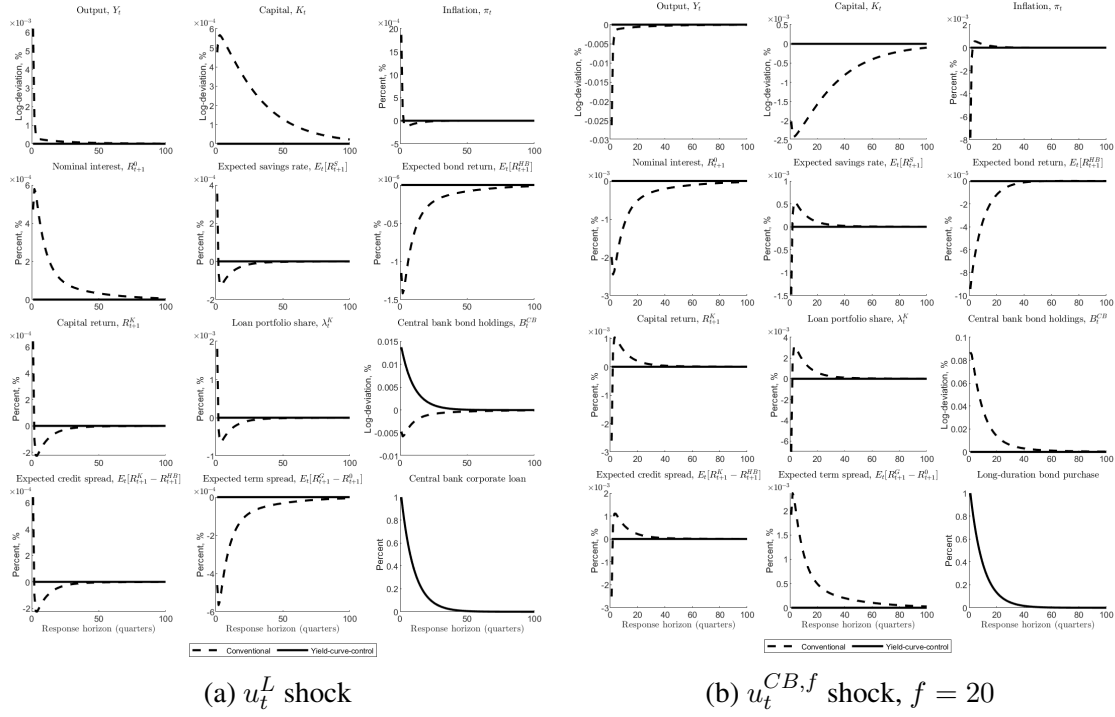


Figure 9: Impulse response to u_t^L and u_t^f , $f = 20$, without ZLB

Proof. See Online Appendix C. ■

Based on the welfare criterion provided in Proposition 1, we compare three monetary policy regimes: (i) conventional policy, represented by (37); (ii) yield-curve control (YCC) policy, represented by (38); and (iii) a *mixed* policy in which the central bank adopts YCC only when the policy rate is constrained by the ZLB.⁶⁶ We evaluate the three regimes using unconditional per-period welfare, mean and median ZLB duration, and ZLB frequency.⁶⁷

	Conventional	Yield-Curve-Control	Mixed Policy
Mean ZLB duration	4.425 quarters	2.375 quarters	2.9402 quarters
Median ZLB duration	2 quarters	2 quarters	1 quarters
ZLB frequency	14.303%	0.19192%	8.9394%
Welfare	-1.3425%	-0.99039%	-1.4067%

Table 1: Policy Comparisons

⁶⁶Upon exiting the ZLB, the central bank immediately resets its holdings of $f \geq 2$ -maturity bonds to their steady-state levels. This approach differs from Karadi and Nakov (2021), who argue that a gradual exit from QE is optimal given that banks face reduced recapitalization incentives without additional QE.

⁶⁷The reported welfare measure in Table 1 includes terms independent of policy (*t.i.p.*).

Table 1 shows that YCC delivers a substantial improvement in welfare relative to conventional monetary policy, increasing the welfare measure by 0.36 percentage points. Also, by enhancing macroeconomic stabilization, YCC decreases the unconditional frequency of ZLB episodes and shortens their average duration. This occurs even though, for some individual shocks, YCC can prolong the duration of a ZLB episode, as illustrated in Figure 7.

Compared with conventional policy, both the frequency and average duration of ZLB episodes are lower under the mixed policy. This is because YCC is activated whenever the short-term policy rate reaches the ZLB, providing an additional stabilization mechanism. The expectation of this policy support improves macroeconomic stabilization even before the ZLB is reached, reducing the likelihood that the economy enters a ZLB episode in the first place.

Despite these improvements, welfare is slightly lower than under conventional policy. The reason is that, for certain shocks, activating YCC can prolong the ZLB episode, making it more difficult for the economy to exit the ZLB. Although such episodes occur with relatively low probability, the associated welfare losses can be substantial, more than offsetting the benefits from the reduction in the overall frequency and duration of ZLB episodes.

6 Conclusion

This paper develops a New Keynesian model that incorporates the term structure of financial markets, as well as the size and maturity composition of government and central bank balance sheets, as active forces shaping the business cycle. The model underscores the importance of market segmentation—captured through households’ finite-elasticity portfolio demand across asset classes and maturities—and of endogenous portfolio reallocation in understanding unconventional monetary policy.

We show that government debt issuance and central bank purchases of bonds with different maturities are the major determinants of the yield curve’s level and slope. Furthermore, the issuance of risk-free government bonds can stimulate the economy when conventional monetary policy is constrained by the ZLB, aligning with the literature on safe-asset shortages. We compare three policy regimes: (i) conventional policy, (ii) yield-curve control (YCC), where the central bank actively manipulates the entire yield curve, and (iii) a mixed regime, where YCC is employed only when the ZLB binds. Our results suggest that YCC interventions offer greater stabilization than conventional policy, both during normal

periods and at the ZLB, though they also sometimes extend the duration of ZLB episodes under particular shocks such as shocks to z_t^1 .

We believe that our model offers a foundation for future research on quantitative tightening in high-inflation environments, particularly in the aftermath of the COVID-19 crisis and under geopolitical uncertainties. In addition, it enables an analysis of the political economy of the expansion of central bank balance sheets. Finally, extending the framework to an international macroeconomic context would be a valuable direction, allowing for a re-examination of issues such as global imbalances (e.g., [Caballero et al. \(2008, 2021\)](#)) and global monetary cycles ([Miranda-Agrippino and Rey, 2021](#)) while incorporating endogenous movements in the term structure of interest rates.

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I Shock Processes

We assume that the household portfolio preference shocks $\{z_t^f\}_{f=1}^F$ and z_t^K follow $AR(1)$ processes, given by $z_t^f = \rho_z z_{t-1}^f + \varepsilon_t^{z,f}$ with $\text{Var}(\varepsilon_t^{z,f}) = \sigma_z^2$, and $z_t^K = \rho_z^K z_{t-1}^K + \varepsilon_t^{z,K}$ with $\text{Var}(\varepsilon_t^{z,K}) = (\sigma_z^K)^2$.

For the government spending ratio ζ_t^G and the revenue ratio ζ_t^T , we assume the following shock processes:

$$\zeta_t^G = \frac{1}{1 + a^G \exp(-u_t^G)}, \quad \zeta_t^T = \frac{1}{1 + a^T \exp(-u_t^T)}, \quad (\text{I.1})$$

where a^G and a^T are constants, and u_t^G and u_t^T follow $AR(1)$ processes $u_t^G = \rho_G u_{t-1}^G + \varepsilon_t^G$ and $u_t^T = \rho_T u_{t-1}^T + \varepsilon_t^T$, with ε_t^G and ε_t^T being i.i.d. shocks.

As the government's bond shares across maturities, $\{\lambda_t^{G,f}\}_{f=1}^F$, are exogenously given, we specify their processes as follows:

$$\lambda_t^{G,1} = \frac{1}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \lambda_t^{G,f} = \frac{a^{B,f} \exp(\tilde{u}_t^{B,f})}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \forall f > 1, \quad (\text{I.2})$$

where $a^{B,f}$ for all $f > 1$ are constants. When F is large, we reduce the number of independent shocks to $J \leq F$ by assuming

$$\tilde{u}_t^{B,f} = \sum_{j=2}^J \tau_{fj}^B u_t^{B,j}, \quad (\text{I.3})$$

where $u_t^{B,j}$ follows an independent $AR(1)$ process, and τ_{fj}^B is a constant for all j, f .

The process for the central bank's bond holdings, $\tilde{u}_t^{CB,f}$, follows a similar construction, with shocks to each maturity written as a linear combination of $J^{CB} \leq F - 1$ independent factors indexed by $h = 1, \dots, J^{CB}$. Since the one-period bond is excluded, factor h maps to maturity $f = h + 1$ when the central-bank shocks are maturity specific:

$$\tilde{u}_t^{CB,f} = \sum_{h=1}^{J^{CB}} \tau_{fh}^{CB} u_t^{CB,h}, \quad (\text{I.4})$$

where $u_t^{CB,h}$ follows an independent $AR(1)$ process, and τ_{fh}^{CB} is constant for all h, f .

Similarly, we reduce the state space of shocks in the YCC regime by representing the monetary policy shocks $\{\tilde{\varepsilon}_t^{YD^f}\}_{f=1}^F$ for different maturities as linear combinations of several factors:

$$\tilde{\varepsilon}_t^{YD^f} = \sum_{l=1}^L \tau_{f,l}^{YD} \varepsilon_t^{YD^l}, \quad (\text{I.5})$$

where $\tau_{f,l}^{YD}$ (for all l, f) are constants, $\varepsilon_t^{YD^l}$ are i.i.d. shocks, and $L \leq F$.

Finally, we assume that the shock to the corporate loan portfolio of the central bank u_t^L follows an AR(1) process:

$$u_t^L = \rho_L u_{t-1}^L + \varepsilon_t^L, \quad (\text{I.6})$$

where ε_t^L is an i.i.d. shock.

II Calibration

Steady-state values		
$\frac{C}{AN}$	9.0108	Normalized consumption
$\frac{Y}{AN}$	16.3301	Normalized output
$\frac{K}{AN}$	192.6051	Normalized capital
$\frac{C}{Y}$	0.5518	Consumption per GDP
$\frac{K}{Y}$	11.7945	Capital per GDP
$\frac{P^K}{P}$	0.0335	Normalized rental price of capital
$\lambda^{HB,f}$	See Figure 1	Household's bond portfolio
λ^K	0.5314	Household's loan share out of total savings
R^K	1.0203	Household's loan rates (quarterly)
YD^f	See Figure 1	Equilibrium yield curve

Table II.1: Steady-state values derived from parameter calibration in Table II.2

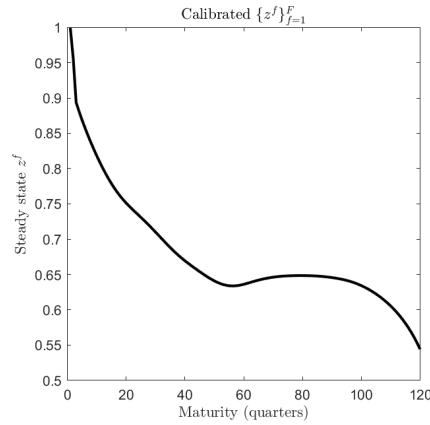


Figure II.1: Calibrated scale parameters of the Fréchet distribution, $\{z^f\}_{f=1}^F$

Households		
(β, η)	(0.998, 1)	Discount factor and Frisch labor elasticity
GN	1.00275	Gross population growth factor
Intermediate good firms		
μ	0.00375	Technology growth rate (log)
γ_L	3	Loan-to-GDP ratio
α	0.4	Capital income share
ϵ	10	Elasticity of substitution between differentiated goods
θ	0.45	Calvo price stickiness parameter
σ_A	0.0027	Standard deviation of technology shock
δ	0.025	Capital depreciation rate
Term structure		
$\{z^f\}_{f=1}^F$	See Figure II.1	Bond-maturity preference scale parameters (Appendix B.1)
z^K	1.00841	Capital-preference scale parameter (Appendix B.1)
κ_B	10	Bond maturity shape (volatility) parameter (Appendix B.2)
κ_S	6	Capital shape parameter: Kekre and Lenel (2023)
(ρ_z, ρ_z^K)	(0.9, 0.9)	Autoregressive coefficient: z_t^f and z_t^K
(σ_z, σ_z^K)	3×10^{-9}	Standard deviation for both z_t^f and z_t^K
Government		
ζ^F	0.11111	Government's optimal subsidy to firms
ζ^G	0.07889	Government expenditure per GDP (net subsidy ζ^F)
a^G	11.67606	Government expenditure coefficient
$\zeta^F + \zeta^G - \zeta^T$	0.017	Government deficit per GDP
ζ^T	0.173	Government tax revenue per GDP
(ρ_G, ρ_T)	(0.9, 0.9)	Autoregressive coefficient: expenditure and tax shocks
(σ_G, σ_T)	0.0011	Standard deviation for both expenditure and tax shocks
Central bank		
ζ^{CB}	-0.18	Central bank's balance sheet per issued government bond
γ_{CB}	0.01	Central bank corporate loan target (fraction of BGP output)
$\bar{\pi}$	0.00575	Trend inflation (steady-state inflation), quarterly rate
γ_{π}^1	1.5	Taylor coefficient of YD_t^1 : responsiveness to inflation
$\gamma_{\pi}^{f \geq 2}$	1.5	Taylor coefficient of $YD_t^{f \geq 2}$: responsiveness to inflation
γ_y	0.5	Taylor coefficient: responsiveness to output
$\gamma_y^{f \geq 2}$	0	Taylor coefficient of $YD_t^{f \geq 2}$: responsiveness to output
(ρ_1, ρ_2)	(1.05, -0.13)	Autocorrelation in monetary policy: Coibion et al. (2012)
$\sigma_{\varepsilon^{YD1}}$	0.00072	Standard deviation: monetary policy shock (for YD_t^1)
$\sigma_{\varepsilon^{YDf}}, f \geq 2$	3×10^{-9}	Standard deviation: monetary policy shock (for $YD_t^{f \geq 2}$)
$(\rho_{CB}, \sigma_{\varepsilon^{CB}})$	(0.9, 3×10^{-9})	Autoregressive coeff. and std. dev.: central bank bond-portfolio shock
$(\rho_L, \sigma_{\varepsilon^L})$	(0.9, 3×10^{-9})	Autoregressive coeff. and std. dev.: corporate loan shock
τ^{YD}	$I_{F \times F}$	State reduction matrix (for $YD_t^{f \geq 2}$)

Table II.2: Parameter values

III Additional Figures

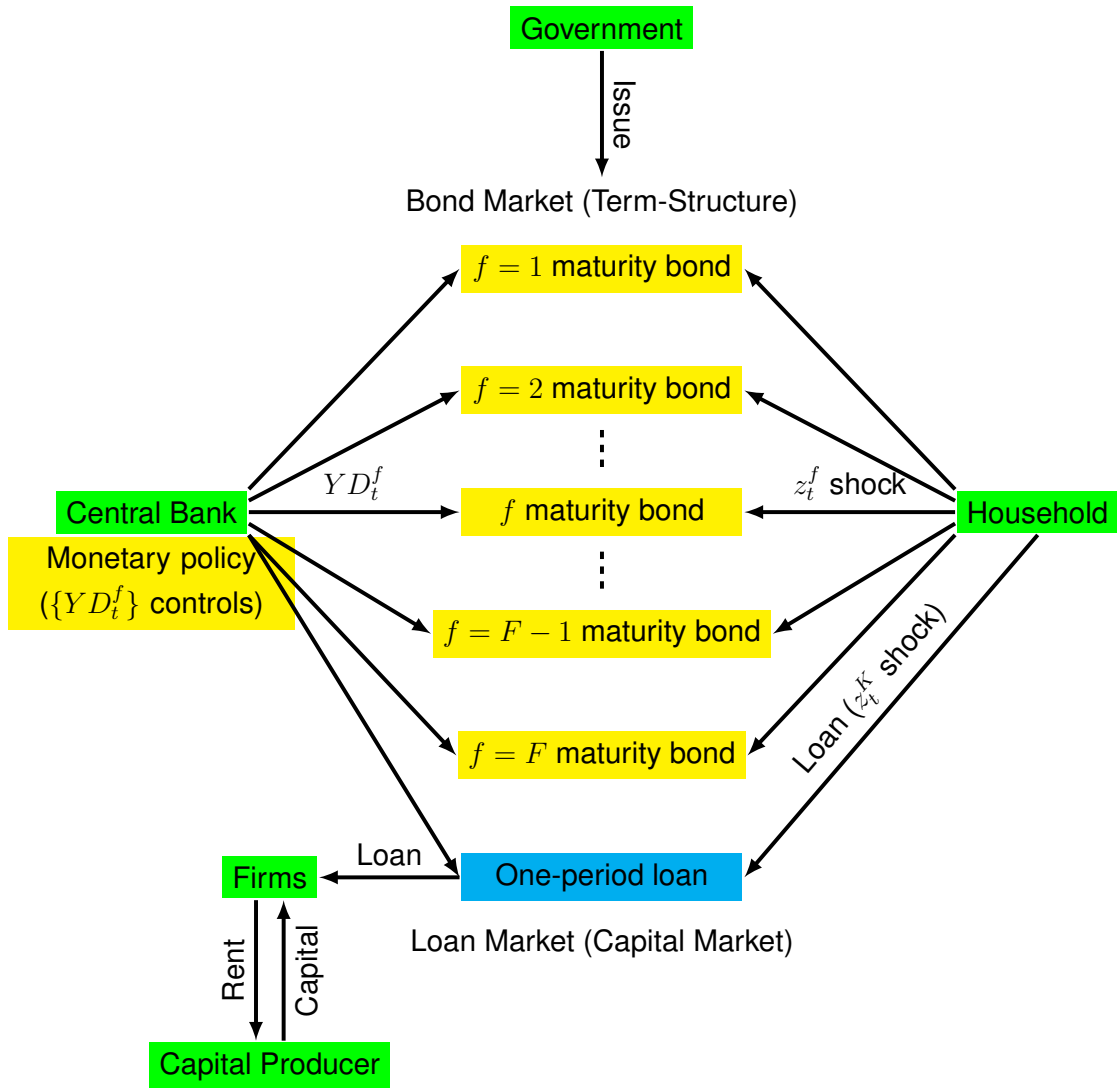


Figure III.2: Markets, agents, and mechanisms. Households allocate wealth between bonds and one-period loans to intermediate goods producers, the government issues bonds across maturities, and the central bank targets either the short rate alone or the full yield curve.

A Derivation and Proofs

A.1 Detailed Derivations in Section 3

As in (29), an intermediate firm ν resetting its price at period t maximizes ¹

$$\max \sum_{j=0}^{\infty} \mathbb{E}_t \left[\theta^j Q_{t,t+j} \cdot \left[\left[1 - \gamma_L \cdot \left(\tilde{R}_{t+j+1}^K - 1 \right) \right] \cdot (1 + \zeta^F) \cdot P_{t+j}(\nu) Y_{t+j}(\nu) \right. \right. \\ \left. \left. - W_{t+j}(\nu) N_{t+j}(\nu) - P_{t+j}^K K_{t+j}(\nu) \right] \right], \quad (\text{A.1})$$

where $Q_{t,t+j} = \beta^j \left(\frac{P_{t+j}}{P_t} \frac{C_{t+j}}{C_t} \right)^{-1}$ is the stochastic discount factor between periods t and $t+j$ and ζ^F is a production subsidy. Also, we define $\tilde{R}_{t+j+1}^K \equiv \mathbb{E}_{t+j}[Q_{t+j,t+j+1} R_{t+j+1}^K]$. Solving for the optimal resetting price at period t , P_t^* , we obtain

$$\frac{P_t^*}{P_t} = \frac{\mathbb{E}_t \left[\sum_{j=0}^{\infty} \theta^j Q_{t,t+j} \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon+1} Y_{t+j} \left(\frac{(1 + \zeta^F)^{-1} \epsilon}{\epsilon - 1} \right) \left(\frac{MC_{t+j|t}(\nu)}{P_{t+j}} \right) \right]}{\mathbb{E}_t \left[\sum_{j=0}^{\infty} \theta^j Q_{t,t+j} \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon} Y_{t+j} \left[1 - \gamma_L \cdot \left(\tilde{R}_{t+j+1}^K - 1 \right) \right] \right]}, \quad (\text{A.2})$$

where subindex $t+j|t$ represents the value of the variable conditional on the firm having reset its price last time at period t , and $MC_{t+j|t}(\nu)/P_t$ is the real marginal cost of production, defined as

$$\frac{MC_{t+j|t}(\nu)}{P_{t+j}} = \left(\frac{P_{t+j}^K}{P_{t+j}} \right)^{\alpha} \left(\frac{W_{t+j|t}(\nu)}{P_{t+j} A_{t+j}} \right)^{1-\alpha}. \quad (\text{A.3})$$

The expression for the one-period ahead stochastic discount factor can be expressed as

$$Q_{t,t+1} = \frac{\beta}{\Pi_{t+1} G A_{t+1} G N} \frac{\left(\frac{C_t}{A_t N_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} N_{t+1}} \right)} \quad (\text{A.4})$$

where Π_{t+1} , $G A_{t+1}$, and $G N$ stand the gross growth rates of inflation, TFP and population, respectively.

¹The factor θ^j is the probability that the price chosen at period t remains in effect through period $t+j$, so the resetting-price problem conditions on the path along which the firm does not reoptimize.

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Combining the expression for $Q_{t,t+1}$ in (A.4) with the household's Euler equation (23), we obtain

$$\mathbb{E}_t [Q_{t,t+1} R_{t+1}^S] = 1. \quad (\text{A.5})$$

A.1.1 Detailed Derivation in Section 3.9

Using (22), (26), (27), and (A.3), we can express firm-specific marginal costs as a function of the aggregate variables as

$$\frac{MC_{t+j|t}(\nu)}{P_{t+j}} = (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{C_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{P_{t+j}^K}{P_{t+j}} \right)^{\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \left(\frac{P_t^*}{P_{t+j}} \right)^{-\left(\frac{\epsilon(1-\alpha)}{\eta+\alpha} \right)}.$$

Similarly, we integrate the firm-level capital and labor demand—equation (30), with the wage and firm output substituted out via equations (22) and (27)—across the continuum of firms. We obtain the following capital and labor aggregation conditions.

$$\frac{K_t}{A_{t-1}\bar{N}_{t-1}} = \alpha(1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA_t \cdot GN \cdot \left(\frac{C_t}{A_t\bar{N}_t} \right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_t}{A_t\bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha} \right)} \Delta_t, \quad (\text{A.6})$$

$$\frac{N_t}{\bar{N}_t} = (1 - \alpha)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{C_t}{A_t\bar{N}_t} \right)^{-\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t\bar{N}_t} \right)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \Delta_t^{\frac{\eta}{\eta+1}}, \quad (\text{A.7})$$

where Δ_t is a measure of price-dispersion that can be recursively defined as

$$\Delta_t = (1 - \theta) \left(\frac{P_t^*}{P_t} \right)^{-\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} + \theta \Pi_t^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \Delta_{t-1}. \quad (\text{A.8})$$

Note that N_t is the aggregate labor index in (13), so the integration $\int_0^1 N_t(\nu)^{(\eta+1)/\eta} d\nu$ leading to (A.7) rescales the firm-level price-dispersion Δ_t by $\eta/(\eta+1)$.

From (A.2), we obtain

$$\begin{aligned} & \left(\frac{P_t^*}{P_t} \right)^{1+\epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right)} \\ &= \frac{\mathbb{E}_t \left[\sum_{j=0}^{\infty} (\theta\beta)^j (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \frac{(1 + \varsigma_F)^{-1}\epsilon}{\epsilon - 1} \left(\frac{C_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{-\alpha \frac{\eta+1}{\eta+\alpha}} \left(\frac{Y_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon \frac{\eta+1}{\eta+\alpha}} \left(\frac{P_{t+j}^K}{P_{t+j}} \right)^{\alpha \frac{\eta+1}{\eta+\alpha}} \right]}{\mathbb{E}_t \left[\sum_{j=0}^{\infty} (\theta\beta)^j \left(\frac{P_{t+j}}{P_t} \right)^{\epsilon-1} \left(\frac{C_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right)^{-1} \left(\frac{Y_{t+j}}{A_{t+j}\bar{N}_{t+j}} \right) \left[1 - \gamma_L \cdot (\tilde{R}_{t+j+1}^K - 1) \right] \right]}. \end{aligned} \quad (\text{A.9})$$

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We can simplify this expression as

$$\frac{P_t^*}{P_t} = \left(\frac{F_t}{H_t} \right)^{\frac{1}{1+\epsilon(\frac{1-\alpha}{\eta+\alpha})}}, \quad (\text{A.10})$$

where F_t and H_t are recursively written as

$$\begin{aligned} F_t &= (1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1} \right) \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-\alpha(\frac{\eta+1}{\eta+\alpha})} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{\alpha(\frac{\eta+1}{\eta+\alpha})} + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon(\frac{\eta+1}{\eta+\alpha})} F_{t+1} \right], \\ H_t &= \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-1} \frac{Y_t}{A_t \bar{N}_t} \left[1 - \gamma_L \cdot (\tilde{R}_{t+1}^K - 1) \right] + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon-1} H_{t+1} \right]. \end{aligned} \quad (\text{A.11})$$

Using (A.11), we obtain the following equilibrium price-resetting condition:

$$\frac{F_t}{H_t} = \left(\frac{1-\theta}{1-\theta \Pi_t^{\epsilon-1}} \right)^{\left(\frac{1}{\epsilon-1} \right) [1+\epsilon(\frac{1-\alpha}{\eta+\alpha})]}. \quad (\text{A.12})$$

Combining the market clearing condition (42), the definition of the government spending share (34), and the capital accumulation equation (24), we obtain the following expression:

$$\frac{C_t}{A_t \bar{N}_t} = (1 - \zeta_t^G) \left(\frac{Y_t}{A_t \bar{N}_t} \right) + \left(\frac{1-\delta}{G A_t \cdot G N} \right) \left(\frac{K_t}{A_{t-1} \bar{N}_{t-1}} \right) - \left(\frac{K_{t+1}}{A_t \bar{N}_t} \right). \quad (\text{A.13})$$

We now rewrite equation (23) as

$$1 = \beta \cdot \mathbb{E}_t \left[\frac{R_{t+1}^S}{\Pi_{t+1} G A_{t+1} G N} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \right]. \quad (\text{A.14})$$

Since R_{t+1}^S depends on bonds return R_{t+1}^{HB} and loans return R_{t+1}^K while shares of savings that flow into bonds $(1 - \lambda_t^K)$ and loans (λ_t^K) are endogenous, we start from analyzing R_{t+1}^{HB} . We can rewrite the aggregate return indices as functions of the bond yields $\{Y D_t^f\}_{f=1}^F$ as

$$R_t^j = \sum_{f=0}^{F-1} \lambda_{t-1}^{j,f+1} \frac{\left(Y D_t^f \right)^{-f}}{\left(Y D_{t-1}^{f+1} \right)^{-(f+1)}}, \quad j \in \{HB, G, CB\},$$

and also the household's bond portfolio share as

$$\lambda_t^{HB,f} = \frac{\mathbb{E}_t \left[\frac{\beta \cdot z_t^f}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \cdot \frac{(YD_{t+1}^{f-1})^{-(f-1)}}{(YD_t^f)^{-f}} \right]}{\Phi_t^B}^{\kappa_B}, \quad \forall f,$$

$$\Phi_t^B = \left[\sum_{j=1}^F \mathbb{E}_t \left[\frac{\beta \cdot z_t^j}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \cdot \frac{(YD_{t+1}^{j-1})^{-(j-1)}}{(YD_t^j)^{-j}} \right] \right]^{\kappa_B \frac{1}{\kappa_B}}.$$

Now we find the equilibrium condition for the bond shares of the agents. Using the bond market equilibrium condition (31), we obtain

$$\lambda_t^{HB,f} = \frac{B_t^{G,f} + B_t^{CB,f}}{B_t^G + B_t^{CB}} = \frac{\lambda_t^{G,f} B_t^G + \lambda_t^{CB,f} B_t^{CB}}{B_t^G + B_t^{CB}}. \quad (\text{A.15})$$

We can rearrange the previous expression as

$$\lambda_t^{CB,f} = \lambda_t^{HB,f} + \left(\lambda_t^{HB,f} - \lambda_t^{G,f} \right) \cdot \frac{B_t^G}{B_t^{CB}}. \quad (\text{A.16})$$

Summing from $f = 2$ to F , and using $\sum_{f=2}^F \lambda_t^{j,f} = 1 - \lambda_t^{j,1}$, $j \in \{HB, G, CB\}$ we obtain

$$\sum_{f=2}^F \lambda_t^{CB,f} = 1 - \lambda_t^{HB,1} + \left(\lambda_t^{G,1} - \lambda_t^{HB,1} \right) \cdot \frac{B_t^G}{B_t^{CB}}. \quad (\text{A.17})$$

Plugging (A.17) into (A.16) and after some rearrangements, we obtain

$$\lambda_t^{CB,f} = \frac{\lambda_t^{HB,f} \left(\lambda_t^{CB,1} - \lambda_t^{G,1} \right) - \lambda_t^{G,f} \left(\lambda_t^{CB,1} - \lambda_t^{HB,1} \right)}{\lambda_t^{HB,1} - \lambda_t^{G,1}}, \quad f > 1. \quad (\text{A.18})$$

Combining the household's relative allocation between loans and bonds (46) with the cor-

porate loan rule (40), we obtain

$$\frac{B_t^H}{A_t \bar{N}_t P_t} = \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left[\gamma_L \cdot (1 + \zeta^F) \left(\frac{Y_t}{A_t \bar{N}_t} \right) - \overline{\left(\frac{Y}{A \bar{N}} \right)} \gamma_{CB} \cdot \exp(u_t^L) \right]. \quad (\text{A.19})$$

where we assume that the central bank's balanced-growth-path real corporate loan target is a fraction γ_{CB} of balanced-growth-path real output, $\overline{\left(\frac{L^{CB}}{A \bar{N} P} \right)} = \overline{\left(\frac{Y}{A \bar{N}} \right)} \gamma_{CB}$.

Using the bond market equilibrium (31) with $\sum_{f=2}^F \lambda_t^{HB,f} = 1 - \lambda_t^{HB,1}$, we get

$$B_t^H = - \frac{\sum_{i=2}^F \left(B_t^{G,i} + B_t^{CB,i} \right)}{1 - \lambda_t^{HB,1}}. \quad (\text{A.20})$$

Combining (A.20) with (31), we obtain the equilibrium set of equations,

$$\frac{\lambda_t^{HB,f}}{1 - \lambda_t^{HB,1}} = \frac{\frac{B_t^{G,f}}{A_t \bar{N}_t P_t} + \frac{B_t^{CB,f}}{A_t \bar{N}_t P_t}}{\sum_{i=2}^F \left(\frac{B_t^{G,i}}{A_t \bar{N}_t P_t} + \frac{B_t^{CB,i}}{A_t \bar{N}_t P_t} \right)}, \quad \forall f > 1. \quad (\text{A.21})$$

Combining (A.19) and (A.20) yields the following equilibrium equation:

$$- \frac{\sum_{i=2}^F \left(\frac{B_t^{G,i}}{A_t \bar{N}_t P_t} + \frac{B_t^{CB,i}}{A_t \bar{N}_t P_t} \right)}{1 - \lambda_t^{HB,1}} = \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left[\gamma_L \cdot (1 + \zeta^F) \left(\frac{Y_t}{A_t \bar{N}_t} \right) - \overline{\left(\frac{Y}{A \bar{N}} \right)} \gamma_{CB} \cdot \exp(u_t^L) \right], \quad (\text{A.22})$$

This equilibrium relation is accounting-based and does not impose a policy-specific rule for the central bank's bond positions. Under conventional policy, the normalized central-bank bond positions follow the exogenous process in (37c); under YCC, they adjust endogenously to implement the yield target. Combining (A.21) and (A.22), we obtain

$$- \left(\frac{B_t^{G,f}}{A_t \bar{N}_t P_t} + \frac{B_t^{CB,f}}{A_t \bar{N}_t P_t} \right) \cdot (\lambda_t^{HB,f})^{-1} = \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left[\gamma_L \cdot (1 + \zeta^F) \left(\frac{Y_t}{A_t \bar{N}_t} \right) - \overline{\left(\frac{Y}{A \bar{N}} \right)} \gamma_{CB} \cdot \exp(u_t^L) \right] \quad (\text{A.23})$$

for all $f > 1$.

A.1.2 Steady-State Derivations in Section 4.1

At the steady state, the central bank decides the level of bond holdings $B^{CB,f}$ that it wants to hold, which can be calibrated to match the data of the central bank's balance sheet. Given $\{\lambda^{CB,f}\}$ and the size of its portfolio B^{CB} , which is ζ^{CB} fraction of total government bond issuance satisfying $B^{CB} = \zeta^{CB} \cdot B^G$, we obtain the steady-state households bond shares as in the main-text relation (48):

$$\lambda^{HB,f} = \frac{\lambda^{G,f} + \lambda^{CB,f} \cdot \zeta^{CB}}{1 + \zeta^{CB}}.$$

Throughout this subsection we adopt the remaining-maturity-at-realization indexing for bond returns of the main text. Concretely, R^{f-1} is the steady-state value of R_{t+1}^{f-1} in (16) of the main text—the per-period return realized one period after holding an f -maturity bond. Equivalently, $R^{f-1} = (YD^f)^f / (YD^{f-1})^{f-1}$ at steady state, with $YD^0 = 1$.

From the definition of R^{HB} we have

$$\sum_{f=1}^F \lambda^{HB,f} \cdot \left(\frac{R^{f-1}}{R^{HB}} \right) = 1,$$

which together with equation (15) can be rearranged as:

$$\lambda^{HB,f} = \left(\frac{z^f \cdot \frac{R^{f-1}}{R^{HB}}}{\tilde{\Phi}^B} \right)^{\kappa_B}, \quad \forall f, \quad \text{with } \tilde{\Phi}^B = \left[\sum_{j=1}^F \left[z^j \cdot \frac{R^{j-1}}{R^{HB}} \right]^{\kappa_B} \right]^{\frac{1}{\kappa_B}}. \quad (\text{A.24})$$

The above (48) and (A.24) jointly determine the steady state yields and household shares. Unfortunately, there is no analytical expression for them and we have to solve for the steady state values numerically. How we proceed, relying on simple iterations:

Step 1 Assume some initial guess for $\left\{ \frac{R^{f-1,guess}}{R^{HB}} \right\}_{f=1}^F$.

Step 2 Construct $\tilde{\Phi}^{B,old}$ using previous guess with $\tilde{\Phi}^B$ in equation (A.24).

Step 3 Update estimates on $\left\{ \frac{R^{f-1}}{R^{HB}} \right\}_{f=1}^F$ in two steps. First, for each $f > 1$, use $\tilde{\Phi}^{B,old}$ to compute

$$\frac{R^{f-1,new}}{R^{HB}} = (\lambda^{HB,f})^{\frac{1}{\kappa_B}} (z^f)^{-1} \tilde{\Phi}^{B,old}, \quad f > 1.$$

Then, using the just-updated $\left\{ \frac{R^{f-1,new}}{R^{HB}} \right\}_{f=2}^F$, compute

$$\frac{R^{0,new}}{R^{HB}} = \frac{1 - \sum_{f=2}^F \lambda^{HB,f} \left(\frac{R^{f-1,new}}{R^{HB}} \right)}{\lambda^{HB,1}}.$$

Step 4 Construct new shares $\lambda^{HB,f,new}$ by plugging $\left\{ \frac{R^{f-1,new}}{R^{HB}} \right\}_{f=1}^F$ into (A.24). Compute the discrepancy between these shares and the true ones found in (48). If the error is big, set $\frac{R^{f-1,guess}}{R^{HB}} = \frac{R^{f-1,new}}{R^{HB}}$ and repeat from Step 2 until convergence.

Using (18) and (23), we obtain

$$R^{HB} = \frac{\beta^{-1}\Pi \cdot GA \cdot GN}{1 - \lambda^K} - \frac{\lambda^K}{1 - \lambda^K} R^K. \quad (\text{A.25})$$

Combining the steady-state expression for R^S derived from equation (18) together with (A.25), we obtain

$$R^S = \beta^{-1}\Pi \cdot GA \cdot GN. \quad (\text{A.26})$$

From the Euler equation (A.5) and (A.26), we obtain an expression for the steady-state stochastic discount factor as

$$Q = \frac{1}{\beta^{-1}\Pi \cdot GA \cdot GN}. \quad (\text{A.27})$$

Therefore, by equation (A.27), the steady-state one-period ahead expected discounted return on capital, defined as $\tilde{R}_{t+1}^K \equiv \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$, is given by

$$\tilde{R}^K = \frac{R^K}{\beta^{-1}\Pi \cdot GA \cdot GN}. \quad (\text{A.28})$$

We can rewrite R^G as

$$R^G = \chi^G \cdot R^{HB}, \quad \chi^G = \sum_{f=1}^F \lambda^{G,f} \cdot \left(\frac{R^{f-1}}{R^{HB}} \right), \quad (\text{A.29})$$

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and from (A.25), it becomes

$$R^G = \chi^G \cdot \left[\frac{\beta^{-1} \Pi \cdot GA \cdot GN}{1 - \lambda^K} - \frac{\lambda^K}{1 - \lambda^K} R^K \right]. \quad (\text{A.30})$$

Using (17), we obtain an expression for the steady-state share of loans as

$$\lambda^K = \frac{\left(z^K \cdot \frac{R^K}{R^{HB}} \right)^{\kappa_S}}{1 + \left(z^K \cdot \frac{R^K}{R^{HB}} \right)^{\kappa_S}}. \quad (\text{A.31})$$

Further combining (A.25) and (A.31), we obtain

$$\frac{\lambda^K}{1 - \lambda^K} \equiv \left(z^K \cdot \frac{R^K}{R^{HB}} \right)^{\kappa_S} = \frac{\beta^{-1} \cdot \Pi \cdot GA \cdot GN - R^{HB}}{R^K - \beta^{-1} \cdot \Pi \cdot GA \cdot GN} \quad (\text{A.32})$$

The equilibrium government bonds are obtained from its budget constraint (34) and written as in the main-text relation (49):

$$\frac{B^G}{ANP} = - \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right)^{-1} [\zeta^G + \zeta^F - \zeta^T] \left(\frac{Y}{AN} \right). \quad (\text{A.33})$$

The government is a net borrower with $B^G < 0$ at the steady-state. Also, we would like to match the data in which the government runs primary deficit, i.e., $\zeta^G + \zeta^F - \zeta^T > 0$. The only way to achieve that is by having $R^G < \Pi \cdot GA \cdot GN$. Combining $B^{CB} = \zeta^{CB} \cdot B^G$, $B_t^H = -(B_t^G + B_t^{CB})$, (A.19), (A.29), (A.32), and (A.33) yields

$$\gamma_L(1 + \zeta^F) - \gamma_{CB} = (1 + \zeta^{CB}) \cdot [\zeta^G + \zeta^F - \zeta^T] \cdot \left(1 - \frac{\chi^G}{\Pi \cdot GA \cdot GN} \cdot R^{HB} \right)^{-1} \cdot \left[\frac{\beta^{-1} \cdot \Pi \cdot GA \cdot GN - R^{HB}}{R^K - \beta^{-1} \cdot \Pi \cdot GA \cdot GN} \right]. \quad (\text{A.34})$$

(A.32) and (A.34) form a non-linear system of equations on the unknown steady-states of R^K and R^{HB} . After then, we can then simply back out bond returns as

$$R^{f-1} = R^{HB} \cdot \left(\frac{R^{f-1}}{R^{HB}} \right), \quad f = 1, \dots, F.$$

Now that we have characterized the bond returns, we can recursively obtain the bond

yields using

$$YD^f = \left[R^{f-1} \cdot (YD^{f-1})^{f-1} \right]^{\frac{1}{f}},$$

where $YD^0 = 1$. The price dispersion is given by

$$\Delta = \left[\frac{1 - \theta}{1 - \theta \Pi^{\epsilon(\frac{\eta+1}{\eta+\alpha})}} \right] \left(\frac{1 - \theta \Pi^{\epsilon-1}}{1 - \theta} \right)^{\left(\frac{\epsilon-1}{\epsilon-1}\right)\left(\frac{\eta+1}{\eta+\alpha}\right)}. \quad (\text{A.35})$$

From the capital producer's optimization (25) and the steady-state expression for the stochastic discount factor (A.27), we obtain an expression for P^K

$$\frac{P^K}{P} = \beta^{-1} \cdot GA \cdot GN - (1 - \delta). \quad (\text{A.36})$$

The steady state representation of firms' pricing (A.11) can be written as

$$F = \xi^F \cdot \left(\frac{C}{AN} \right)^{-\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)} \left(\frac{Y}{AN} \right)^{\frac{\eta+1}{\eta+\alpha}}, \quad (\text{A.37})$$

$$H = \xi^H \cdot \left(\frac{C}{AN} \right)^{-1} \left(\frac{Y}{AN} \right), \quad (\text{A.38})$$

$$\begin{aligned} \text{with } \xi^F &= (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left[1 - \theta \beta \Pi^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)} \right]^{-1} \left(\frac{(1 + \xi^F)^{-1} \epsilon}{\epsilon - 1} \right) \left(\frac{P^K}{P} \right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)}, \\ \xi^H &= \left[1 - \gamma_L \cdot \left(\frac{R^K}{R^S} - 1 \right) \right] \cdot [1 - \theta \beta \cdot \Pi^{\epsilon-1}]^{-1}. \end{aligned}$$

Using (A.37), (A.38) and (A.12), we obtain

$$\left(\frac{C}{AN} \right)^{\left(\frac{1-\alpha}{\eta+\alpha}\right)\eta} = \xi^Y \cdot \left(\frac{Y}{AN} \right)^{-\left(\frac{1-\alpha}{\eta+\alpha}\right)}, \quad \text{with } \xi^Y = \left(\frac{\xi^H}{\xi^F} \right) \left(\frac{1 - \theta}{1 - \theta \Pi^{\epsilon-1}} \right)^{\left(\frac{1}{\epsilon-1}\right)\left[1 + \epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]}. \quad (\text{A.39})$$

Combining (A.39) and (A.6), we obtain

$$\frac{K}{AN} = \xi^K \cdot \left(\frac{Y}{AN} \right), \quad \text{with } \xi^K = \alpha \cdot (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA \cdot GN \cdot \left(\frac{P^K}{P} \right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha}\right)} \Delta \cdot \xi^Y. \quad (\text{A.40})$$

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Henceforth, $C/A\bar{N}$ and $Y/A\bar{N}$ stand for the steady-states of $C_t/(A_t\bar{N}_t)$ and $Y_t/(A_t\bar{N}_t)$, while $K/A\bar{N}$ stands for $K_t/(A_{t-1}\bar{N}_{t-1})$, matching the timing convention in (A.6).

Plugging (A.40) into the aggregate resource constraint, we obtain

$$\frac{C}{A\bar{N}} = \xi^C \cdot \left(\frac{Y}{A\bar{N}} \right), \quad \text{with} \quad \xi^C = (1 - \zeta^G) - \xi^K \cdot \left[1 - \left(\frac{1 - \delta}{GA \cdot GN} \right) \right]. \quad (\text{A.41})$$

Combining (A.39) and (A.41), we obtain

$$\frac{Y}{A\bar{N}} = (\xi^Y)^{\left(\frac{1}{1-\alpha}\right)\left(\frac{\eta+\alpha}{\eta+1}\right)} (\xi^C)^{-\left(\frac{\eta}{\eta+1}\right)}. \quad (\text{A.42})$$

A.1.3 Log-linearization

We start by log-linearizing the equations that are common to the conventional policy model and the yield-curve-control model, then derive the ones that are different. Log-linearize the gross labor-productivity growth rate, defined as $GA_t \equiv \exp(\mu + \varepsilon_t^A)$, and the spending and revenue ratio expressions (I.1):

$$\hat{g}a_t = \hat{\varepsilon}_t^A, \quad \hat{\zeta}_t^G = \frac{a^G}{1 + a^G} \cdot \hat{u}_t^G, \quad \hat{\zeta}_t^T = \frac{a^T}{1 + a^T} \cdot \hat{u}_t^T. \quad (\text{A.43})$$

Equations (A.13) and (A.14), with the help of (A.43), can be linearized as

$$\hat{c}_t = \left[(1 - \zeta^G) \cdot \frac{Y}{C} \right] \left[\hat{y}_t - \frac{1}{1 + a^G} \cdot \hat{u}_t^G \right] + \left[\frac{1 - \delta}{GA \cdot GN} \frac{K}{C} \right] (\hat{k}_t - \hat{\varepsilon}_t^A) - \frac{K}{C} \hat{k}_{t+1}, \quad (\text{A.44})$$

$$\hat{c}_t = \mathbb{E}_t [\hat{c}_{t+1} - (\hat{r}_{t+1}^S - \hat{\pi}_{t+1})], \quad (\text{A.45})$$

where we use (A.43) to substitute for $\hat{\zeta}_t^G$ and $\hat{g}a_t$. The term $\mathbb{E}_t \hat{g}a_{t+1}$ drops out of the Euler equation because technology growth is i.i.d. Plugging (A.44) into (A.45), we obtain the following dynamic IS equation for output $\hat{y}_t$.

$$\begin{aligned} \hat{y}_t = \mathbb{E}_t \left[\hat{y}_{t+1} - \left[\frac{(1 - \zeta^G)^{-1}(1 - \delta)}{GA \cdot GN} \cdot \frac{K}{Y} \right] (\hat{k}_t - \hat{\varepsilon}_t^A) + (1 - \zeta^G)^{-1} \left[1 + \frac{1 - \delta}{GA \cdot GN} \right] \frac{K}{Y} \hat{k}_{t+1} \right. \\ \left. - (1 - \zeta^G)^{-1} \frac{K}{Y} \hat{k}_{t+2} - (1 - \zeta^G)^{-1} \frac{C}{Y} (\hat{r}_{t+1}^S - \hat{\pi}_{t+1}) + \frac{1 - \rho_G}{1 + a^G} \cdot \hat{u}_t^G \right]. \end{aligned} \quad (\text{A.46})$$

where the coefficient $(1 - \rho_G)/(1 + a^G)$ is from substituting the AR(1) process for u_{t+1}^G into the expression.

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Substituting the expression for the stochastic discount factor (A.4) into the household's bond portfolio conditions (15) and log-linearizing the resulting expression yields

$$\hat{\lambda}_t^{HB,f} = \kappa^B \mathbb{E}_t \left[\hat{z}_t^f - \hat{\pi}_{t+1} + \hat{c}_t - \hat{c}_{t+1} - (f-1) \hat{y} d_{t+1}^{f-1} + f \hat{y} d_t^f - \hat{\phi}_t^B \right], \quad (\text{A.47})$$

where

$$\begin{aligned} \hat{\phi}_t^B &= \mathbb{E}_t (-\hat{\pi}_{t+1} + \hat{c}_t - \hat{c}_{t+1}) + \sum_{j=1}^F \left[\frac{\beta z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \hat{z}_t^j \\ &+ \sum_{j=1}^F j \left[\frac{\beta z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \hat{y} d_t^j - \sum_{j=0}^{F-1} j \left[\frac{\beta z^{j+1} (YD^j)^{-j}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^{j+1})^{-(j+1)}} \right]^{\kappa^B} \mathbb{E}_t (\hat{y} d_{t+1}^j). \end{aligned} \quad (\text{A.48})$$

These expressions use $\mathbb{E}_t \hat{g} a_{t+1} = 0$, following from the i.i.d. assumption of the technology-growth shock. The final sum is indexed by $j = 0, \dots, F-1$ because it collects the future-yield terms $\hat{y} d_{t+1}^j$ associated with $j+1$ -period bonds.

Combining (A.47) and (A.48), we obtain the following expression for $\hat{\lambda}_t^{HB,f}$:

$$\hat{\lambda}_t^{HB,f} = \sum_{j=1}^F \Psi_{fj}^1 \hat{z}_t^j + \sum_{j=1}^F \Psi_{fj}^2 \hat{y} d_t^j + \sum_{j=1}^F \Psi_{fj}^3 \mathbb{E}_t [\hat{y} d_{t+1}^j],$$

where

$$\begin{aligned} \Psi_{fj}^1 &= \begin{cases} \left[1 - \left[\frac{\beta \cdot z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \right] \cdot \kappa^B & , \text{ if } f = j, \\ - \left[\frac{\beta \cdot z^j (YD^{j-1})^{-(j-1)}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^j)^{-j}} \right]^{\kappa^B} \cdot \kappa^B & , \text{ if } f \neq j, \end{cases} \\ \Psi_{fj}^2 &= j \cdot \Psi_{fj}^1, \\ \Psi_{fj}^3 &= \begin{cases} -j \cdot \left[1 - \left[\frac{\beta \cdot z^{j+1} (YD^j)^{-j}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^{j+1})^{-(j+1)}} \right]^{\kappa^B} \right] \cdot \kappa^B & , \text{ if } j = f-1, \\ j \cdot \left[\frac{\beta \cdot z^{j+1} (YD^j)^{-j}}{\Pi \cdot GA \cdot GN \cdot \Phi^B (YD^{j+1})^{-(j+1)}} \right]^{\kappa^B} \cdot \kappa^B & , \text{ if } j \neq f-1, \\ 0 & , \text{ if } j = F. \end{cases} \end{aligned}$$

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We can put the system of F equations in matrix format as

$$\overrightarrow{\hat{\lambda}_t^{HB}} = \Psi^1 \cdot \overrightarrow{\hat{z}}_t + \Psi^2 \cdot \overrightarrow{\hat{y}d}_t + \Psi^3 \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d}_{t+1} \right], \quad (\text{A.49})$$

where $\{\Psi^1, \Psi^2, \Psi^3\}$ are $F \times F$ matrices containing elements of $\{\Psi_{fj}^1, \Psi_{fj}^2, \Psi_{fj}^3\}$, with f representing rows and j columns. Linearizing equations (I.2) and (I.3) yields

$$\overrightarrow{\hat{\lambda}_t^G} = \tilde{\Xi} \cdot \overrightarrow{\hat{u}_t^B}, \quad \overrightarrow{\hat{u}_t^B} = \mathcal{T}^B \cdot \overrightarrow{\hat{u}_t^B},$$

where $\overrightarrow{\hat{u}_t^B}$ is the vector of transformed maturity-specific shocks, while $\overrightarrow{\hat{u}_t^B}$ is the reduced shock vector. $\tilde{\Xi}$ is a matrix whose elements $\tilde{\Xi}_{fj}$ (f -rows, j -columns) are

$$\tilde{\Xi}_{fj} = \begin{cases} 0 & , \text{ if } f = 1 \text{ \& } j = f, \\ 1 - \lambda^{G,f} & , \text{ if } f \geq 2 \text{ \& } j = f, \\ -\lambda^{G,j} & , \text{ if } j \neq f, \end{cases}$$

and similarly $\mathcal{T}^B$ is a matrix containing elements τ_{fj}^B from (I.3). By defining $\Xi = \tilde{\Xi} \cdot \mathcal{T}^B$, we can combine the previous two equations to obtain

$$\overrightarrow{\hat{\lambda}_t^G} = \Xi \cdot \overrightarrow{\hat{u}_t^B}. \quad (\text{A.50})$$

Let $\tilde{B}^{G,f} = -B^{G,f}$ denote the positive amount of government debt, so hats below are log deviations of $\tilde{B}_t^{G,f}$. Combining this definition with (A.50) gives

$$\overrightarrow{\hat{b}_t^G} = \Xi \cdot \overrightarrow{\hat{u}_t^B} + \overrightarrow{1}_{F \times 1} \cdot \hat{b}_t^G, \quad (\text{A.51})$$

where $\overrightarrow{1}_{F \times 1}$ is a unit vector of size F . Log-linearizing the household's stochastic discount factor (A.4) yields:

$$\hat{q}_{t,t+1} = \hat{c}_t - \hat{c}_{t+1} - \hat{\pi}_{t+1} - \hat{g}a_{t+1}. \quad (\text{A.52})$$

Log-linearizing Φ_t^S in the household's portfolio between loans and bonds (17), we obtain

$$\hat{\phi}_t^S = \mathbb{E}_t [\hat{q}_{t,t+1}] + \frac{(R^{HB})^{\kappa^S}}{(R^{HB})^{\kappa^S} + (z^K R^K)^{\kappa^S}} \mathbb{E}_t [\hat{r}_{t+1}^{HB}] + \frac{(z^K R^K)^{\kappa^S}}{(R^{HB})^{\kappa^S} + (z^K R^K)^{\kappa^S}} (\hat{z}_t^K + \mathbb{E}_t [\hat{r}_{t+1}^K]). \quad (\text{A.53})$$

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Log-linearizing the household's portfolio decision between loans and bonds (17) and making use of the previous expression (A.53), we obtain the main-text relation (54):

$$\begin{aligned}\hat{\lambda}_t^K &= \kappa^S \cdot \left[\frac{(R^{HB})^{\kappa^S}}{(R^{HB})^{\kappa^S} + (z^K R^K)^{\kappa^S}} \right] (\hat{z}_t^K + \mathbb{E}_t [\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]) \\ &= \kappa^S (1 - \lambda^K) (\hat{z}_t^K + \mathbb{E}_t [\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}]).\end{aligned}$$

where the second equality uses the steady-state portfolio share in (A.31).

By linearizing the formula for the effective savings rate of the household (18), we obtain

$$\hat{r}_t^S = \frac{\lambda^K (R^K - R^{HB})}{R^S} \hat{\lambda}_{t-1}^K + \frac{(1 - \lambda^K) R^{HB}}{R^S} \hat{r}_t^{HB} + \frac{\lambda^K R^K}{R^S} \hat{r}_t^K. \quad (\text{A.54})$$

Log-linearizing the effective bond rates of households, government, and central bank,

$$\hat{r}_t^j = \sum_{f=1}^F \frac{\lambda^{j,f} (YD^{f-1})^{-(f-1)}}{R^j (YD^f)^{-f}} \cdot [\hat{\lambda}_{t-1}^{j,f} - (f-1) \hat{y}d_t^{f-1} + f \hat{y}d_{t-1}^f], \quad j \in \{HB, G, CB\},$$

with which we can express these equations on matrix format as

$$\hat{r}_t^j = \Psi^{j,4} \cdot \overrightarrow{\hat{\lambda}_{t-1}^j} - \Psi^{j,5} \cdot \overrightarrow{\hat{y}d_t} + \Psi^{j,6} \cdot \overrightarrow{\hat{y}d_{t-1}}, \quad j \in \{HB, G, CB\}, \quad (\text{A.55})$$

where $\{\Psi^{j,4}, \Psi^{j,5}, \Psi^{j,6}\}$ are $1 \times F$ -sized matrices whose elements are defined as:

$$\begin{aligned}\Psi_{1f}^{j,4} &= \frac{\lambda^{j,f} (YD^{f-1})^{-(f-1)}}{R^j (YD^f)^{-f}}, \\ \Psi_{1f}^{j,5} &= \begin{cases} \frac{\lambda^{j,f+1} (YD^f)^{-f}}{R^j (YD^{f+1})^{-(f+1)}} f & , \text{ if } f < F, j \in \{HB, G, CB\}, \\ 0 & , \text{ if } f = F, \end{cases} \\ \Psi_{1f}^{j,6} &= \frac{\lambda^{j,f} (YD^{f-1})^{-(f-1)}}{R^j (YD^f)^{-f}} f.\end{aligned}$$

where the vector $\overrightarrow{\hat{y}d_t}$ in equation (A.55) is indexed by current remaining maturity, so the current yield term for an $f + 1$ -period bond enters position f in $\Psi^{j,5}$.

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By plugging (A.50) into $\hat{r}^G$ in (A.55), we obtain

$$\hat{r}_t^G = \Psi^{G,4} \cdot \Xi \cdot \overrightarrow{\hat{u}_{t-1}^B} - \Psi^{G,5} \cdot \overrightarrow{\hat{y}d_t} + \Psi^{G,6} \cdot \overrightarrow{\hat{y}d_{t-1}}. \quad (\text{A.56})$$

By plugging (A.49) into $\hat{r}^{HB}$ in (A.55), we obtain

$$\hat{r}_t^{HB} = \Psi^{HB,4} \Psi^1 \cdot \overrightarrow{\hat{z}_{t-1}} + [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}] \cdot \overrightarrow{\hat{y}d_{t-1}} + \Psi^{HB,4} \Psi^3 \cdot \mathbb{E}_{t-1} \left[\overrightarrow{\hat{y}d_t} \right] - \Psi^{HB,5} \cdot \overrightarrow{\hat{y}d_t}. \quad (\text{A.57})$$

Taking the expectation operator $\mathbb{E}_t$ on the previous equation (A.57), we obtain

$$\mathbb{E}_t [\hat{r}_{t+1}^{HB}] = \Psi^{HB,4} \Psi^1 \cdot \overrightarrow{\hat{z}_t} + [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}] \cdot \overrightarrow{\hat{y}d_t} + [\Psi^{HB,4} \Psi^3 - \Psi^{HB,5}] \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right]. \quad (\text{A.58})$$

Plugging equations (54) and (A.58) into the $t+1$ expression of equation (A.54) and applying the expectations operator $\mathbb{E}_t$, we obtain the expected effective savings rate as:

$$\mathbb{E}_t [\hat{r}_{t+1}^S] = \Psi^7 \overrightarrow{\hat{z}_t} + \Psi^8 \overrightarrow{\hat{y}d_t} + \Psi^9 \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right] + \Psi^{10} \hat{r}_{t+1}^K + \Psi^{11} \hat{z}_t^K, \quad (\text{A.59})$$

where

$$\begin{aligned} \Psi^7 &= \Psi^{HB,4} \Psi^1 \left[\frac{(1 + \kappa^S \lambda^K)(1 - \lambda^K) R^{HB} - \kappa^S (1 - \lambda^K) \lambda^K R^K}{R^S} \right], \\ \Psi^8 &= [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}] \left[\frac{(1 + \kappa^S \lambda^K)(1 - \lambda^K) R^{HB} - \kappa^S (1 - \lambda^K) \lambda^K R^K}{R^S} \right], \\ \Psi^9 &= [\Psi^{HB,4} \Psi^3 - \Psi^{HB,5}] \left[\frac{(1 + \kappa^S \lambda^K)(1 - \lambda^K) R^{HB} - \kappa^S (1 - \lambda^K) \lambda^K R^K}{R^S} \right], \\ \Psi^{10} &= \frac{[1 + \kappa^S (1 - \lambda^K)] \lambda^K R^K - \kappa^S (1 - \lambda^K) \lambda^K R^{HB}}{R^S}, \\ \Psi^{11} &= \frac{\kappa^S \lambda^K (1 - \lambda^K) (R^K - R^{HB})}{R^S}. \end{aligned}$$

Plugging back the expression of the household's expected bonds rate (A.58) into her portfolio decision between loans and bonds (54), we obtain

$$\hat{\lambda}_t^K = \kappa^S (1 - \lambda^K) (\hat{z}_t^K + \hat{r}_{t+1}^K) - \Psi^{12} \cdot \overrightarrow{\hat{z}_t} - \Psi^{13} \cdot \overrightarrow{\hat{y}d_t} - \Psi^{14} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right], \quad (\text{A.60})$$

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where $\Psi^{12} = \kappa^S(1 - \lambda^K)\Psi^{HB,4}\Psi^1$, $\Psi^{13} = \kappa^S(1 - \lambda^K)[\Psi^{HB,4}\Psi^2 + \Psi^{HB,6}]$, and $\Psi^{14} = \kappa^S(1 - \lambda^K)[\Psi^{HB,4}\Psi^3 - \Psi^{HB,5}]$. Linearizing the capital aggregation equation (A.6) and use (A.43), we obtain

$$\hat{k}_t = \hat{\varepsilon}_t^A + \left(\frac{\eta + 1}{\eta + \alpha} \right) \hat{y}_t - \left(\frac{\eta(1 - \alpha)}{\eta + \alpha} \right) \cdot [\hat{p}_t^K - \hat{c}_t]. \quad (\text{A.61})$$

Combining (A.44) and the above (A.61), we obtain

$$\begin{aligned} \hat{p}_t^K &= \left[(1 - \zeta^G) \frac{Y}{C} + \frac{\eta + 1}{\eta(1 - \alpha)} \right] \cdot \hat{y}_t - \left[(1 - \zeta^G) \cdot \frac{Y}{C} \right] \left(\frac{1}{1 + a^G} \right) \cdot \hat{u}_t^G \\ &\quad + \left[\frac{1 - \delta}{GA \cdot GN} \cdot \frac{K}{C} - \frac{\eta + \alpha}{\eta(1 - \alpha)} \right] \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] - \frac{K}{C} \cdot \hat{k}_{t+1}. \end{aligned} \quad (\text{A.62})$$

If we linearize the supply block (A.11) and (A.12), we obtain

$$\begin{aligned} \hat{f}_t &= \left[1 - \theta\beta\Pi^{\epsilon(\frac{\eta+1}{\eta+\alpha})} \right] \left(\frac{\eta + 1}{\eta + \alpha} \right) [\hat{y}_t + \alpha\mathbb{E}_t[\hat{q}_{t,t+1} + \hat{p}_t^K - \hat{c}_t]] \\ &\quad + \theta\beta\Pi^{\epsilon(\frac{\eta+1}{\eta+\alpha})}\mathbb{E}_t \left[\epsilon \left(\frac{\eta + 1}{\eta + \alpha} \right) \hat{\pi}_{t+1} + \hat{f}_{t+1} \right], \end{aligned} \quad (\text{A.63})$$

$$\begin{aligned} \hat{h}_t &= [1 - \theta\beta\Pi^{\epsilon-1}] \left[\hat{y}_t - \hat{c}_t - \left(\frac{\gamma_L \cdot R^K}{R^S - \gamma_L \cdot (R^K - R^S)} \right) \cdot [\hat{r}_{t+1}^K + \mathbb{E}_t[\hat{q}_{t,t+1}]] \right] \\ &\quad + \theta\beta\Pi^{\epsilon-1}\mathbb{E}_t \left[(\epsilon - 1)\hat{\pi}_{t+1} + \hat{h}_{t+1} \right], \end{aligned} \quad (\text{A.64})$$

leading to

$$\hat{f}_t - \hat{h}_t = \left[1 + \epsilon \left(\frac{1 - \alpha}{\eta + \alpha} \right) \right] \left(\frac{\theta\Pi^{\epsilon-1}}{1 - \theta\Pi^{\epsilon-1}} \right) \hat{\pi}_t. \quad (\text{A.65})$$

Combining (A.45), (A.52), and (A.61) with equations (A.64) and (A.63), we obtain:

$$\begin{aligned} \hat{f}_t &= -\Psi^{16} \cdot \overrightarrow{\hat{z}}_t - \Psi^{17} \cdot \hat{z}_t^K - \Psi^{18} \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] + \Psi^{19} \cdot \hat{y}_t - \Psi^{20} \cdot \overrightarrow{\hat{y}d}_t \\ &\quad - \Psi^{21} \cdot \hat{r}_{t+1}^K - \Psi^{22} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d}_{t+1} \right] + \Psi^{23} \cdot \mathbb{E}_t[\hat{\pi}_{t+1}] + \Psi^{24} \cdot \mathbb{E}_t[\hat{f}_{t+1}]. \end{aligned} \quad (\text{A.66})$$

$$\begin{aligned} \hat{h}_t &= \Psi^{26} \cdot \overrightarrow{\hat{z}}_t + \Psi^{27} \cdot \hat{z}_t^K - \Psi^{28} \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] + \Psi^{29} \cdot \hat{u}_t^G + \Psi^{30} \cdot \hat{y}_t + \Psi^{31} \cdot \overrightarrow{\hat{y}d}_t \\ &\quad - \Psi^{32} \cdot \hat{r}_{t+1}^K + \Psi^{33} \cdot \hat{k}_{t+1} + \Psi^{34} \cdot \mathbb{E}_t \left[\overrightarrow{\hat{y}d}_{t+1} \right] + \Psi^{35} \cdot \mathbb{E}_t[\hat{\pi}_{t+1}] + \Psi^{36} \cdot \mathbb{E}_t[\hat{h}_{t+1}]. \end{aligned} \quad (\text{A.67})$$

where

$$\begin{aligned}
 \Psi^{15} &= \left[1 - \theta\beta\Pi^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)} \right] \left(\frac{\eta+1}{\eta+\alpha} \right), & \Psi^{26} &= \Psi^{25}\Psi^7, \\
 \Psi^{16} &= \alpha\Psi^{15}\Psi^7, & \Psi^{27} &= \Psi^{25}\Psi^{11}, \\
 \Psi^{17} &= \alpha\Psi^{15}\Psi^{11}, & \Psi^{28} &= [1 - \theta\beta\Pi^{\epsilon-1}] \left[\frac{1-\delta}{GA \cdot GN} \cdot \frac{K}{C} \right], \\
 \Psi^{18} &= \Psi^{15} \left(\frac{\alpha}{1-\alpha} \right) \left(\frac{\eta+\alpha}{\eta} \right), & \Psi^{29} &= [1 - \theta\beta\Pi^{\epsilon-1}] (1 - \zeta^G) \frac{Y}{C} \left(\frac{1}{1+a^G} \right), \\
 \Psi^{19} &= \Psi^{15} \left[1 + \left(\frac{\alpha}{1-\alpha} \right) \left(\frac{\eta+1}{\eta} \right) \right], & \Psi^{30} &= [1 - \theta\beta\Pi^{\epsilon-1}] \left[1 - (1 - \zeta^G) \frac{Y}{C} \right], \\
 \Psi^{20} &= \alpha\Psi^{15}\Psi^8, & \Psi^{31} &= \Psi^{25}\Psi^8, \\
 \Psi^{21} &= \alpha\Psi^{15}\Psi^{10}, & \Psi^{32} &= \Psi^{25} \cdot (1 - \Psi^{10}), \\
 \Psi^{22} &= \alpha\Psi^{15}\Psi^9, & \Psi^{33} &= [1 - \theta\beta\Pi^{\epsilon-1}] \frac{K}{C}, \\
 \Psi^{23} &= \theta\beta\Pi^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)} \epsilon \left(\frac{\eta+1}{\eta+\alpha} \right), & \Psi^{34} &= \Psi^{25}\Psi^9, \\
 \Psi^{24} &= \theta\beta\Pi^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)}, & \Psi^{35} &= \theta\beta\Pi^{\epsilon-1}(\epsilon - 1), \\
 \Psi^{25} &= [1 - \theta\beta\Pi^{\epsilon-1}] \left(\frac{\gamma_L \cdot R^K}{R^S - \gamma_L \cdot (R^K - R^S)} \right), & \Psi^{36} &= \theta\beta\Pi^{\epsilon-1}.
 \end{aligned}$$

Linearizing the government's budget constraint (34) yields

$$\begin{aligned}
 \hat{b}_t^G &= \frac{R^G}{\Pi \cdot GA \cdot GN} \left[\hat{r}_t^G - \hat{\pi}_t - \hat{g}a_t + \hat{b}_{t-1}^G \right] \\
 &\quad - \left[\zeta^G + \zeta^F - \zeta^T \right] \left(\frac{Y}{B/P} \right) \left[\hat{y}_t + \left(\frac{\zeta^G}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^G}{1+a^G} \right) \hat{u}_t^G - \left(\frac{\zeta^T}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^T}{1+a^T} \right) \hat{u}_t^T \right].
 \end{aligned} \tag{A.68}$$

Here, steady-state ratios use the original government bond position B/P , while hatted debt variables use the positive-debt variable $\tilde{B}^G = -B^G$.

Using the steady-state equilibrium condition (49), with (A.43) and (A.56), we express the previous (A.68) as

$$\begin{aligned}
 \hat{b}_t^G &= \frac{R^G}{\Pi \cdot GA \cdot GN} \left[\Psi^{G,4} \cdot \Xi \cdot \overrightarrow{\hat{u}_{t-1}^B} - \Psi^{G,5} \cdot \overrightarrow{\hat{y}_t^d} + \Psi^{G,6} \cdot \overrightarrow{\hat{y}_{t-1}^d} - \hat{\pi}_t - \hat{\varepsilon}_t^A + \hat{b}_{t-1}^G \right] \\
 &\quad + \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right) \left[\hat{y}_t + \left(\frac{\zeta^G}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^G}{1+a^G} \right) \hat{u}_t^G - \left(\frac{\zeta^T}{\zeta^G + \zeta^F - \zeta^T} \right) \left(\frac{a^T}{1+a^T} \right) \hat{u}_t^T \right],
 \end{aligned} \tag{A.69}$$

where the coefficient on the fiscal-output term follows directly from substituting the steady-state debt-output relation in (49).

Linearizing the capital producer's optimization condition (25) yields

$$0 = \mathbb{E}_t \left[\hat{q}_{t,t+1} + \hat{\pi}_{t+1} + \left(\frac{P^K/P}{1 - \delta + P^K/P} \right) \hat{p}_{t+1}^K \right]. \quad (\text{A.70})$$

By plugging (A.45) and (A.52) into the previous (A.70) and rearranging, we get

$$\mathbb{E}_t [\hat{r}_{t+1}^S - \hat{\pi}_{t+1}] = \left(\frac{P^K/P}{1 - \delta + P^K/P} \right) \mathbb{E}_t [\hat{p}_{t+1}^K]. \quad (\text{A.71})$$

This rearrangement uses $\mathbb{E}_t \hat{g}_{t+1} = 0$, because technology growth is i.i.d.

Plugging expressions on the effective savings rate (A.59) and the rental price of capital (A.62) into (A.71) we obtain

$$\begin{aligned} \hat{r}_{t+1}^K = & -\Psi^{37} \cdot \vec{\hat{z}}_t - \Psi^{38} \cdot \hat{z}_t^K - \Psi^{39} \cdot \vec{\hat{y}d}_t - \Psi^{40} \cdot \mathbb{E}_t [\vec{\hat{y}d}_{t+1}] + \Psi^{41} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] \\ & + \Psi^{42} \cdot \mathbb{E}_t [\hat{y}_{t+1}] + \Psi^{43} \cdot \hat{k}_{t+1} - \Psi^{44} \cdot \mathbb{E}_t [\hat{k}_{t+2}] - \Psi^{45} \cdot \hat{u}_t^G, \end{aligned} \quad (\text{A.72})$$

where we define

$$\begin{aligned} \Psi^{37} &= (\Psi^{10})^{-1} \Psi^7, & \Psi^{42} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) \left[(1 - \zeta^G) \frac{Y}{C} + \left(\frac{\eta + 1}{\eta(1 - \alpha)} \right) \right], \\ \Psi^{38} &= (\Psi^{10})^{-1} \Psi^{11}, & \Psi^{43} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) \left[\frac{1 - \delta}{GA \cdot GN} \frac{K}{C} - \left(\frac{\eta + \alpha}{\eta(1 - \alpha)} \right) \right], \\ \Psi^{39} &= (\Psi^{10})^{-1} \Psi^8, & \Psi^{44} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) \frac{K}{C}, \\ \Psi^{40} &= (\Psi^{10})^{-1} \Psi^9, & \Psi^{45} &= (\Psi^{10})^{-1} \left(\frac{\frac{P^K}{P}}{1 - \delta + \frac{P^K}{P}} \right) (1 - \zeta^G) \frac{Y}{C} \left(\frac{\rho^G}{1 + a^G} \right). \\ \Psi^{41} &= (\Psi^{10})^{-1}, \end{aligned}$$

The coefficient Ψ^{45} uses the AR(1) expectation of the government spending shock, so $\mathbb{E}_t \hat{u}_{t+1}^G = \rho^G \hat{u}_t^G$. Finally, plugging the effective savings rate (A.59) into the Euler equation (A.46), we obtain

$$\begin{aligned} \hat{y}_t = & \mathbb{E}_t \left[\hat{y}_{t+1} + \Psi^{46} \cdot \hat{\pi}_{t+1} - \Psi^{47} \cdot \vec{\hat{z}}_t - \Psi^{48} \cdot \hat{z}_t^K - \Psi^{49} \cdot \vec{\hat{y}d}_t - \Psi^{50} \cdot \mathbb{E}_t [\vec{\hat{y}d}_{t+1}] \right. \\ & \left. - \Psi^{51} \cdot \hat{r}_{t+1}^K - \Psi^{52} \cdot (\hat{k}_t - \hat{\varepsilon}_t^A) + \Psi^{53} \cdot \hat{k}_{t+1} - \Psi^{54} \cdot \hat{k}_{t+2} + \Psi^{55} \cdot \hat{u}_t^G \right], \end{aligned} \quad (\text{A.73})$$

where we define

$$\begin{aligned}
 \Psi^{46} &= (1 - \zeta^G)^{-1} \frac{C}{Y}, & \Psi^{52} &= \frac{(1 - \zeta^G)^{-1} (1 - \delta) K}{GA \cdot GN} \frac{K}{Y}, \\
 \Psi^{47} &= \Psi^{46} \Psi^7, & \Psi^{53} &= (1 - \zeta^G)^{-1} \left[1 + \frac{1 - \delta}{GA \cdot GN} \right] \frac{K}{Y}, \\
 \Psi^{48} &= \Psi^{46} \Psi^{11}, & \Psi^{54} &= (1 - \zeta^G)^{-1} \frac{K}{Y}, \\
 \Psi^{49} &= \Psi^{46} \Psi^8, & \Psi^{55} &= \frac{1 - \rho_G}{1 + a^G}. \\
 \Psi^{50} &= \Psi^{46} \Psi^9, & & \\
 \Psi^{51} &= \Psi^{46} \Psi^{10}, & &
 \end{aligned}$$

Linearizing the labor aggregation condition (A.7) yields

$$\hat{n}_t = -\alpha \left(\frac{\eta}{\eta + \alpha} \right) \hat{c}_t + \left(\frac{\eta}{\eta + \alpha} \right) \hat{y}_t + \alpha \left(\frac{\eta}{\eta + \alpha} \right) \cdot \hat{p}_t^K, \quad (\text{A.74})$$

where we rely on that price dispersion has no first-order effect around the symmetric steady state.

Plugging equations (A.44) and (A.62) into equation (A.74), we obtain

$$\hat{n}_t = \left(\frac{\eta}{\eta + \alpha} \right) \left[1 + \left(\frac{\alpha}{1 - \alpha} \right) \left(\frac{\eta + 1}{\eta} \right) \right] \cdot \hat{y}_t - \left(\frac{\alpha}{1 - \alpha} \right) \cdot \left[\hat{k}_t - \hat{\varepsilon}_t^A \right], \quad (\text{A.75})$$

where the fiscal-shock and future-capital terms present in (A.44) and (A.62) cancel algebraically after the substitution.

Linearizing the bond market clearing condition (31), we obtain:

$$\Psi^{56} \hat{b}_t^{CB} + \Psi^{57} \hat{b}_t^H = \hat{b}_t^G,$$

where we defined $\Psi^{56} = \left(\frac{B^{CB}}{B^{CB} + B^H} \right)$ and $\Psi^{57} = \left(\frac{B^H}{B^{CB} + B^H} \right)$.

We can rearrange the previous equation as:

$$\hat{b}_t^{CB} = \Psi^{58} \hat{b}_t^H + \Psi^{59} \hat{b}_t^G, \quad (\text{A.76})$$

where we defined $\Psi^{58} = -(\Psi^{56})^{-1} \Psi^{57}$ and $\Psi^{59} = (\Psi^{56})^{-1}$.

Linearizing equation (A.20) we obtain:

$$\hat{b}_t^H = \Psi^{60} \overrightarrow{\hat{\lambda}_t^{HB}} + \Psi^{61} \overrightarrow{\hat{b}_t^G} + \Psi^{62} \overrightarrow{\hat{b}_t^{CB}}, \quad (\text{A.77})$$

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where Ψ^{60} , Ψ^{61} , and Ψ^{62} are $1 \times F$ -sized matrices with elements Ψ_{1f}^{60} , Ψ_{1f}^{61} , and Ψ_{1f}^{62} (row 1, column f) defined as:

$$\Psi_{1f}^{60} = \begin{cases} \frac{\lambda^{HB,1}}{1-\lambda^{HB,1}} & , \text{ if } f = 1, \\ 0 & , \text{ otherwise,} \end{cases} \quad \Psi_{1f}^{62} = \begin{cases} -\left(\frac{1}{1-\lambda^{HB,1}}\right) \left(\frac{B^{CB,f}}{B^H}\right) & , \text{ if } f \geq 2, \\ 0 & , \text{ otherwise.} \end{cases}$$

$$\Psi_{1f}^{61} = \begin{cases} -\left(\frac{1}{1-\lambda^{HB,1}}\right) \left(\frac{B^{G,f}}{B^H}\right) & , \text{ if } f \geq 2, \\ 0 & , \text{ otherwise,} \end{cases}$$

Plugging equation (A.77) into (A.76), we obtain:

$$\hat{b}_t^{CB} = \Psi^{59} \hat{b}_t^G + \Psi^{63} \overrightarrow{\hat{\lambda}_t^{HB}} + \Psi^{64} \overrightarrow{\hat{b}_t^G} + \Psi^{65} \overrightarrow{\hat{b}_t^{CB}}, \quad (\text{A.78})$$

where we defined $\Psi^{63} = \Psi^{58} \Psi^{60}$, $\Psi^{64} = \Psi^{58} \Psi^{61}$ and $\Psi^{65} = \Psi^{58} \Psi^{62}$. Further substituting equations (A.49) and (A.51) into equation (A.78) we obtain:

$$\hat{b}_t^{CB} = \Psi^{66} \hat{b}_t^G + \Psi^{67} \overrightarrow{\hat{z}_t} + \Psi^{68} \overrightarrow{\hat{y}d_t} + \Psi^{69} \mathbb{E}_t \left[\overrightarrow{\hat{y}d_{t+1}} \right] + \Psi^{70} \overrightarrow{\hat{u}_t^B} + \Psi^{65} \overrightarrow{\hat{b}_t^{CB}}, \quad (\text{A.79})$$

where we defined

$$\begin{aligned} \Psi^{66} &= \Psi^{59} + \Psi^{64} \cdot \overrightarrow{1}_{F \times 1}, & \Psi^{69} &= \Psi^{63} \Psi^3, \\ \Psi^{67} &= \Psi^{63} \Psi^1, & \Psi^{70} &= \Psi^{64} \Xi. \\ \Psi^{68} &= \Psi^{63} \Psi^2, \end{aligned}$$

Linearizing the central bank aggregate bond holding equation, $B_t^{CB} = \sum_{f=1}^F B_t^{CB,f}$:

$$\hat{b}_t^{CB} = \sum_{f=1}^F \lambda^{CB,f} \hat{b}_t^{CB,f}. \quad (\text{A.80})$$

Linearizing the bond market equilibrium condition (A.23), we obtain

$$\begin{aligned} \hat{\lambda}_t^{HB,f} &= \left(\frac{B^{G,f}}{B^{G,f} + B^{CB,f}} \right) \hat{b}_t^{G,f} + \left(\frac{B^{CB,f}}{B^{G,f} + B^{CB,f}} \right) \hat{b}_t^{CB,f} - \left(\frac{\gamma_L(1 + \zeta^F)}{\gamma_L(1 + \zeta^F) - \gamma_{CB}} \right) \hat{y}_t \\ &\quad + \frac{1}{1 - \lambda^K} \cdot \hat{\lambda}_t^K + \left(\frac{\gamma_{CB}}{\gamma_L(1 + \zeta^F) - \gamma_{CB}} \right) \hat{u}_t^L, \quad f \geq 2. \end{aligned} \quad (\text{A.81})$$

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As above, steady-state ratios use the original bond positions, while hatted government debt uses $\tilde{B}^{G,f} = -B^{G,f}$.

From $\lambda_t^{HB,1} = 1 - \sum_{f=2}^F \lambda_t^{HB,f}$ we obtain

$$\hat{\lambda}_t^{HB,1} = - \sum_{f=2}^F \frac{\lambda_t^{HB,f}}{\lambda_t^{HB,1}} \hat{\lambda}_t^{HB,f}. \quad (\text{A.82})$$

We can rearrange equations (A.81) and (A.82) in the matrix form as

$$\Psi^{71} \cdot \overrightarrow{\hat{\lambda}_t^{HB}} = \Psi^{72} \cdot \overrightarrow{\hat{b}_t^G} + \Psi^{73} \cdot \overrightarrow{\hat{b}_t^{CB}} - \Psi^{74} \cdot \hat{y}_t + \Psi^{75} \cdot \hat{\lambda}_t^K + \Psi^{76} \hat{u}_t^L,$$

where $\{\Psi^{71}, \Psi^{72}, \Psi^{73}\}$ are $F \times F$ -sized matrices with elements Ψ_{jf}^{71} , Ψ_{jf}^{72} and Ψ_{jf}^{73} (row j , column f) and $\{\Psi^{74}, \Psi^{75}, \Psi^{76}\}$ are $F \times 1$ vectors with Ψ_{j1}^{74} (j th element). We define their elements as

$$\begin{aligned} \Psi_{jf}^{71} &= \begin{cases} 1 & , \text{ if } j = f, \\ \frac{\lambda_t^{HB,f}}{\lambda_t^{HB,1}} & , \text{ if } j = 1 \text{ \& } f > 1, \\ 0 & , \text{ otherwise,} \end{cases} & \Psi_{j1}^{74} &= \begin{cases} 0 & , \text{ if } j = 1, \\ \left(\frac{\gamma_L(1+\zeta^F)}{\gamma_L(1+\zeta^F) - \gamma_{CB}} \right) & , \text{ otherwise,} \end{cases} \\ \Psi_{jf}^{72} &= \begin{cases} \frac{B^{G,f}}{B^{G,f} + B^{CB,f}} & , \text{ if } j > 1 \text{ \& } j = f, \\ 0 & , \text{ otherwise,} \end{cases} & \Psi_{j1}^{75} &= \begin{cases} 0 & , \text{ if } j = 1, \\ \frac{1}{1 - \lambda^K} & , \text{ otherwise,} \end{cases} \\ \Psi_{jf}^{73} &= \begin{cases} \frac{B^{CB,f}}{B^{G,f} + B^{CB,f}} & , \text{ if } j > 1 \text{ \& } j = f, \\ 0 & , \text{ otherwise,} \end{cases} & \Psi_{j1}^{76} &= \begin{cases} 0 & , \text{ if } j = 1, \\ \left(\frac{\gamma_{CB}}{\gamma_L(1+\zeta^F) - \gamma_{CB}} \right) & , \text{ otherwise.} \end{cases} \end{aligned}$$

We can rearrange the previous equation as:

$$\overrightarrow{\hat{\lambda}_t^{HB}} = \Psi^{77} \cdot \overrightarrow{\hat{b}_t^G} + \Psi^{78} \cdot \overrightarrow{\hat{b}_t^{CB}} - \Psi^{79} \cdot \hat{y}_t + \Psi^{80} \cdot \hat{\lambda}_t^K + \Psi^{81} \cdot \hat{u}_t^L, \quad (\text{A.83})$$

where we defined

$$\begin{aligned} \Psi^{77} &= (\Psi^{71})^{-1} \Psi^{72}, & \Psi^{80} &= (\Psi^{71})^{-1} \Psi^{75}, \\ \Psi^{78} &= (\Psi^{71})^{-1} \Psi^{73}, & \Psi^{81} &= (\Psi^{71})^{-1} \Psi^{76}, \\ \Psi^{79} &= (\Psi^{71})^{-1} \Psi^{74}, \end{aligned}$$

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Plugging equations (A.49), (A.51) and (A.60) into equation (A.83), we obtain:

$$\begin{aligned} \Psi^{82} \cdot \vec{\hat{z}}_t + \Psi^{83} \cdot \vec{\hat{y}d}_t + \Psi^{84} \cdot \mathbb{E}_t \left[\vec{\hat{y}d}_{t+1} \right] &= \Psi^{85} \cdot \vec{\hat{u}}_t^B + \Psi^{86} \cdot \hat{b}_t^G + \Psi^{78} \cdot \vec{\hat{b}}_t^{CB} - \Psi^{79} \cdot \hat{y}_t \\ &\quad + \Psi^{87} \cdot (\hat{z}_t^K + \hat{r}_{t+1}^K) + \Psi^{81} \cdot \hat{u}_t^L, \end{aligned} \quad (\text{A.84})$$

where we defined

$$\begin{aligned} \Psi^{82} &= \Psi^1 + \Psi^{80} \Psi^{12}, & \Psi^{85} &= \Psi^{77} \Xi, \\ \Psi^{83} &= \Psi^2 + \Psi^{80} \Psi^{13}, & \Psi^{86} &= \Psi^{77} \cdot \vec{1}_{F \times 1}, \\ \Psi^{84} &= \Psi^3 + \Psi^{80} \Psi^{14}, & \Psi^{87} &= \Psi^{80} \kappa^S (1 - \lambda^K). \end{aligned}$$

Log-linearization: the conventional policy specific derivations Linearizing the central bank bond-holding rule in (37c) gives:

$$\hat{b}_t^{CB,f} = \hat{u}_t^{CB,f}, \quad \forall f = 2, \dots, F. \quad (\text{A.85})$$

We stack the central-bank bond holdings for maturities $f = 2, \dots, F$ into $\vec{\hat{b}}_t^{CB} = [\hat{b}_t^{CB,2}, \dots, \hat{b}_t^{CB,F}]'$ and the underlying factors into $\vec{\hat{u}}_t^{CB} = [\hat{u}_t^{CB,1}, \dots, \hat{u}_t^{CB,J^{CB}}]'$. Substituting the shock mapping (I.4) into the conventional-policy holding rule (A.85) and stacking over maturities gives:

$$\vec{\hat{b}}_t^{CB} = \mathcal{T}^{CB} \vec{\hat{u}}_t^{CB}, \quad (\text{A.86})$$

where $\mathcal{T}^{CB}$ is the $(F-1) \times J^{CB}$ matrix with elements $\mathcal{T}_{f,l}^{CB} = \tau_{f+1,l}^{CB}$ (row f , column l).

Log-linearization: the Yield-Curve-Control (YCC) target Linearizing the policy rule in (38) for an f -maturity bond, the YCC target satisfies

$$\hat{y}d_t^{YCC,f} = \rho_1 \hat{y}d_{t-1}^{YCC,f} + \rho_2 \hat{y}d_{t-2}^{YCC,f} + (1 - \rho_1 - \rho_2) \left[\gamma_\pi^f \hat{\pi}_t + \gamma_y^f \hat{y}_t + \tilde{\varepsilon}_t^{YDf} \right], \quad f \geq 2. \quad (\text{A.87})$$

To stack this across maturities, we define the vectors $\vec{\gamma}_\pi = [\gamma_\pi^2, \dots, \gamma_\pi^F]'$ and $\vec{\gamma}_y = [\gamma_y^2, \dots, \gamma_y^F]'$, and from (I.5) the $(F-1) \times L$ matrix $\mathcal{T}_{(f \geq 2)}^{YD}$ with elements $\mathcal{T}_{(f \geq 2),f,l}^{YD} = \tau_{f+1,l}^{YD}$ (row f , column l). With the shock vector $\vec{\varepsilon}_t^{YD} = [\varepsilon_t^{YD,1}, \dots, \varepsilon_t^{YD,L}]'$ and the stacked

YCC target $\overrightarrow{\hat{y}d_t^{YCC}}_{(f \geq 2)} = \left[\hat{y}d_t^{YCC,2}, \dots, \hat{y}d_t^{YCC,F} \right]'$, the target rule in vector form reads

$$\begin{aligned} \overrightarrow{\hat{y}d_t^{YCC}}_{(f \geq 2)} &= \rho_1 \overrightarrow{\hat{y}d_{t-1}^{YCC}}_{(f \geq 2)} + \rho_2 \overrightarrow{\hat{y}d_{t-2}^{YCC}}_{(f \geq 2)} \\ &\quad + (1 - \rho_1 - \rho_2) \left[\overrightarrow{\gamma}_{\pi(f \geq 2)} \hat{\pi}_t + \overrightarrow{\gamma}_{y(f \geq 2)} \hat{y}_t + \mathcal{T}_{(f \geq 2)}^{YD} \overrightarrow{\varepsilon}_t^{YD} \right]. \end{aligned} \quad (\text{A.88})$$

Log-linearization: the general policy The general-policy (GP) yield blends the counterfactual conventional-policy yield with the YCC target,

$$\hat{y}d_t^{GP,f} = \gamma_{CP}^f \hat{y}d_t^{CP,f} + (1 - \gamma_{CP}^f) \hat{y}d_t^{YCC,f}, \quad f \geq 2. \quad (\text{A.89})$$

Let Γ^{CP} be the $(F-1) \times (F-1)$ diagonal matrix with $\Gamma_{ff}^{CP} = \gamma_{CP}^{f+1}$ for $f = 1, \dots, F-1$ and $\Gamma_{ij}^{CP} = 0$ for $i \neq j$, and stack the GP and conventional-policy yields into $\overrightarrow{\hat{y}d_t^{GP}}_{(f \geq 2)} = \left[\hat{y}d_t^{GP,2}, \dots, \hat{y}d_t^{GP,F} \right]'$ and $\overrightarrow{\hat{y}d_t^{CP}}_{(f \geq 2)} = \left[\hat{y}d_t^{CP,2}, \dots, \hat{y}d_t^{CP,F} \right]'$. The general policy rule in vector form is

$$\overrightarrow{\hat{y}d_t^{GP}}_{(f \geq 2)} = \Gamma^{CP} \overrightarrow{\hat{y}d_t^{CP}}_{(f \geq 2)} + (I - \Gamma^{CP}) \overrightarrow{\hat{y}d_t^{YCC}}_{(f \geq 2)}, \quad (\text{A.90})$$

where I is the identity matrix of size $F-1$, and $\overrightarrow{\hat{y}d_t^{CP}}$ and $\overrightarrow{\hat{y}d_t^{YCC}}$ are the yield vectors that prevail in the counterfactual scenarios where the current yield is determined by conventional monetary policy and YCC policy, respectively. Setting $\Gamma^{CP} = 0$ recovers the pure yield-curve-control system, in which the realized yields coincide with the YCC targets, $\overrightarrow{\hat{y}d_t^{GP}} = \overrightarrow{\hat{y}d_t^{YCC}}$.

B Calibration and Estimation Strategy

B.1 Calibrating $\{z^f\}_{f=1}^F$ and z^K at the Steady State

Calibration of $\{z^f\}_{f=1}^F$ This section explains the calibration of $\{z^f\}_{f=1}^F$ in matching the steady-state yield curve. Following the convention of Appendix A.1.2, R^{f-1} denotes the per-period return on an f -maturity bond at the steady state (i.e., the steady-state value of R_{t+1}^{f-1} in (16)). Based on the data on yields of bonds with different maturities, we calculate f -maturity bond's average holding returns $\{R^{f-1}\}_{f=1}^F$, which are our calibration target.

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Step 1 Calculate steady-state bond shares $\{\lambda^{HB,f}\}_{f=1}^F$ using equation (48), given the data on $\{\lambda^{G,f}\}_{f=1}^F$, $\{\lambda^{CB,f}\}_{f=1}^F$, and ζ^{CB} .

Step 2 Aggregate the household's bond-portfolio return as $R^{HB} = \sum_{f=1}^F \lambda^{HB,f} \cdot R^{f-1}$ and compute the return ratios $\left\{\frac{R^{f-1}}{R^{HB}}\right\}_{f=1}^F$.

Step 3 Normalize $z^1 = 1$ and obtain initial guess for $\{z^{j,guess}\}$. Set $z^{j,old} = z^{j,guess}$ in the iteration below.

Step 4 Construct $\tilde{\Phi}^{B,old}$ based on the following formula, where the return ratios $\left\{\frac{R^{f-1}}{R^{HB}}\right\}_{f=1}^F$ across maturities are obtained from the data:

$$\tilde{\Phi}^{B,old} = \left[\sum_{f=1}^F \left[z^{f,old} \left(\frac{R^{f-1}}{R^{HB}} \right) \right]^{\kappa_B} \right]^{\frac{1}{\kappa_B}}.$$

Step 5 Back out new $z^{f,new}$, $f = 2, \dots, F$ estimates using:

$$z^{f,new} = (\lambda^{HB,f})^{\frac{1}{\kappa_B}} \left(\frac{R^{f-1}}{R^{HB}} \right)^{-1} \tilde{\Phi}^{B,old}.$$

Step 6 If $\{z^{f,new}\}_{f=2}^F$ differs from $\{z^{f,old}\}_{f=2}^F$ above a tolerance, set $z^{f,old} = z^{f,new}$ and return to Step 4.

Calibration of z^K We calibrate z^K such that the model's steady-state return on the household's bond portfolio, R^{HB} , matches with the observed data average. This alignment is achieved by finding the steady-state value of R^K from (A.34),² and subsequently replacing the values of R^{HB} and R^K into equation (A.32), which enables us to recover the z^K value consistent with the model's moment.

B.2 Elasticity Estimation, κ_B

We estimate the parameter κ_B using two empirical measures of the return spread and two instrumental-variable strategies for each measure. This allows us to assess how sensitive the implied elasticity is to the construction of the regressor and to the exclusion restriction

²In the context of our present calibration, the derived R^K is situated within the range corresponding to the average corporate debt rate across varied ratings.

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used for identification. Taking logs on both sides of equation (15) and differencing across maturities, we obtain

$$\log \left(\lambda_t^{HB,f} \right) - \log \left(\lambda_t^{HB,l} \right) = \alpha^{fl} + \kappa_B \cdot \left(\log E_t \left[Q_{t,t+1} R_{t+1}^{f-1} \right] - \log E_t \left[Q_{t,t+1} R_{t+1}^{l-1} \right] \right) + \varepsilon_t^{fl}, \quad (\text{B.1})$$

where α^{fl} denotes a constant containing the steady-state difference in the bond maturity demand preferences $z^f - z^l$, and $\varepsilon_t^{fl} \equiv (z_t^f - z^f) - (z_t^l - z^l)$ is a residual term capturing the discrepancy between the preference shocks for bond maturities f and l at time t . To a first-order approximation, the stochastic discount factor $Q_{t,t+1}$ on the right-hand side of (B.1) drops out, and it can be written as the difference in expected holding returns across maturities:

$$\log \left(\lambda_t^{HB,f} \right) - \log \left(\lambda_t^{HB,l} \right) = \alpha^{fl} + \kappa_B \cdot E_t \left[r_{t+1}^{f-1} - r_{t+1}^{l-1} \right] + \varepsilon_t^{fl}. \quad (\text{B.2})$$

The expected bond return spread $E_t \left[r_{t+1}^{f-1} - r_{t+1}^{l-1} \right]$ is not directly observable. The first empirical measure therefore uses an approximation centered on the bond yield spread:

$$\begin{aligned} E_t \left[r_{t+1}^{f-1} - r_{t+1}^{l-1} \right] &= yd_t^f - yd_t^l \\ &\quad - \underbrace{(f-1) \cdot E_t \left[yd_{t+1}^f - yd_t^f \right]}_{\text{add as control}} + \underbrace{(l-1) \cdot E_t \left[yd_{t+1}^l - yd_t^l \right]}_{\text{add as control}} \\ &\quad + \underbrace{(f-1) \cdot E_t \left[yd_{t+1}^f - yd_{t+1}^{f-1} \right]}_{\approx 0 \text{ (by assumption)}} - \underbrace{(l-1) \cdot E_t \left[yd_{t+1}^l - yd_{t+1}^{l-1} \right]}_{\approx 0 \text{ (by assumption)}}, \end{aligned} \quad (\text{B.3})$$

where the spread $yd_t^f - yd_t^l$ in the first line is directly observable, the expected yield change in the second line is approximated by using the realized changes as control variables, and the terms in the final line are assumed close to zero over short time intervals. This last condition is the identifying assumption that allows the level spread $yd_t^f - yd_t^l$ to serve as the regressor: it requires the expected term-premium differences between adjacent maturities to be small and, conditional on the controls, uncorrelated with the regressor. The equation used to estimate the elasticity parameter κ_B becomes:

$$\log \left(\lambda_{t+h}^{HB,f} \right) - \log \left(\lambda_{t+h}^{HB,l} \right) = \alpha_h^{fl} + \kappa_{B,h} \cdot \left[yd_t^f - yd_t^l \right] + \mathbf{x}_t' \beta_{\mathbf{h}}^{\mathbf{fl}} + \varepsilon_{t+h}^{fl}, \quad h \geq 0, \quad (\text{B.4})$$

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where $\mathbf{x}_t$ denotes a vector of control variables with corresponding coefficients β_h^{fl} , and the time subindex $h \geq 0$ allows for a lagged response of the bond portfolio composition to the yield spread.

The second empirical measure constructs the expected SDF-weighted returns that appear in (B.1). We proceed in the following three steps. First, we obtain a model-consistent estimate of the realized stochastic discount factor $Q_{t,t+1} \equiv \beta \frac{P_t C_t}{P_{t+1} C_{t+1}}$ by proxying the ratio with the nominal PCE index. Second, we construct the quarterly return to holding a bond of a given maturity from the daily estimates of the [Svensson \(1994\)](#) yield curve model provided by [Gürkaynak et al. \(2007\)](#); the implied bond price at any maturity and date follows from the model parameters and yields an approximation of realized returns. Finally, we estimate $E_t [Q_{t,t+1} R_{t+1}^{f-1}]$ by predicting realized SDF-weighted returns from their own lagged values, lagged nominal PCE growth, and the three principal components of the same fitted yield curve. This measure follows the exact model equation more closely, without the linearization and short-maturity assumptions in (B.3), at the cost of a more structural stance on the stochastic discount factor, the yield curve, and the quarterly holding period used to estimate expected returns.

For either measure, identification requires either the expected SDF-weighted return spread $\log E_t [Q_{t,t+1} R_{t+1}^{f-1}] - \log E_t [Q_{t,t+1} R_{t+1}^{l-1}]$ or the bond yield spread $yd_t^f - yd_t^l$, to be uncorrelated with the relative preference shocks captured by ε_t^{fl} . In the model, ε_t^{fl} contains the contemporaneous demand shocks for maturities f and l that directly move their portfolio shares, beyond any indirect effect operating through bond returns and yields. Since both return spreads are themselves affected by supply and demand shocks, the main threat to identification is omitted-variable bias from the unobserved demand shocks z_t^f and z_t^l : these shocks enter ε_t^{fl} and affect portfolio shares directly as well as indirectly through returns and yields. Aggregate shocks that shift the regressor but are uncorrelated with current and future household maturity-demand preferences are therefore natural candidate instruments. We use two such strategies:

1. Instrument the regressor with its own first lag, including lags of the dependent variable as controls to absorb serial correlation in the demand preference shocks;
2. Instrument the regressor with the Treasury supply shocks of [Phillot \(2025\)](#), identified from high-frequency movements in Treasury futures prices around auction announcements.

The dependent variable is the log difference between the aggregate household portfolio

share of a long-maturity Treasury group and that of a short-maturity Treasury group. The long-maturity group contains bonds with remaining maturity between 5 and 10 years, while the short-maturity group contains bonds with remaining maturity above 15 days and up to 90 days. For both groups, aggregate household Treasury holdings are computed monthly by subtracting U.S. Treasury securities held by the Federal Reserve from the government's outstanding Treasury amounts within the selected maturity ranges.³ In the yield-spread specification, the regressor is the spread between the 7-year and 3-month constant-maturity Treasury yields, which lie within the maturity bands used to construct the dependent variable. We also control for the one-month-ahead changes in those yields and for the first three lags of the dependent variable. The regressions are estimated over 2003m3 to 2019m3.⁴

The validity of the own-lag instrument follows the logic of [Arellano and Bond \(1991\)](#): conditional on the controls, lagged values of the regressor are excluded from the equation for future portfolio shares. Lagged dependent variables absorb serial correlation in the demand preference shocks that move portfolio shares independently of the regressor, while the lagged regressor retains predictive power for its current value through the persistence of other shocks that drive it. The identifying assumption is analogous to timing restrictions commonly used in dynamic panel data and VAR models ([Christiano et al., 1999](#)): the exclusion restriction holds as long as a shock that affects only the regressor does not generate future bond portfolio demand shocks (for example, via general equilibrium effects) within one month, the frequency of the data.

Figure [B.1](#) reports the own-lag IV and OLS estimates from Jordà local projections over a fifty-month horizon, together with the 95% and 68% robust confidence bands for the IV estimates. Both panels are consistent with the model's predicted response of aggregate household portfolio shares to a higher return spread. For calibration purposes, we select $\kappa_B = 10$, which is consistent with the short-horizon response (roughly $h \leq 6$ months) in Figure [B.1](#).

The second approach uses Treasury supply shocks constructed from Treasury auction announcement data by [Phillot \(2025\)](#) as an IV.⁵ The shocks are identified from 30-minute changes in front-month Treasury futures prices following auction announcements. Under

³The outstanding amounts of Government Treasuries are reported in the U.S. Treasury Monthly Statement of the Public Debt (MSPD).

⁴The lower bound of the sample period is dictated by the availability of maturity-disaggregated statistics concerning the Federal Reserve's Treasury bond portfolio, commencing from 2002m12; the first three months are absorbed by the lag controls L^1 to L^3 on the dependent variable.

⁵[Phillot \(2025\)](#) isolates the component of futures-price movements attributable to U.S. Treasury announcements over July 1998 to March 2020.

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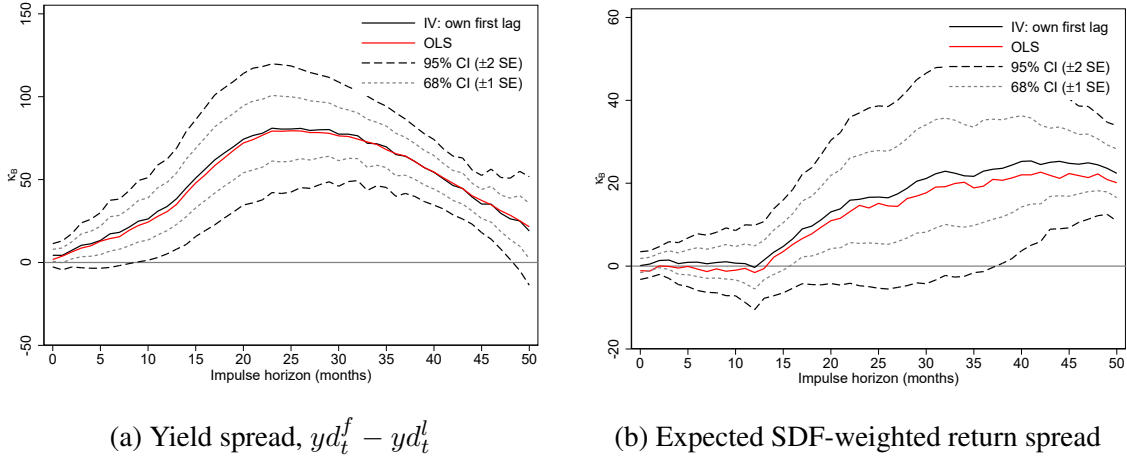


Figure B.1: Impulse response to a shock in the regressor of equation (B.4), instrumented with the regressor’s own first lag. Panel (a) uses the yield spread $yd_t^f - yd_t^l$ and panel (b) the expected SDF-weighted return spread as the regressor. The solid black line is the instrumental variables (IV) estimate of the bond portfolio elasticity, κ_B , with long and short-dashed lines indicating the 95 and 68 percent robust confidence intervals, respectively. The red line reports the OLS estimate. The sample period is from 2003m3 to 2019m3.

the assumptions that Treasury demand is fixed within the window and that the announcement is the only systematic news, a no-arbitrage argument maps an announcement-driven price decline, equivalently a yield increase, into higher-than-expected supply. The paper constructs separate shock series from the 2-, 5-, 10-, and 30-year Treasury futures contracts; we use the 5-year and 10-year supply shocks, which bracket the 5-to-10-year long leg of the benchmark pair, as separate instruments. These shocks are relevant because an unexpected placement of long-maturity debt requires investors to absorb additional duration risk. The price must fall and the yield must rise to clear the market, lifting the term premium and widening both the yield spread $yd_t^f - yd_t^l$ and the expected SDF-weighted return spread. The same variation is plausibly orthogonal to the relative preference shocks ε_t^{fl} because it is identified from government financing announcements rather than household maturity demand: the 30-minute window is designed to isolate the announcement surprise, while the exclusion restriction requires these issuance surprises not to forecast future relative demand by households.

Figure B.2 reports the elasticities obtained by using the 5- and 10-year supply shocks jointly as an IV. The estimates are broadly consistent with the own-lag IV results in Figure B.1 and with the sign predicted by the model. The supply shocks are, however, weaker instruments. According to the [Olea and Pflueger \(2013\)](#) effective F -statistic, the appropri-

ate weak-instrument diagnostic under the Newey-West standard errors used throughout, the first stage for the expected-return regression in panel B.2b remains below the conventional critical values for 10% and 20% worst-case bias of the 2SLS estimator and is close to, but generally below, the 30% benchmark. The yield specification in panel B.2a is weaker still. Effective F -statistics at this level imply that the supply-IV estimates may be materially biased toward the OLS benchmark in the worst case, so these estimates are best read as a corroborating cross-check rather than as sharp identification. We therefore retain the own-lag IV as our preferred specification, while noting that the two approaches deliver broadly consistent values of κ_B .

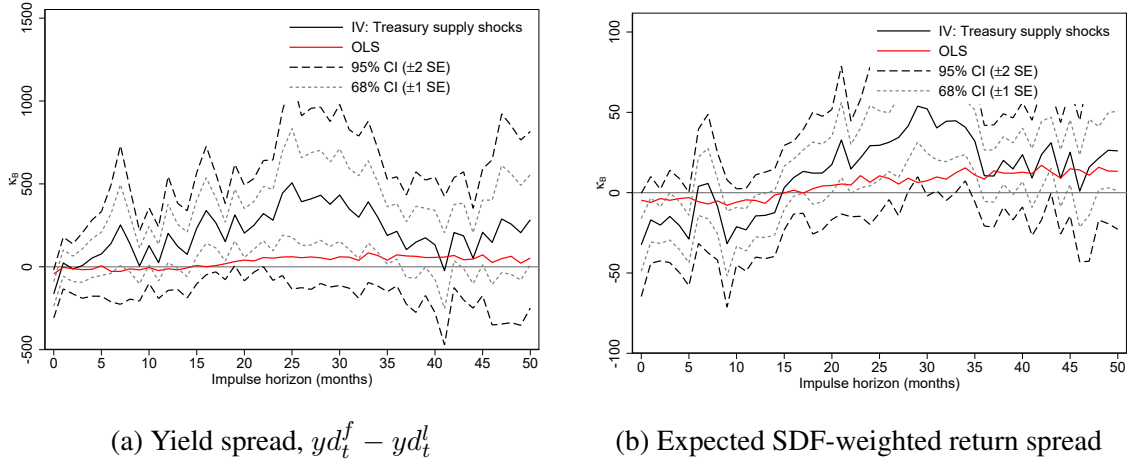


Figure B.2: Impulse response to a shock in the regressor of equation (B.4), instrumented with the Phillot (2025) 5- and 10-year Treasury supply shocks. Panel (a) uses the yield spread $yd_t^f - yd_t^l$ and panel (b) the expected SDF-weighted return spread as the regressor. The solid black line is the instrumental variables (IV) estimate of the elasticity κ_B , with long and short-dashed lines indicating the 95 and 68 percent robust confidence intervals, respectively. The red line reports the OLS estimate. The sample period is from 2003m3 to 2019m3.

C Welfare

C.1 Deriving a second-order welfare approximation

In order to approximate welfare up to a second-order, due to the presence of trend inflation, we cannot discard $\hat{\Delta}_t$, which is the price dispersion term's log-deviation from its steady-state value. For this issue, see Coibion et al. (2012).

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Step 1: Following the literature, for any variable X , we define $\bar{X}$ as its steady-state value (with the positive trend inflation $\bar{\Pi} > 1$) and $\bar{X}^F$ as its flexible price (efficient) steady-state value. Also define (small) letter $\tilde{x}$ as log-deviation of X around $\bar{X}^F$, and $\hat{x}$ as log-deviation of X around $\bar{X}$.

We first characterize the constrained efficient (i.e., flexible-price) steady state. With the optimal production subsidy $\zeta^F = (\epsilon - 1)^{-1}$ that eliminates the monopolistic competition distortion, there is no distortion other than the firms' financing constraint in the flexible-price steady state economy anymore.⁶ In particular, each firm's price resetting condition (A.2) becomes

$$1 = \frac{P_t^*}{P_t} = \underbrace{\left(\frac{(1 + \zeta^F)^{-1} \epsilon}{\epsilon - 1} \right)}_{=1} \left[1 - \gamma_L (\tilde{R}_{t+1}^K - 1) \right]^{-1} \cdot \frac{MC_t}{P_t},$$

which can be simplified to

$$\frac{MC_t}{P_t} = 1 - \gamma_L (\tilde{R}_{t+1}^K - 1), \quad (\text{C.1})$$

where we use that all firms become identical, and thus $MC_t(\nu) = MC_t$ for all $\nu \in [0, 1]$. Therefore, the real marginal cost becomes the same for all firms. Plugging the real marginal cost (A.3) into the individual firm's labor demand (30) with $W_t(\nu) = W_t$ for $\forall \nu$, and defining $n_t = \frac{N_t}{\bar{N}_t}$ and $y_t = \frac{Y_t}{A_t \bar{N}_t}$, we obtain

$$n_t = (1 - \alpha) y_t \left(\frac{P_t^K}{P_t} \right)^\alpha \left(\frac{W_t}{P_t A_t} \right)^{-\alpha} = (1 - \alpha) \left[1 - \gamma_L (\tilde{R}_{t+1}^K - 1) \right] y_t \left(\frac{W_t}{P_t A_t} \right)^{-1}, \quad (\text{C.2})$$

which, with the household's intra-temporal consumption-labor decision (22), becomes:

$$\frac{n_t^{\frac{1}{\eta}}}{c_t^{-1}} = (1 - \alpha) \left[1 - \gamma_L (\tilde{R}_{t+1}^K - 1) \right] \frac{y_t}{n_t}, \quad (\text{C.3})$$

which is the efficiency condition that ensures that the household's marginal rate of substitution matches with the marginal rate of technical substitution. Therefore, at the flexible-price steady state, the new constant Φ , which will turn out to enter in the per-period welfare later,

⁶The capital producing firm is competitive and thus our economy features no friction other than the firms' financing constraint if it were not nominal rigidity nor trend inflation. Still, the constraint on loan issuance does not affect firms' marginal decisions on labor and capital.

can be calculated as

$$\Phi \equiv (\bar{n}^F)^{1+\frac{1}{\eta}} = (1-\alpha) \left[1 - \gamma_L \left(\frac{R^K}{\beta^{-1}\Pi \cdot GA \cdot GN} - 1 \right) \right] \frac{\bar{y}^F}{\bar{c}^F}, \quad (\text{C.4})$$

where the steady state expression for $\tilde{R}^K$ has been replaced using equation (A.28), and $\bar{n}^F$, $\bar{y}^F$, and $\bar{c}^F$ denote values of normalized labor, output, and consumption, respectively.

Step 2: With aggregation equations (A.6) and (A.7), we obtain

$$\left(\frac{N_t}{\bar{N}_t} \right)^{1-\alpha} \left(\frac{K_t}{A_{t-1}\bar{N}_{t-1}} \right)^\alpha = \alpha^\alpha (1-\alpha)^{1-\alpha} (GA_t \cdot GN)^\alpha \left(\frac{Y_t}{A_t \bar{N}_t} \right) \Delta_t^{(1-\alpha)\left[\frac{\eta}{\eta+1} + \frac{\alpha}{1-\alpha}\right]}, \quad (\text{C.5})$$

which is the aggregate production function with the price dispersion Δ_t . Plugging steady-state (with trend-inflation) capital (A.40) and output (A.42) into equation (C.5) yields

$$\frac{N}{\bar{N}} = \left[\alpha^\alpha (1-\alpha)^{1-\alpha} (GA \cdot GN)^\alpha (\xi^K)^{-\alpha} \right]^{\frac{1}{1-\alpha}} \Delta^{\frac{\eta+\alpha}{(\eta+1)(1-\alpha)}} (\xi^Y)^{\frac{1}{1-\alpha} \left(\frac{\eta+\alpha}{\eta+1} \right)} (\xi^C)^{-\frac{\eta}{\eta+1}} \quad (\text{C.6})$$

where ξ^K in (A.40), ξ^Y in (A.39), and ξ^C in (A.41) all depend on θ and the trend inflation Π . Therefore, we see that $\bar{n} \neq \bar{n}^F$ and define $\log X_n \equiv \log \bar{n} - \log \bar{n}^F$, which will turn out to be useful later when we calculate the household's first-order labor cost.

Step 3: Price dispersion with positive trend inflation

Delta method We use the following approximation throughout this section. For a random variable X with $\mathbb{E}[X] = \mu_X$,

$$\text{Var}(f(X)) = f'(\mu_X)^2 \cdot \text{Var}(X) + \text{h.o.t.} \quad (\text{C.7})$$

where h.o.t. denotes higher-order terms. In the remainder of this section, displays derived from Taylor or delta-method expansions omit such terms unless they are shown explicitly.

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Price dispersion We use lower-case p_t and $p_t(\nu)$ as logarithms of P_t and $P_t(\nu)$. By applying the above delta method to $P_t^{1-\epsilon} = \mathbb{E}_\nu (P_t(\nu)^{1-\epsilon})$, we obtain

$$p_t = \underbrace{\int_0^1 p_t(\nu) d\nu}_{\equiv \bar{p}_t} + \frac{1}{2} \left(\frac{1}{1-\epsilon} \right) \frac{\text{Var}_\nu (P_t(\nu)^{1-\epsilon})}{\mathbb{E}_\nu (P_t(\nu)^{1-\epsilon})^2} + \text{h.o.t.} \quad (\text{C.8})$$

where we define $\bar{p}_t \equiv \mathbb{E}_\nu (p_t(\nu))$. Applying the delta method to $\text{Var}_\nu (P_t(\nu)^{1-\epsilon})$, we have

$$\text{Var}_\nu (P_t(\nu)^{1-\epsilon}) = (1-\epsilon)^2 \cdot [\exp((1-\epsilon)\bar{p}_t)]^2 \cdot \text{Var}_\nu (p_t(\nu)), \quad (\text{C.9})$$

where we define $D_t \equiv \text{Var}_\nu (p_t(\nu))$. Applying the delta method to $\mathbb{E}_\nu (P_t(\nu)^{1-\epsilon})$, we obtain

$$\mathbb{E}_\nu (P_t(\nu)^{1-\epsilon}) = \exp((1-\epsilon)\bar{p}_t) \left[1 + \frac{(1-\epsilon)^2}{2} D_t \right]. \quad (\text{C.10})$$

Plugging (C.9) and (C.10) into (C.8), we obtain

$$p_t = \bar{p}_t + \frac{1-\epsilon}{2} \cdot \frac{D_t}{\left[1 + \frac{(1-\epsilon)^2}{2} D_t \right]^2}, \quad (\text{C.11})$$

which we linear-approximate around $D_t = \bar{D}$ and obtain

$$p_t - \bar{p}_t = \underbrace{\frac{1-\epsilon}{2} \cdot \frac{\bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D} \right]^2}}_{\equiv \Theta_1^p} + \underbrace{\frac{1-\epsilon}{2} \cdot \frac{1 - \frac{(1-\epsilon)^2}{2} \bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D} \right]^3}}_{\equiv \Theta_2^p} \cdot (D_t - \bar{D}) = \Theta_1^p + \Theta_2^p (D_t - \bar{D}). \quad (\text{C.12})$$

Now from our original definition of the price dispersion Δ_t (i.e., (44)), we take logarithm on both sides, linear-approximate around $\bar{D}$, and plug (C.12) into it to attain

$$\begin{aligned} \ln \Delta_t &= \ln \int_0^1 \left(\frac{P_t(\nu)}{P_t} \right)^{\frac{-\epsilon(\eta+1)}{\eta+\alpha}} d\nu \\ &= \frac{\epsilon(\eta+1)}{\eta+\alpha} (p_t - \bar{p}_t) + \ln \left(1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha} \right)^2 \bar{D} \right) + \frac{\frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha} \right)^2}{1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha} \right)^2 \bar{D}} (D_t - \bar{D}) \\ &= \Theta_1^\Delta + \Theta_2^\Delta \cdot (D_t - \bar{D}) + \text{h.o.t.}, \end{aligned}$$

where

$$\Theta_1^\Delta \equiv \frac{\epsilon(\eta+1)}{\eta+\alpha} \cdot \frac{1-\epsilon}{2} \cdot \frac{\bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D}\right]^2} + \ln \left(1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha}\right)^2 \bar{D}\right), \quad (\text{C.13})$$

$$\Theta_2^\Delta \equiv \frac{\epsilon(\eta+1)}{\eta+\alpha} \cdot \frac{1-\epsilon}{2} \cdot \frac{1 - \frac{(1-\epsilon)^2}{2} \bar{D}}{\left[1 + \frac{(1-\epsilon)^2}{2} \bar{D}\right]^3} + \frac{\frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha}\right)^2}{1 + \frac{1}{2} \left(\frac{\epsilon(\eta+1)}{\eta+\alpha}\right)^2 \bar{D}}. \quad (\text{C.14})$$

If we define b_t as the logarithm of the newly price-resetting firm's relative price P_t^*/P_t and $\bar{b}$ as its steady state value, we have $\bar{b} \neq 0$ due to the trend inflation. Combining (A.10) and (A.12) and linearizing, we obtain

$$b_t \equiv p_t^* - p_t = \bar{b} + \underbrace{\frac{\theta \Pi^{\epsilon-1}}{1 - \theta \Pi^{\epsilon-1}}}_{\equiv M} \hat{\pi}_t = \bar{b} + M \cdot \hat{\pi}_t, \quad \text{with } \bar{b} = \frac{1}{\epsilon-1} \ln \left(\frac{1-\theta}{1 - \theta \Pi^{\epsilon-1}} \right). \quad (\text{C.15})$$

With $D_t = \text{Var}_\nu(p_t(\nu)) = \mathbb{E}_\nu((p_t(\nu) - p_t + p_t - \bar{p}_t)^2)$, we can write it as

$$D_t = \int_0^{1-\theta} (p_t^* - p_t)^2 d\nu + 2 \left(\int_0^{1-\theta} (p_t^* - p_t) d\nu \right) (p_t - \bar{p}_t) \quad (\text{C.16})$$

$$\begin{aligned} &+ (1-\theta)(p_t - \bar{p}_t)^2 + \int_{1-\theta}^1 (p_{t-1}(\nu) - \bar{p}_t)^2 d\nu \\ &= (1-\theta)(p_t^* - p_t)^2 + 2(1-\theta)(p_t^* - p_t)(p_t - \bar{p}_t) + (1-\theta)(p_t - \bar{p}_t)^2 + \theta D_{t-1} + \theta(\bar{p}_t - \bar{p}_{t-1})^2, \end{aligned} \quad (\text{C.17})$$

where we use

$$\int_{1-\theta}^1 (p_{t-1}(\nu) - \bar{p}_t)^2 d\nu = \theta D_{t-1} + \theta(\bar{p}_{t-1} - \bar{p}_t)^2. \quad (\text{C.18})$$

Conjecture Following Coibion et al. (2012), we conjecture the dynamics of D_t up to second order as⁷

$$D_t - \bar{D} = \kappa_D \hat{\pi}_t + Z_D (\hat{\pi}_t)^2 + F_D (D_{t-1} - \bar{D}) + G_D (D_{t-1} - \bar{D}) \hat{\pi}_t + H_D (D_{t-1} - \bar{D})^2. \quad (\text{C.19})$$

⁷Following Coibion et al. (2012), we treat κ_D as being of the same order as the shocks. Hence $\kappa_D \hat{\pi}_t$ is a second-order term, since $\hat{\pi}_t$ is a first-order deviation. This allows the first-order log-linearized model to omit price dispersion while retaining the effect of price dispersion in the second-order welfare expansion.

with

$$\begin{aligned}
 \bar{D} &= (\bar{b} + \Theta_1^p)^2 + \frac{\theta}{1-\theta}(\bar{\pi})^2 \quad (\text{i.e., steady state value of } D_t), \\
 \kappa_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [2(1-\theta)M(\bar{b} + \Theta_1^p) + 2\theta\bar{\pi}], \\
 Z_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [(1-\theta)M^2 + 2(1-\theta)M\Theta_2^p\kappa_D + (\Theta_2^p)^2(\kappa_D)^2 + \theta - 2\theta\Theta_2^p\kappa_D], \\
 F_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [\theta + 2\theta\Theta_2^p\bar{\pi}], \\
 G_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [2(1-\theta)M\Theta_2^pF_D + 2(\Theta_2^p)^2\kappa_DF_D - 2\theta\Theta_2^pF_D \\
 &\quad + 2\theta\Theta_2^p - 2\theta(\Theta_2^p)^2\kappa_D], \\
 H_D &= [1 - 2(1-\theta)\Theta_2^p(\bar{b} + \Theta_1^p) + 2\theta\Theta_2^p\bar{\pi}]^{-1} [(\Theta_2^p)^2(F_D)^2 + \theta(\Theta_2^p)^2 - 2\theta(\Theta_2^p)^2F_D].
 \end{aligned} \tag{C.20}$$

In the absence of trend inflation, we would have $\bar{\pi} = 0$ and $\bar{D} = 0$, implying that D_t is a second-order variable around zero and $\kappa_D = 0$. By contrast, when steady-state inflation is positive ($\bar{\pi} > 0$) and the price dispersion measure satisfies $\bar{D} > 0$, the linear response of price dispersion to inflation is generally nonzero, so κ_D is treated as a first-order coefficient. This does not make $\kappa_D\hat{\pi}_t$ a first-order object: because $\hat{\pi}_t$ is itself a first-order deviation, the product $\kappa_D\hat{\pi}_t$ remains second order and belongs in the welfare expansion.

Our goal is to derive (C.19) from firms' optimal pricing decisions and the impact of price dispersion on the aggregate price level. Substituting (C.12) and (C.15) into (C.16), and replacing $(D_t - \bar{D})$ with its conjectured form in (C.19) up to second order,⁸ and then matching coefficients yields the set of parameters reported in (C.20).

Consumption utility We can second-order approximate the utility of consumption as

$$\begin{aligned}
 u(c_t) &= \log c_t = u(\bar{c}^F) + u'_{\bar{c}^F} \cdot \bar{c}^F \cdot \underbrace{\left(\frac{c_t - \bar{c}^F}{\bar{c}^F} \right)}_{=\tilde{c}_t + \frac{1}{2}(\tilde{c}_t)^2} + \frac{1}{2}u''_{\bar{c}^F} \cdot (\bar{c}^F)^2 \cdot \underbrace{\left(\frac{c_t - \bar{c}^F}{\bar{c}^F} \right)^2}_{=(\tilde{c}_t)^2} + \text{h.o.t} \\
 &= u(\bar{c}^F) + \tilde{c}_t + \text{h.o.t.}
 \end{aligned} \tag{C.21}$$

⁸On the right-hand side, $(p_t - \bar{p}_t)^2$ appears and contains a second-order component, $(D_t - \bar{D})^2$, from (C.12). We use (C.19) to express this term in terms of $(\hat{\pi}_t)^2$, $(D_{t-1} - \bar{D})^2$, and $\hat{\pi}_t(D_{t-1} - \bar{D})$.

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Step 4: Labor aggregation and cost By applying the delta method—see (C.7)—to the labor aggregator, we can obtain⁹

$$\tilde{n}_t - \mathbb{E}_\nu(\tilde{n}_t(\nu)) = \underbrace{\frac{\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{\nabla}}{\left[1 + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right)^2 \bar{\nabla} \right]^2}}_{\equiv \Theta_1^n} + \underbrace{\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \frac{1 - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right)^2 \bar{\nabla}}{\left[1 + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right)^2 \bar{\nabla} \right]^3}}_{\equiv \Theta_2^n} \cdot (\nabla_t - \bar{\nabla}) \quad (\text{C.22})$$

where $\nabla_t \equiv \text{Var}_\nu(\log n_t(\nu))$. The second-order approximation to the firm ν -specific labor cost around the flexible-price steady state yields

$$\frac{\eta}{\eta+1} \left(\frac{N_t(\nu)}{\bar{N}_t} \right)^{\frac{\eta+1}{\eta}} = \frac{\eta}{\eta+1} (\bar{n}^F)^{\frac{\eta+1}{\eta}} + \Phi \left[\tilde{n}_t(\nu) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \tilde{n}_t(\nu)^2 \right] + h.o.t \quad (\text{C.23})$$

where the constant Φ is from (C.4). Aggregating (C.23) over firms $\nu \in [0, 1]$ and plugging (C.22) results in

$$\begin{aligned} \frac{\eta}{\eta+1} \int_0^1 \left(\frac{N_t(\nu)}{\bar{N}_t} \right)^{\frac{\eta+1}{\eta}} d\nu - \frac{\eta}{\eta+1} (\bar{n}^F)^{\frac{\eta+1}{\eta}} &= \Phi \left[\mathbb{E}_\nu(\tilde{n}_t(\nu)) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \int_0^1 \tilde{n}_t(\nu)^2 d\nu \right] \\ &= -\Phi \left(\Theta_1^n - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_1^n)^2 \right) + \Phi \left[\left(1 - \left(\frac{\eta+1}{\eta} \right) \Theta_1^n \right) \bar{n}_t + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{n}_t^2 \right. \\ &\quad \left. + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_2^n)^2 (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla})^2 - \frac{\eta+1}{\eta} \Theta_2^n \bar{n}_t (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) \right. \\ &\quad \left. + \left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n - \Theta_2^n) (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{\nabla} \right] \right]. \end{aligned} \quad (\text{C.24})$$

Labor dispersion From individual firm's labor and capital demand (30) and the household's intra-marginal condition (22), we obtain

$$\tilde{k}_t(\nu) = \left(1 + \frac{1}{\eta} \right) \tilde{n}_t(\nu) + \text{aggregate}, \quad (\text{C.25})$$

where ‘aggregate’ stands for aggregate variables. Therefore, we obtain

$$\tilde{y}_t(\nu) = \left(1 + \frac{\alpha}{\eta} \right) \tilde{n}_t(\nu) + \text{aggregate}, \quad (\text{C.26})$$

⁹In the flexible-price steady-state, there is no heterogeneity among firms, i.e., $\bar{n}^F(\nu) = \bar{n}^F$ for $\forall \nu$.

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by plugging (C.25) into each firm's production function $\tilde{y}_t(\nu) = \alpha \tilde{k}_t(\nu) + (1 - \alpha) \tilde{n}_t(\nu)$. From the Dixit-Stiglitz good demand (27) and with (C.26), we can get

$$\text{Var}_\nu(\tilde{n}_t(\nu)) = \left(\frac{\epsilon}{1 + \frac{\alpha}{\eta}} \right)^2 \text{Var}_\nu(p_t(\nu)), \quad \text{with } \bar{\nabla} = \left(\frac{\epsilon}{1 + \frac{\alpha}{\eta}} \right)^2 \bar{D}. \quad (\text{C.27})$$

Step 5: Constructing a welfare function: Combining consumption utility (C.21) and labor disutility (C.24), we construct welfare as

$$\begin{aligned} \mathbb{E}U_t - \bar{U}^F = \mathbb{E} & \left[\tilde{c}_t + \Phi \left(\Theta_1^n - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_1^n)^2 \right) - \Phi \left\{ \left(1 - \left(\frac{\eta+1}{\eta} \right) \Theta_1^n \right) \tilde{n}_t + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \tilde{n}_t^2 \right. \right. \\ & + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_2^n)^2 (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla})^2 - \frac{\eta+1}{\eta} \Theta_2^n \tilde{n}_t (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) \\ & \left. \left. + \left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n \right) (\text{Var}_\nu(\tilde{n}_t(\nu)) - \bar{\nabla}) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \bar{\nabla} \right\} \right], \end{aligned}$$

where from (A.41), (A.42), and (C.6) the flexible-price steady-state utility is given as

$$\begin{aligned} \bar{U}^F &= \log(\bar{c}^F) - \frac{\eta}{\eta+1} \left(\frac{N^F}{\bar{N}} \right)^{\frac{\eta+1}{\eta}} \\ &= \frac{1}{\eta+1} \left[\log(\xi^{C,f}) + \frac{\eta+\alpha}{1-\alpha} \log(\xi^{Y,f}) \right] - \frac{\eta}{\eta+1} \left[\alpha^\alpha (1-\alpha)^{1-\alpha} (GA \cdot GN)^\alpha (\xi^{K,f})^{-\alpha} \right]^{\frac{\eta+1}{(1-\alpha)\eta}} (\xi^{Y,f})^{\frac{\eta+\alpha}{(1-\alpha)\eta}} (\xi^{C,f})^{-1}, \end{aligned}$$

where

$$\begin{aligned} \xi^{F,f} &= (1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left[\beta^{-1} \cdot GA \cdot GN - (1-\delta) \right]^\alpha \left(\frac{\eta+1}{\eta+\alpha} \right), \quad \xi^{H,f} = 1 - \gamma_L \left(\frac{R^K}{R^S} - 1 \right), \quad \xi^{Y,f} = \frac{\xi^{H,f}}{\xi^{F,f}}, \\ \xi^{K,f} &= \alpha(1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA \cdot GN \cdot \left[\beta^{-1} \cdot GA \cdot GN - (1-\delta) \right]^{-\frac{\eta(1-\alpha)}{\eta+\alpha}} \xi^{Y,f}, \\ \xi^{C,f} &= 1 - \zeta^G - \xi^{K,f} \left(1 - \frac{1-\delta}{GA \cdot GN} \right). \end{aligned}$$

Note that $\xi^{C,f}$, $\xi^{Y,f}$, and $\xi^{K,f}$ are levels of ξ^C , ξ^Y , and ξ^K when $\theta = 0$ —corresponding to the flexible price steady state. With

$$\log \frac{\xi^Y}{\xi^{Y,f}} = \log \left(\frac{1 - \theta \beta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)}}{1 - \theta \beta \Pi^{\epsilon-1}} \right) + \frac{1}{\epsilon-1} \left[1 + \epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right) \right] \log \left(\frac{1-\theta}{1 - \theta \Pi^{\epsilon-1}} \right),$$

and

$$\log \frac{\xi^K}{\xi^{K,f}} = \log \left(\frac{\xi^Y}{\xi^{Y,f}} \right) + \log \Delta,$$

$$\log \frac{\xi^C}{\xi^{C,f}} = \log \frac{1 - \zeta^G - \xi^K \left(1 - \frac{1 - \delta}{GA \cdot GN} \right)}{1 - \zeta^G - \xi^{K,f} \left(1 - \frac{1 - \delta}{GA \cdot GN} \right)},$$

where Δ at the steady state with trend inflation is defined in (A.35), which gives

$$\log \Delta = \log \left(\frac{1 - \theta}{1 - \theta \Pi^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)}} \right) + \frac{\epsilon}{\epsilon - 1} \left(\frac{\eta + 1}{\eta + \alpha} \right) \log \left(\frac{1 - \theta \Pi^{\epsilon-1}}{1 - \theta} \right).$$

If we define $\log X_c = \tilde{c}_t - \hat{c}_t$ as the log-difference in consumption between our steady state (with trend-inflation) and the flexible price steady state, we obtain

$$\log X_c \equiv \log \bar{c} - \log \bar{c}^F = \frac{1}{\eta + 1} \left[\log \left(\frac{\xi^C}{\xi^{C,f}} \right) + \frac{\eta + \alpha}{1 - \alpha} \log \left(\frac{\xi^Y}{\xi^{Y,f}} \right) \right]. \quad (\text{C.28})$$

For labor, we define $\log X_n$ as the log-difference in labor between our steady state with trend inflation and the flexible price steady state, which is, with the help of (C.6), given by

$$\log X_n \equiv \log \bar{n} - \log \bar{n}^F = -\frac{\alpha}{1 - \alpha} \log \frac{\xi^K}{\xi^{K,f}} + \frac{1}{1 - \alpha} \left(\frac{\eta + \alpha}{\eta + 1} \right) \log \frac{\xi^Y}{\xi^{Y,f}} - \frac{\eta}{\eta + 1} \log \frac{\xi^C}{\xi^{C,f}} + \frac{\eta + \alpha}{(\eta + 1)(1 - \alpha)} \log \Delta. \quad (\text{C.29})$$

With $\tilde{c}_t = \hat{c}_t + \log X_c$, $\tilde{n}_t = \hat{n}_t + \log X_n$, and the stationarity assumption (Coibion et al., 2012; Dordal i Carreras et al., 2016), we can get

$$\mathbb{E} \left[\tilde{c}_t - \Phi \left(1 - \left(\frac{\eta + 1}{\eta} \right) \Theta_1^n \right) \tilde{n}_t \right] = \log X_c - \Phi \left(1 - \left(\frac{\eta + 1}{\eta} \right) \Theta_1^n \right) \log X_n.$$

Second order terms: With $\tilde{n}_t = \hat{n}_t + \log X_n$, second-order terms can be collected as

$$\begin{aligned} & -\Phi \left[\frac{\eta + 1}{2\eta} \mathbb{E} (\hat{n}_t^2) + \frac{\eta + 1}{2\eta} (\Theta_2^n)^2 \mathbb{E} \left((\text{Var}_\nu(\hat{n}_t(\nu)) - \bar{\nabla})^2 \right) - \frac{\eta + 1}{\eta} \Theta_2^n \mathbb{E} (\hat{n}_t (\text{Var}_\nu(\hat{n}_t(\nu)) - \bar{\nabla})) \right. \\ & \left. + \left(\frac{1}{2} \left(\frac{\eta + 1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n - \frac{\eta + 1}{\eta} \Theta_2^n \log X_n \right) \mathbb{E} (\text{Var}_\nu(\hat{n}_t(\nu)) - \bar{\nabla}) \right], \end{aligned}$$

which, after we can plug (C.27) into, becomes

$$\begin{aligned}
 & -\Phi \left[\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \text{Var}(\hat{n}_t) + \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_2^n)^2 \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^4 \mathbb{E}(D_t - \bar{D})^2 \right. \\
 & \quad \left. - \frac{\eta+1}{\eta} \Theta_2^n \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \text{Cov}(\hat{n}_t, D_t) \right. \\
 & \quad \left. + \left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n \left(1 + \frac{\eta+1}{\eta} \log X_n \right) \right) \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \mathbb{E}(D_t - \bar{D}) \right]. \quad (\text{C.30})
 \end{aligned}$$

Finally, by plugging (C.19) into (C.30), we get the following proposition. Since κ_D is of the same order as shock processes, up to a second-order, we can ignore covariance terms and the square term of D_t . Taking expectations of (C.19) under stationarity then gives

$$\mathbb{E}(D_t - \bar{D}) = F_D \mathbb{E}(D_t - \bar{D}) + Z_D \text{Var}(\hat{\pi}_t),$$

so that

$$\mathbb{E}(D_t - \bar{D}) = \frac{Z_D}{1 - F_D} \text{Var}(\hat{\pi}_t).$$

Therefore, a 2nd-order approximation to the expected per-period welfare would be given as

$$\mathbb{E}U_t - \bar{U}^F = \Omega_0 + \Omega_n \text{Var}(\hat{n}_t) + \Omega_\pi \text{Var}(\hat{\pi}_t) + h.o.t.,$$

with

$$\begin{aligned}
 \Omega_0 &= \log X_c - \Phi \left(1 - \left(\frac{\eta+1}{\eta} \right) \Theta_1^n \right) \log X_n + \Phi \left(\Theta_1^n - \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (\Theta_1^n)^2 \right) - \Phi \frac{1}{2} \frac{\eta+1}{\eta} (\log X_n)^2 \\
 & \quad - \Phi \frac{1}{2} \left(\frac{\eta+1}{\eta} \right) \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \bar{D}, \\
 \Omega_n &= -\Phi \frac{1}{2} \left(\frac{\eta+1}{\eta} \right), \\
 \Omega_\pi &= -\Phi \left[\left(\frac{1}{2} \left(\frac{\eta+1}{\eta} \right) (1 + 2\Theta_1^n \Theta_2^n) - \Theta_2^n \left(1 + \frac{\eta+1}{\eta} \log X_n \right) \right) \left(\frac{\epsilon}{1+\frac{\alpha}{\eta}} \right)^2 \frac{Z_D}{1 - F_D} \right],
 \end{aligned}$$

where $\log X_c$ and $\log X_n$ are defined in (C.28) and (C.29), respectively; the coefficients Θ_1^n and Θ_2^n are provided in (C.22); and $\bar{D}$ is determined by jointly solving (C.12)—which describes the definition of Θ_1^p —and (C.20). The parameters κ_D , Z_D , F_D , G_D , and H_D are specified in (C.20). Higher-order terms are denoted by *h.o.t.* Finally, the coefficient Ω_0 is part of the policy-independent terms (*t.i.p.*) that appear in Proposition 1.

D Summary of Equilibrium Equations

D.1 Summary of Equilibrium Equations

The equations below collect the non-linear equilibrium conditions of the model. Equations (i)–(xxxvii) are common to the conventional-policy and yield-curve-control regimes and can be reused without modification; the two regimes differ only in the policy closure that occupies slot (xxxviii), reported separately afterward. The bond-market clearing conditions in (xxiv)–(xxv) are accounting relations that hold under both regimes; the regimes differ only in whether the central bank’s long-maturity bond holdings or the long-maturity yields are treated as exogenous.

Common equilibrium equations The following equations hold under both regimes.

$$(i). \quad \frac{C_t}{A_t \bar{N}_t} = (1 - \zeta_t^G) \left(\frac{Y_t}{A_t \bar{N}_t} \right) + \left(\frac{1 - \delta}{GA_t \cdot GN} \right) \left(\frac{K_t}{A_{t-1} \bar{N}_{t-1}} \right) - \left(\frac{K_{t+1}}{A_t \bar{N}_t} \right), \quad (D.1)$$

$$(ii). \quad 1 = \beta \cdot \mathbb{E}_t \left[\frac{R_{t+1}^S}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)} \right], \quad (D.2)$$

$$(iii). \quad \lambda_t^{HB,1} = 1 - \sum_{f=2}^F \lambda_t^{HB,f}, \quad (D.3)$$

$$(iv). \quad YD_t^1 = \max \{ YD_t^{1*}, 1 \}, \quad (D.4)$$

$$(v). \quad \frac{YD_t^{1*}}{YD^1} = \left(\frac{YD_{t-1}^{1*}}{YD^1} \right)^{\rho_1} \left(\frac{YD_{t-2}^{1*}}{YD^1} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\gamma_\pi} \left(\frac{Y_t}{\bar{Y}_t} \right)^{\gamma_y} \exp \left(\tilde{\varepsilon}_t^{YD^1} \right) \right]^{1 - (\rho_1 + \rho_2)}, \quad (D.5)$$

$$(vi). \quad \tilde{\varepsilon}_t^{YD^f} = \sum_{l=1}^L \tau_{f,l}^{YD} \varepsilon_t^{YD^l}, \quad (D.6)$$

$$(vii). \quad \lambda_t^{HB,f} = \left(\frac{\mathbb{E}_t \left[\frac{\beta z_t^f}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right) \left(YD_{t+1}^{f-1} \right)^{-(f-1)}}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right) \left(YD_t^f \right)^{-f}} \right]}{\Phi_t^B} \right)^{\kappa_B}, \quad \forall f \quad (D.7)$$

$$(viii). \quad \Phi_t^B = \left[\sum_{j=1}^F \mathbb{E}_t \left[\frac{\beta z_t^j}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \cdot \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right) \left(Y D_{t+1}^{j-1} \right)^{-(j-1)}}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right) \left(Y D_t^j \right)^{-j}} \right]^{\kappa_B} \right]^{\frac{1}{\kappa_B}}, \quad (\text{D.8})$$

$$(ix). \quad \lambda_t^K = \left(\frac{z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K]}{\Phi_t^S} \right)^{\kappa_S}, \quad (\text{D.9})$$

$$(x). \quad \Phi_t^S = \left[\left(\mathbb{E}_t [Q_{t,t+1} R_{t+1}^{HB}] \right)^{\kappa_S} + \left(z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K] \right)^{\kappa_S} \right]^{\frac{1}{\kappa_S}}, \quad (\text{D.10})$$

$$(xi). \quad R_t^j = \sum_{f=0}^{F-1} \lambda_{t-1}^{j,f+1} \frac{\left(Y D_t^f \right)^{-f}}{\left(Y D_{t-1}^{f+1} \right)^{-(f+1)}}, \quad j \in \{HB, G, CB\}, \quad (\text{D.11})$$

$$(xii). \quad R_t^S = (1 - \lambda_{t-1}^K) R_t^{HB} + \lambda_{t-1}^K R_t^K, \quad (\text{D.12})$$

$$(xiii). \quad 1 = \mathbb{E}_t \left[Q_{t,t+1} \Pi_{t+1} \left[(1 - \delta) + \frac{P_{t+1}^K}{P_{t+1}} \right] \right], \quad (\text{D.13})$$

$$(xiv). \quad F_t = (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{(1 + \varsigma_F)^{-1} \epsilon}{\epsilon - 1} \right) \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta+1}{\eta+\alpha} \right)} \\ + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} F_{t+1} \right], \quad (\text{D.14})$$

$$(xv). \quad H_t = \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-1} \frac{Y_t}{A_t \bar{N}_t} \left[1 - \gamma_L \cdot \left(\tilde{R}_{t+1}^K - 1 \right) \right] + \theta \beta \mathbb{E}_t \left[\Pi_{t+1}^{\epsilon-1} H_{t+1} \right], \quad (\text{D.15})$$

$$(xvi). \quad \frac{F_t}{H_t} = \left(\frac{1 - \theta}{1 - \theta \Pi_t^{\epsilon-1}} \right)^{\left(\frac{1}{\epsilon-1} \right) \left[1 + \epsilon \left(\frac{1-\alpha}{\eta+\alpha} \right) \right]}, \quad (\text{D.16})$$

$$(xvii). \quad \Delta_t = (1 - \theta) \left(\frac{1 - \theta \Pi_t^{\epsilon-1}}{1 - \theta} \right)^{\left(\frac{\epsilon}{\epsilon-1} \right) \left(\frac{\eta+1}{\eta+\alpha} \right)} + \theta \Pi_t^{\epsilon \left(\frac{\eta+1}{\eta+\alpha} \right)} \Delta_{t-1}, \quad (\text{D.17})$$

$$(xviii). \quad \frac{N_t}{\bar{N}_t} = (1 - \alpha)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{C_t}{A_t \bar{N}_t} \right)^{-\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\left(\frac{\eta}{\eta+\alpha} \right)} \left(\frac{P_t^K}{P_t} \right)^{\alpha \left(\frac{\eta}{\eta+\alpha} \right)} \Delta_t^{\frac{\eta}{\eta+1}}, \quad (\text{D.18})$$

$$(xix). \quad \frac{K_t}{A_{t-1} \bar{N}_{t-1}} = \alpha (1 - \alpha)^{\frac{1-\alpha}{\eta+\alpha}} \cdot GA_t \cdot GN \cdot \left(\frac{C_t}{A_t \bar{N}_t} \right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(\frac{Y_t}{A_t \bar{N}_t} \right)^{\frac{\eta+1}{\eta+\alpha}} \left(\frac{P_t^K}{P_t} \right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha} \right)} \Delta_t, \quad (\text{D.19})$$

$$(xx). \quad \frac{B_t^G}{P_t A_t \bar{N}_t} = \frac{R_t^G}{\Pi_t \cdot GA_t \cdot GN} \cdot \frac{B_{t-1}^G}{P_{t-1} A_{t-1} \bar{N}_{t-1}} - [\zeta_t^G + \zeta^F - \zeta_t^T] \left(\frac{Y_t}{A_t \bar{N}_t} \right), \quad (\text{D.20})$$

$$(xxi). \quad Q_{t,t+1} = \frac{\beta}{\Pi_{t+1} \cdot GA_{t+1} \cdot GN} \frac{\left(\frac{C_t}{A_t \bar{N}_t} \right)}{\left(\frac{C_{t+1}}{A_{t+1} \bar{N}_{t+1}} \right)}, \quad (D.21)$$

$$(xxii). \quad \tilde{R}_{t+1}^K \equiv \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K], \quad (D.22)$$

$$(xxiii). \quad YD_t^f \equiv \left(Q_t^f \right)^{-1/f}, \quad \forall f, \quad (D.23)$$

$$(xxiv). \quad - \left(\frac{B_t^{G,f}}{A_t \bar{N}_t P_t} + \frac{B_t^{CB,f}}{A_t \bar{N}_t P_t} \right) \cdot \left(\lambda_t^{HB,f} \right)^{-1} = \left(\frac{1 - \lambda_t^K}{\lambda_t^K} \right) \left[\gamma_L \cdot (1 + \zeta^F) \left(\frac{Y_t}{A_t \bar{N}_t} \right) - \left(\frac{Y}{A \bar{N}} \right) \gamma_{CB} \cdot \exp(u_t^L) \right], \quad \forall f > 1, \quad (D.24)$$

$$(xxv). \quad \lambda_t^{CB,f} = \frac{\lambda_t^{HB,f} \left(\lambda_t^{CB,1} - \lambda_t^{G,1} \right) - \lambda_t^{G,f} \left(\lambda_t^{CB,1} - \lambda_t^{HB,1} \right)}{\lambda_t^{HB,1} - \lambda_t^{G,1}}, \quad f = 2, \dots, F, \quad (D.25)$$

$$(xxvi). \quad \lambda_t^{G,1} = \frac{1}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad (D.26)$$

$$\lambda_t^{G,f} = \frac{a^{B,f} \exp(\tilde{u}_t^{B,f})}{1 + \sum_{l=2}^F a^{B,l} \exp(\tilde{u}_t^{B,l})}, \quad \forall f > 1,$$

$$(xxvii). \quad \tilde{u}_t^{B,f} = \sum_{j=2}^J \tau_{fj}^B u_t^{B,j}, \quad (D.27)$$

$$(xxviii). \quad u_t^{B,j} = \rho_{B,j} u_{t-1}^{B,j} + \varepsilon_t^{B,j}, \quad \forall j = 1, \dots, J, \quad (D.28)$$

$$(xxix). \quad u_t^{CB,h} = \rho_{CB} u_{t-1}^{CB,h} + \varepsilon_t^{CB,h}, \quad \forall h = 1, \dots, J^{CB}, \quad (D.29)$$

$$(xxx). \quad B_t^{G,f} = \lambda_t^{G,f} B_t^G, \quad \forall f = 1, \dots, F, \quad (D.30)$$

$$(xxxi). \quad B_t^{CB} = \sum_{f=1}^F B_t^{CB,f}, \quad (D.31)$$

$$(xxxii). \quad u_t^L = \rho_L \cdot u_{t-1}^L + \varepsilon_t^L, \quad (D.32)$$

$$(xxxiii). \quad GA_t = \exp(\mu + \varepsilon_t^A), \quad (D.33)$$

$$(xxxiv). \quad \zeta_t^G = \frac{1}{1 + a^G \exp(-u_t^G)}, \quad (D.34)$$

$$(xxxv). \quad \zeta_t^T = \frac{1}{1 + a^T \exp(-u_t^T)}, \quad (D.35)$$

$$(xxxvi). \quad u_t^G = \rho_G \cdot u_{t-1}^G + \varepsilon_t^G, \quad (D.36)$$

$$(xxxvii). \quad u_t^T = \rho_T \cdot u_{t-1}^T + \varepsilon_t^T, \quad (D.37)$$

Conventional policy closure Under conventional policy, the central bank's maturity-by-maturity bond holdings, $B_t^{CB,f}$, follow an exogenous process, so the long-maturity yields YD_t^f , $f \geq 2$, are determined endogenously by the common clearing block in (xxiv)–(xxv). Slot (xxxviii) is then closed by the exogenous central-bank holdings:

$$(xxxviii). \quad \frac{B_t^{CB,f}}{A_t \bar{N}_t P_t} = \frac{\overline{B^{CB,f}}}{A \bar{N} P} \exp \left(\sum_{h=1}^{J^{CB}} \tau_{fh}^{CB} u_t^{CB,h} \right), \quad \forall f = 2, \dots, F. \quad (D.38)$$

Yield-curve-control closure Under yield-curve control, the central bank sets the long-maturity yields, YD_t^f , $f \geq 2$, through a maturity-by-maturity rule, and its bond holdings adjust endogenously through the common clearing block in (xxiv)–(xxv). Slot (xxxviii) is then filled by the long-maturity yield rule:

$$(xxxviii). \quad \frac{YD_t^f}{YD^f} = \left(\frac{YD_{t-1}^f}{YD^f} \right)^{\rho_1} \left(\frac{YD_{t-2}^f}{YD^f} \right)^{\rho_2} \left[\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\gamma_\pi^f} \left(\frac{Y_t}{\bar{Y}_t} \right)^{\gamma_y^f} \exp \left(\tilde{\varepsilon}_t^{YD^f} \right) \right]^{1-(\rho_1+\rho_2)}, \quad f \geq 2. \quad (D.39)$$

Equation (xxxi) is the accounting identity that aggregates the central bank's maturity-specific positions into its total bond holdings, $B_t^{CB} = \sum_{f=1}^F B_t^{CB,f}$. The aggregate B_t^{CB} does not appear in the remaining equilibrium conditions directly; it is the level used in the central bank's balance sheet and in the portfolio-share relations $\lambda_t^{CB,f} = B_t^{CB,f} / B_t^{CB}$ that underlie the bond-market block and the steady-state calibration in Appendix A.1.2.

D.2 Summary of Log-Linearized Equations

This subsection collects the log-linearized equilibrium conditions used to solve the model. Equations (i)–(xxvi) are common to the conventional-policy and yield-curve-control (YCC) regimes and can be reused without modification; the two regimes are different only in the closure that occupies slot (xxvii), reported separately afterward. The composite coefficients used below are introduced earlier in this appendix: Ξ in equation (A.50), Ψ^1 – Ψ^3 in equation (A.49), $\Psi^{HB,4}$ – $\Psi^{HB,6}$ in equation (A.55), $\Psi^{G,4}$ – $\Psi^{G,6}$ in equation (A.56), Ψ^{16} – Ψ^{24} in equation (A.66), Ψ^{26} – Ψ^{36} in equation (A.67), Ψ^{37} – Ψ^{45} in equation (A.72), Ψ^{46} – Ψ^{55} in equation (A.73), and Ψ^{56} – Ψ^{87} in equation (A.76), equation (A.77), equation (A.78), and

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equation (A.83). Remaining variables can be recovered from the equations stated earlier in this appendix.

Common linearized equations The following equations hold under both regimes.

$$(i). \quad \hat{y}_t = \mathbb{E}_t \left[\hat{y}_{t+1} + \Psi^{46} \cdot \hat{\pi}_{t+1} - \Psi^{47} \cdot \vec{\hat{z}}_t - \Psi^{48} \cdot \hat{z}_t^K - \Psi^{49} \cdot \vec{\hat{y}}_t - \Psi^{50} \cdot \vec{\hat{y}}_{t+1} \right. \\ \left. - \Psi^{51} \cdot \hat{r}_{t+1}^K - \Psi^{52} \cdot (\hat{k}_t - \hat{\varepsilon}_t^A) + \Psi^{53} \cdot \hat{k}_{t+1} - \Psi^{54} \cdot \hat{k}_{t+2} + \Psi^{55} \cdot \hat{u}_t^G \right], \quad (D.40)$$

$$(ii). \quad \hat{b}_t^{CB} = \Psi^{66} \hat{b}_t^G + \Psi^{67} \vec{\hat{z}}_t + \Psi^{68} \vec{\hat{y}}_t + \Psi^{69} \mathbb{E}_t \left[\vec{\hat{y}}_{t+1} \right] + \Psi^{70} \vec{\hat{u}}_t^B + \Psi^{65} \vec{\hat{b}}_t^{CB}, \quad (D.41)$$

$$(iii). \quad \Psi^{82} \cdot \vec{\hat{z}}_t + \Psi^{83} \cdot \vec{\hat{y}}_t + \Psi^{84} \cdot \mathbb{E}_t \left[\vec{\hat{y}}_{t+1} \right] = \Psi^{85} \cdot \vec{\hat{u}}_t^B + \Psi^{86} \cdot \hat{b}_t^G + \Psi^{78} \cdot \vec{\hat{b}}_t^{CB} - \Psi^{79} \cdot \hat{y}_t \\ + \Psi^{87} \cdot (\hat{z}_t^K + \hat{r}_{t+1}^K) + \Psi^{81} \cdot \hat{u}_t^L, \quad (D.42)$$

$$(iv). \quad \hat{y}_t^1 = \max \left\{ \hat{y}_t^{1*}, 0 \right\}, \quad (D.43)$$

$$(v). \quad \hat{y}_t^{1*} = \rho_1 \hat{y}_{t-1}^{1*} + \rho_2 \hat{y}_{t-2}^{1*} + (1 - \rho_1 - \rho_2) \left[\gamma_\pi \hat{\pi}_t + \gamma_y \hat{y}_t + \hat{\varepsilon}_t^{YD^1} \right], \quad (D.44)$$

$$(vi). \quad \hat{\varepsilon}_t^{YD^f} = \sum_{l=1}^L \tau_{f,l}^{YD} \varepsilon_t^{YD^l}, \quad \forall f = 1, \dots, F, \quad (D.45)$$

$$(vii). \quad \hat{r}_{t+1}^K = -\Psi^{37} \cdot \vec{\hat{z}}_t - \Psi^{38} \cdot \hat{z}_t^K - \Psi^{39} \cdot \vec{\hat{y}}_t - \Psi^{40} \cdot \mathbb{E}_t \left[\vec{\hat{y}}_{t+1} \right] + \Psi^{41} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{42} \cdot \mathbb{E}_t [\hat{y}_{t+1}] \\ + \Psi^{43} \cdot \hat{k}_{t+1} - \Psi^{44} \cdot \mathbb{E}_t [\hat{k}_{t+2}] - \Psi^{45} \cdot \hat{u}_t^G, \quad (D.46)$$

$$(viii). \quad \hat{b}_t^G = \frac{R^G}{\Pi \cdot GA \cdot GN} \cdot \left[\Psi^{G,4} \vec{\hat{u}}_{t-1}^B - \Psi^{G,5} \vec{\hat{y}}_t + \Psi^{G,6} \vec{\hat{y}}_{t-1} - \hat{\pi}_t - \hat{\varepsilon}_t^A + \hat{b}_{t-1}^G \right] \\ + \left(1 - \frac{R^G}{\Pi \cdot GA \cdot GN} \right) \left[\hat{y}_t + \left(\frac{\zeta^G}{\zeta^G + \zeta^F - \zeta^T} \right) \frac{a^G}{1 + a^G} \hat{u}_t^G - \left(\frac{\zeta^T}{\zeta^G + \zeta^F - \zeta^T} \right) \frac{a^T}{1 + a^T} \hat{u}_t^T \right], \quad (D.47)$$

$$(ix). \quad \hat{f}_t = -\Psi^{16} \cdot \vec{\hat{z}}_t - \Psi^{17} \cdot \hat{z}_t^K - \Psi^{18} \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] + \Psi^{19} \cdot \hat{y}_t - \Psi^{20} \cdot \vec{\hat{y}}_t - \Psi^{21} \cdot \hat{r}_{t+1}^K - \Psi^{22} \cdot \mathbb{E}_t \left[\vec{\hat{y}}_{t+1} \right] \\ + \Psi^{23} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{24} \cdot \mathbb{E}_t [\hat{f}_{t+1}] \quad (D.48)$$

$$(x). \quad \hat{h}_t = \Psi^{26} \cdot \vec{\hat{z}}_t + \Psi^{27} \cdot \hat{z}_t^K - \Psi^{28} \cdot [\hat{k}_t - \hat{\varepsilon}_t^A] + \Psi^{29} \cdot \hat{u}_t^G + \Psi^{30} \cdot \hat{y}_t + \Psi^{31} \cdot \vec{\hat{y}}_t - \Psi^{32} \cdot \hat{r}_{t+1}^K \\ + \Psi^{33} \cdot \hat{k}_{t+1} + \Psi^{34} \cdot \mathbb{E}_t \left[\vec{\hat{y}}_{t+1} \right] + \Psi^{35} \cdot \mathbb{E}_t [\hat{\pi}_{t+1}] + \Psi^{36} \cdot \mathbb{E}_t [\hat{h}_{t+1}] \quad (D.49)$$

$$(xi). \quad \hat{f}_t - \hat{h}_t = \left[1 + \epsilon \left(\frac{1 - \alpha}{\eta + \alpha} \right) \right] \left(\frac{\theta \Pi^{\epsilon-1}}{1 - \theta \Pi^{\epsilon-1}} \right) \hat{\pi}_t, \quad (D.50)$$

$$(xii). \quad \hat{c}_t = \left[(1 - \zeta^G) \frac{Y}{C} \right] \left[\hat{y}_t - \frac{1}{1 + a^G} \hat{u}_t^G \right] + \left[\frac{1 - \delta}{GA \cdot GN} \frac{K}{C} \right] \left(\hat{k}_t - \hat{\varepsilon}_t^A \right) - \frac{K}{C} \hat{k}_{t+1}, \quad (D.51)$$

$$(xiii). \quad \hat{n}_t = \left(\frac{\eta}{\eta + \alpha} \right) \left[1 + \left(\frac{\alpha}{1 - \alpha} \right) \left(\frac{\eta + 1}{\eta} \right) \right] \hat{y}_t - \left(\frac{\alpha}{1 - \alpha} \right) \left[\hat{k}_t - \hat{\varepsilon}_t^A \right], \quad (D.52)$$

$$(xiv). \quad \hat{b}_t^{CB} = \sum_{f=1}^F \lambda^{CB,f} \hat{b}_t^{CB,f}, \quad (D.53)$$

$$(xv). \quad \hat{\lambda}_t^K = \kappa^S (1 - \lambda^K) \left(\hat{z}_t^K + \mathbb{E}_t [\hat{r}_{t+1}^K - \hat{r}_{t+1}^{HB}] \right), \quad (D.54)$$

$$(xvi). \quad \mathbb{E}_t [\hat{r}_{t+1}^{HB}] = \Psi^{HB,4} \Psi^1 \cdot \hat{z}_t^{\rightarrow} + [\Psi^{HB,4} \Psi^2 + \Psi^{HB,6}] \cdot \hat{y}_t^{\rightarrow} + [\Psi^{HB,4} \Psi^3 - \Psi^{HB,5}] \cdot \mathbb{E}_t [\hat{y}_{t+1}^{\rightarrow}], \quad (D.55)$$

$$(xvii). \quad \mathbb{E}_t [\hat{r}_{t+1}^S] = \frac{\lambda^K (R^K - R^{HB})}{R^S} \hat{\lambda}_t^K + \frac{(1 - \lambda^K) R^{HB}}{R^S} \mathbb{E}_t [\hat{r}_{t+1}^{HB}] + \frac{\lambda^K R^K}{R^S} \hat{r}_{t+1}^K, \quad (D.56)$$

$$(xviii). \quad \mathbb{E}_t [\hat{r}_{t+1}^G] = \Psi^{G,4} \cdot \hat{u}_t^{\rightarrow} - \Psi^{G,5} \cdot \mathbb{E}_t [\hat{y}_{t+1}^{\rightarrow}] + \Psi^{G,6} \cdot \hat{y}_t^{\rightarrow}, \quad (D.57)$$

$$(xix). \quad \hat{\Delta}_t = 0, \quad (D.58)$$

$$(xx). \quad u_t^{B,j} = \rho_{B,j} \cdot u_{t-1}^{B,j} + \varepsilon_t^{B,j}, \quad \forall j = 1, \dots, J, \quad (D.59)$$

$$(xxi). \quad u_t^{CB,h} = \rho_{CB} \cdot u_{t-1}^{CB,h} + \varepsilon_t^{CB,h}, \quad \forall h = 1, \dots, J^{CB}, \quad (D.60)$$

$$(xxii). \quad u_t^L = \rho_L \cdot u_{t-1}^L + \varepsilon_t^L, \quad (D.61)$$

$$(xxiii). \quad u_t^G = \rho_G \cdot u_{t-1}^G + \varepsilon_t^G, \quad (D.62)$$

$$(xxiv). \quad u_t^T = \rho_T \cdot u_{t-1}^T + \varepsilon_t^T, \quad (D.63)$$

$$(xxv). \quad \hat{z}_t^{\rightarrow} = \rho_z \hat{z}_{t-1}^{\rightarrow} + \hat{\varepsilon}_t^z, \quad (D.64)$$

$$(xxvi). \quad \hat{z}_t^K = \rho_z^K \hat{z}_{t-1}^K + \hat{\varepsilon}_t^{z,K}. \quad (D.65)$$

Conventional policy closure Under conventional policy the central bank sets the short-maturity rate through the common rule in (iv)–(v) and holds its long-maturity bond positions fixed at the exogenous path in equation (A.86). The long-maturity yields $\hat{y}_t^f$, $f \geq 2$, are then determined endogenously by the common block. Slot (xxvii) is closed by the exogenous holding rule:

$$(xxvii). \quad \hat{b}_t^{CB} \xrightarrow{(f \geq 2)} = \mathcal{T}^{CB} \hat{u}_t^{CB}. \quad (D.66)$$

Yield-curve-control closure Under yield-curve control the central bank steers the entire yield curve. The short-maturity rate keeps following the common rule in (iv)–(v), while the long-maturity yields $\hat{y}_t^f$, $f \geq 2$, follow a maturity-by-maturity Taylor rule and the central bank's bond holdings adjust endogenously to implement these targets. Slot (xxvii) is therefore filled by the long-maturity policy rule:

$$\begin{aligned}
 (xxvii). \quad \overrightarrow{\hat{y}_{CC}^f}_{(f \geq 2)} &= \rho_1 \overrightarrow{\hat{y}_{CC}^f}_{(f \geq 2)} + \rho_2 \overrightarrow{\hat{y}_{CC}^f}_{(f \geq 2)} \\
 &\quad + (1 - \rho_1 - \rho_2) \left[\overrightarrow{\gamma_{\pi(f \geq 2)}} \hat{\pi}_t + \overrightarrow{\gamma_{y(f \geq 2)}} \hat{y}_t + \mathcal{T}_{(f \geq 2)}^{YD} \overrightarrow{\varepsilon}_t^{YD} \right].
 \end{aligned}
 \tag{D.67}$$

Note that price dispersion has no first-order effect around the symmetric steady state, hence (D.58) follows. Lead identities such as $\tilde{R}_{t+1}^K \equiv \mathbb{E}_t[Q_{t,t+1} R_{t+1}^K]$ and growth definitions $GA_{t+1} \equiv A_{t+1}/A_t$ apply at all dates and are inherited from the non-linear block. The vector notation for bond-preference shocks is kept for the general derivation. In the numerical simulations and determinacy checks, only the first bond-preference component is active, which is equivalent to setting the volatility of the remaining bond-preference components to zero.

E Additional Results: Steady-State Comparative Statics

Comparative statics with κ_B Figure E.3 shows how the steady-state yield curve changes as κ_B increases. A higher value of κ_B implies that household demand across bond maturities becomes more elastic with respect to $z^f R^{f-1}$. At the steady state, $z^f R^{f-1}$ is generally decreasing in f , primarily because z^f declines with maturity (see Table II.1).

As a result, an increase in κ_B induces households to tilt their portfolios toward shorter-maturity bonds. To maintain the same portfolio allocation at the steady state, the yield curve must become steeper, so that $z^f R^{f-1}$ declines less sharply with maturity f . Consequently, higher κ_B is associated with a steeper steady-state yield curve.

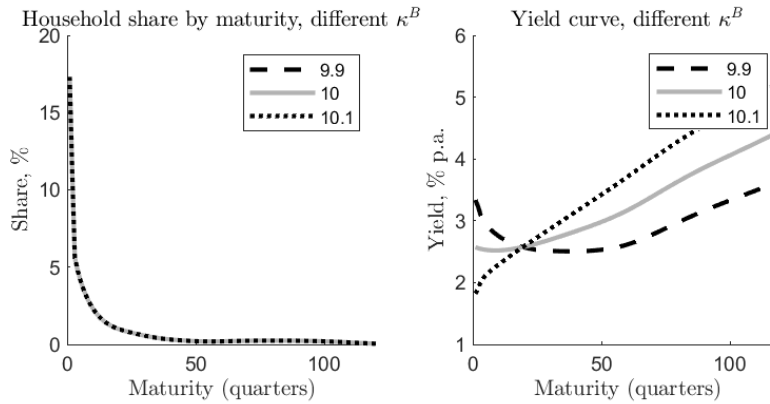


Figure E.3: Variations in κ_B : as $\kappa_B \rightarrow \infty$

Comparative statics with z^K Figure E.4 presents comparative statics with respect to z^K , which affects the allocation of savings between the bonds and loans. Holding the calibrated $\{z^f\}_{f=1}^F$ unchanged and varying $z^K \in [0.8, 1.3]$, a higher z^K means households invest more in loans to firms rather than in the bond market, increasing λ^K . This reallocation increases steady-state capital, output, and consumption.

As households tilt their portfolios toward loans, the average marginal propensity to con-

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sume (MPC) and the equilibrium loan rate R^K both decline. Through portfolio rebalancing, this shifts the entire yield curve downward and, somewhat paradoxically, increases credit spreads. At the same time, the fall in R^G lowers the ratio of government bond holdings to GDP.

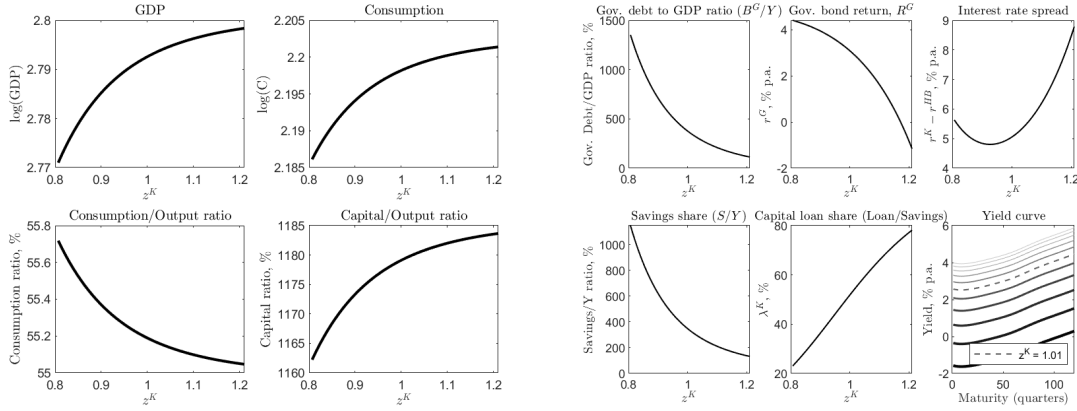


Figure E.4: Variations in z^K

Comparative statics with κ_S Figure E.5 presents comparative statics for the shape parameter κ_S , which affects the savings allocation condition between the bond and loan markets. Holding the calibrated values $\{z^f\}_{f=1}^F$ and z^K fixed, and varying κ_S over the range $[0, 25]$, we find that an increase in κ_S leads to a higher λ^K . Intuitively, since $R^K > R^{HB}$ in the steady state, a larger κ_S makes households' demand for loans more elastic.

This heightened cross-substitution towards loans pushes down the loan interest rate R^K . As borrowing becomes cheaper, firms expand capital demand, which in turn raises output and consumption. At the same time, the average marginal propensity to consume declines as before. Credit spreads widen, and the fall in R^K also reduces the government's effective bond return R^G via portfolio rebalancing, pushing down the entire yield curve further. As a result, the government's debt-to-GDP ratio decreases.

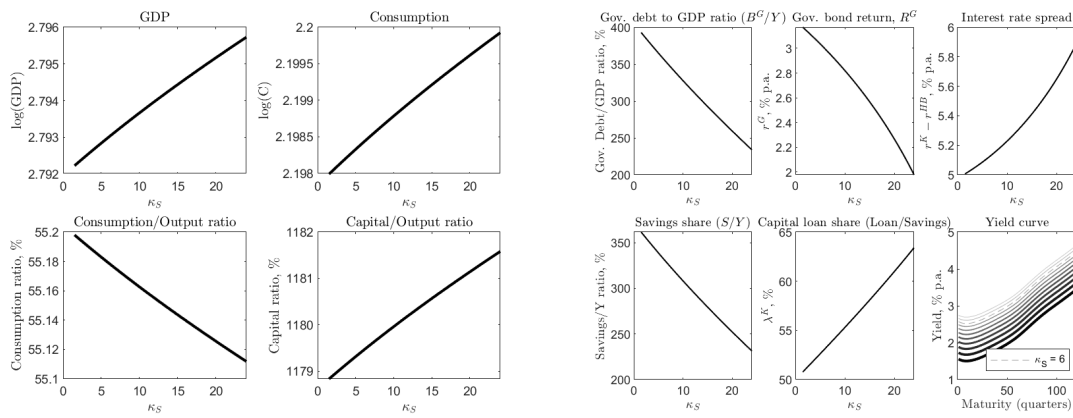


Figure E.5: Variations in κ^S

F Additional Results: Impulse-Responses

F.1 Without ZLB

Technology shock, ε_t^A Figure F.6 displays the impulse-responses to a ε_t^A shock. A positive shock in technology growth GA_t yields similar effects as documented in prior literature under conventional policy,¹⁰ with output increasing¹¹ as inflation decreases. Under the yield-curve-control regime, the normalized output decreases less: since inflation falls, bond yields shift downwards, boosting consumption and reducing both the return on capital and the wage.

Monetary policy shock, $\varepsilon_t^{YD^1}$ Figure F.7 presents the impulse-response to the monetary policy shock $\varepsilon_t^{YD^1}$. Under conventional policy, a contractionary policy shock paradoxically results in an increase in output, inflation, and capital. As we explain in Section 2, a contractionary monetary policy shock induces households to reallocate their portfolios toward short-term bonds, reducing the aggregate bond return R^{HB} . At the same time, as the aggregate bond return R^{HB} decreases, households shift more strongly toward loans, prompting more capital accumulation and output increase.

The YCC policy insulates the economy almost perfectly: in response to the shock, the central bank shifts the entire yield curve, mitigating portfolio rebalancing and raising the effective monetary responsiveness. It prevents input prices—loan rates and wages—from rising too high, and thereby changes in output and inflation.

¹⁰For the effects of technology shocks in a standard New Keynesian framework, see e.g., Ireland (2004).

¹¹Even if the normalized—by trend technology—output drops under our calibration, the output level rises.

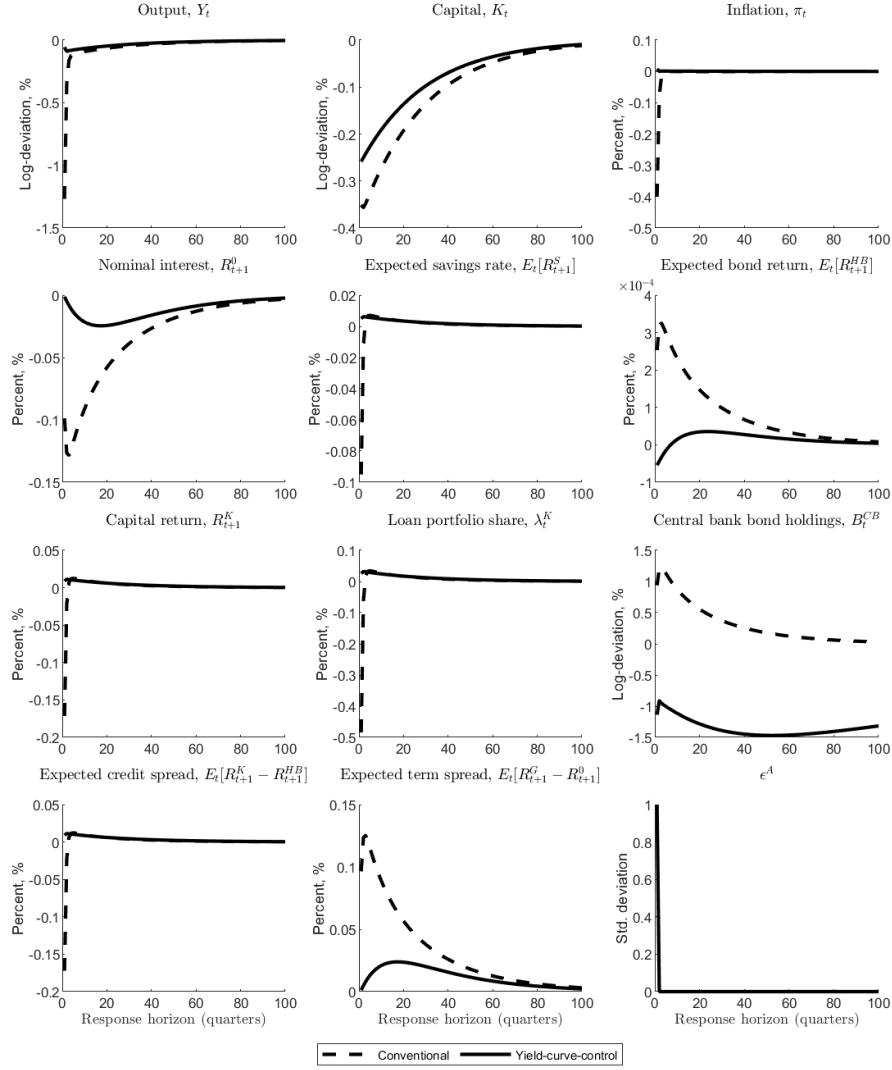


Figure F.6: Impulse response to a ε_t^A shock: A positive technology growth shock generates effects in line with the existing literature, resulting in an increase in output and a decline in inflation. As inflation decreases, all yields undergo a downward shift, which in turn lowers both the capital return and wage relative to the conventional scenario.

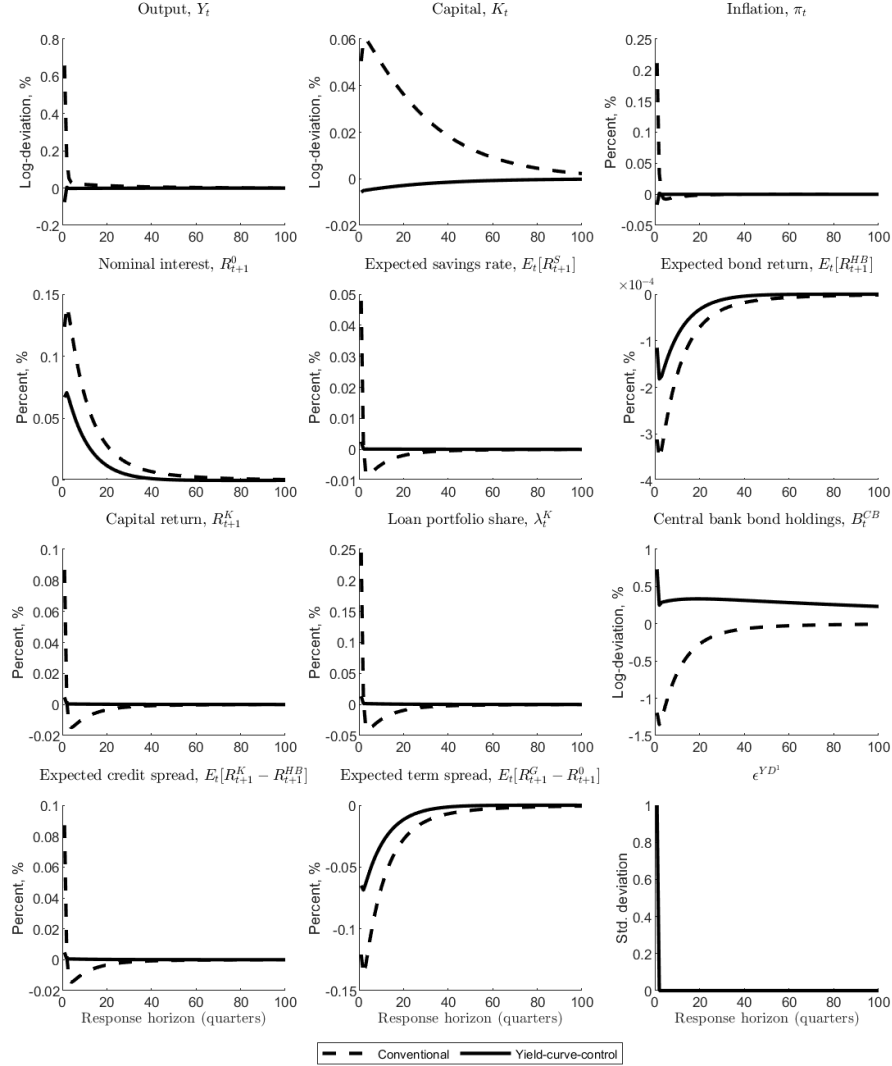


Figure F.7: Impulse response to a $\varepsilon_t^{YD^1}$ shock: A contractionary monetary policy shock paradoxically raises output, inflation, and capital due to household portfolio reallocation towards short-term bonds. The yield-curve-control policy better insulates the economy.

G Additional Results: Model Determinacy

Under the conventional policy (37) and the yield-curve control policy (38), we derive conditions for model determinacy in terms of the Taylor coefficients— (γ_π, γ_y) for the conventional policy and $(\{\gamma_\pi^f\}_{f=1}^F, \{\gamma_y^f\}_{f=1}^F)$ for the yield-curve control policy. For illustration, we assume that $\gamma_\pi^f = \gamma_\pi$ and $\gamma_y^f = \gamma_y$ for all $f = 1, \dots, F$, which can be easily generalized.

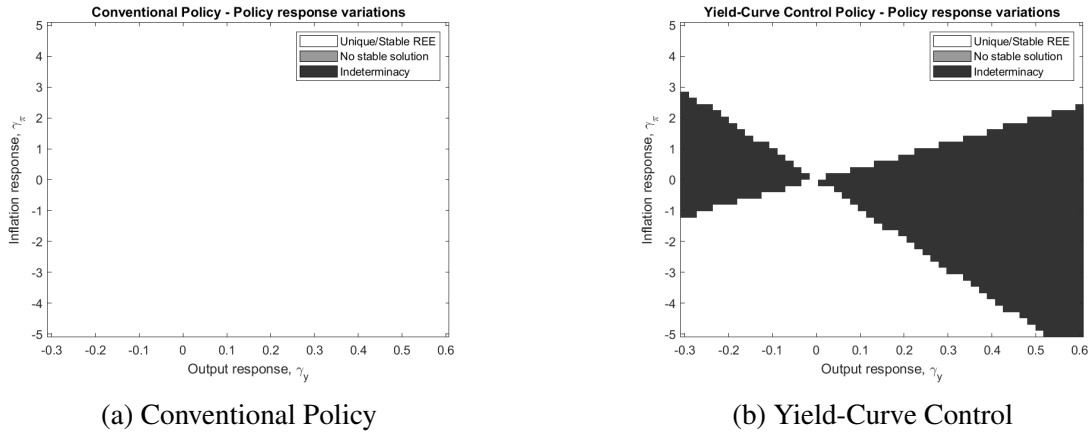


Figure G.8: Determinacy Regions

Figures G.8a and G.8b show the determinacy regions in the (γ_π, γ_y) plane for the conventional and yield-curve control policies, respectively. First, under the conventional policy, determinacy is guaranteed for most values of the Taylor coefficients.¹² When the short-term policy rate rises in response to inflation, households reallocate their portfolios toward short-term bonds, reducing the aggregate bond return R^{HB} . At the same time, households shift more strongly toward loans, raising the effective savings rate R^S . Therefore, the overall transmission of monetary policy is amplified, leading to a substantially higher effective Taylor coefficient.

In contrast, the yield-curve control policy's determinacy depends on both coefficients. Notably, when output targeting is close to zero ($\gamma_y = 0$), the model is determinate regardless of the inflation-targeting coefficient. These findings differ from those of the traditional New Keynesian model, which typically requires an aggressive inflation-targeting response (or a combination of responses to inflation and output) to ensure a unique and stable equilibrium.

¹²Note that it deviates from the traditional New Keynesian model, which typically requires an aggressive inflation-targeting response (or a combination of responses to inflation and output) to ensure a unique, stable equilibrium. It is from that our model introduces a portfolio rebalancing channel on the households' side and an input substitution channel on the firms' side.

librium. Again, this difference arises from the interaction of household portfolio rebalancing and the presence of physical capital. When γ_π is very high, a negative demand shock induces the central bank to push down the entire yield curve. Households respond by re-allocating savings toward loans, which stimulates capital accumulation, raises output and inflation, and helps stabilize the economy.

When γ_π is low, the same negative demand shock produces a smaller downward shift in the yield curve. Nevertheless, households still reallocate part of their portfolios toward loans, leading to higher investment demand, output, and inflation. Therefore, the stabilizing effect from the supply side is qualitatively similar, though weaker in magnitude. Because the adjustment in the yield curve is smaller, the effective return on savings increases less, resulting in a more modest decline in consumption demand and contributing to equilibrium stability.

H Robustness: Alternative Calibration

This section presents the robustness of the main results, especially the impulse responses, to alternative calibrations. Impulse responses are generally robust to alternative calibrations, and especially, our main result that YCC is significantly more effective than conventional policy in stabilizing the economy is robust to alternative calibrations.

H.1 Alternative Calibration of κ_B

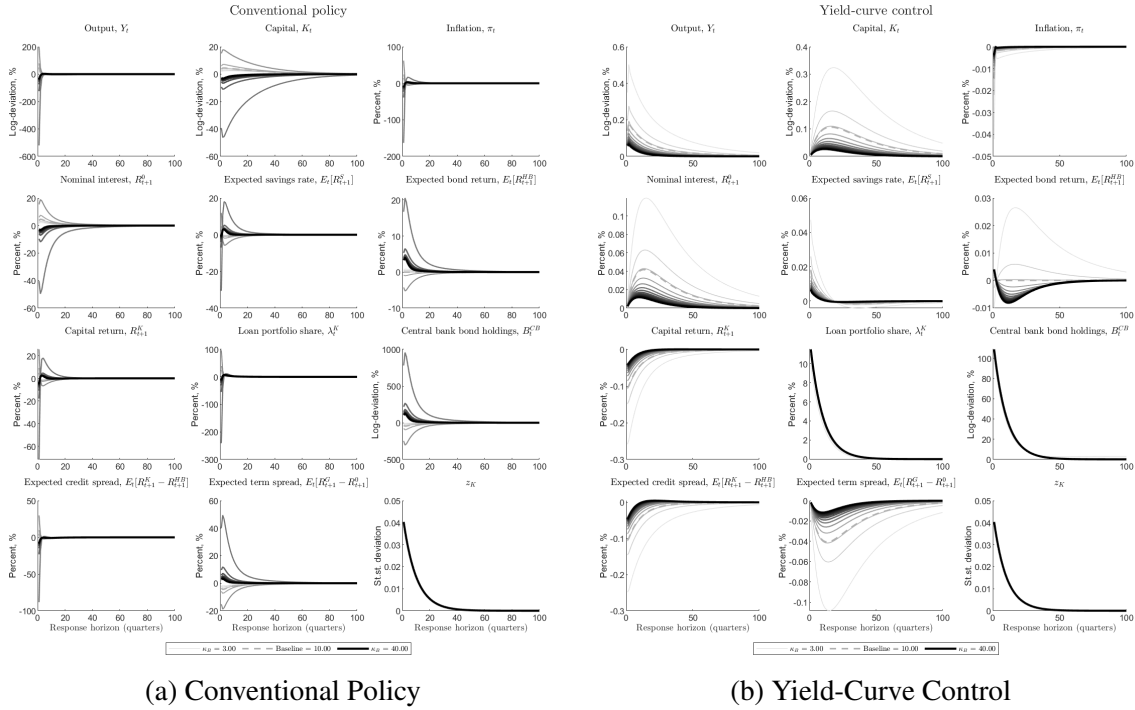


Figure H.9: z_t^K shock under different κ_B

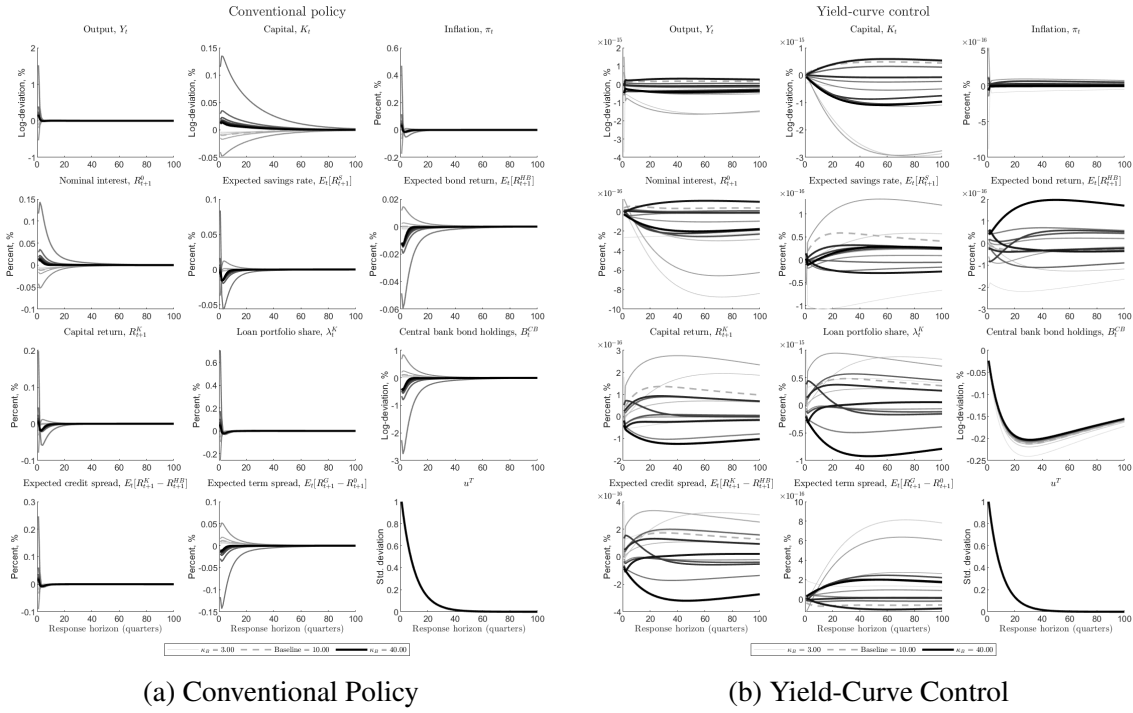


Figure H.10: Tax shock under different κ_B

H.2 Alternative Calibration of Deficit Ratio $\zeta^F + \zeta^G - \zeta^T$

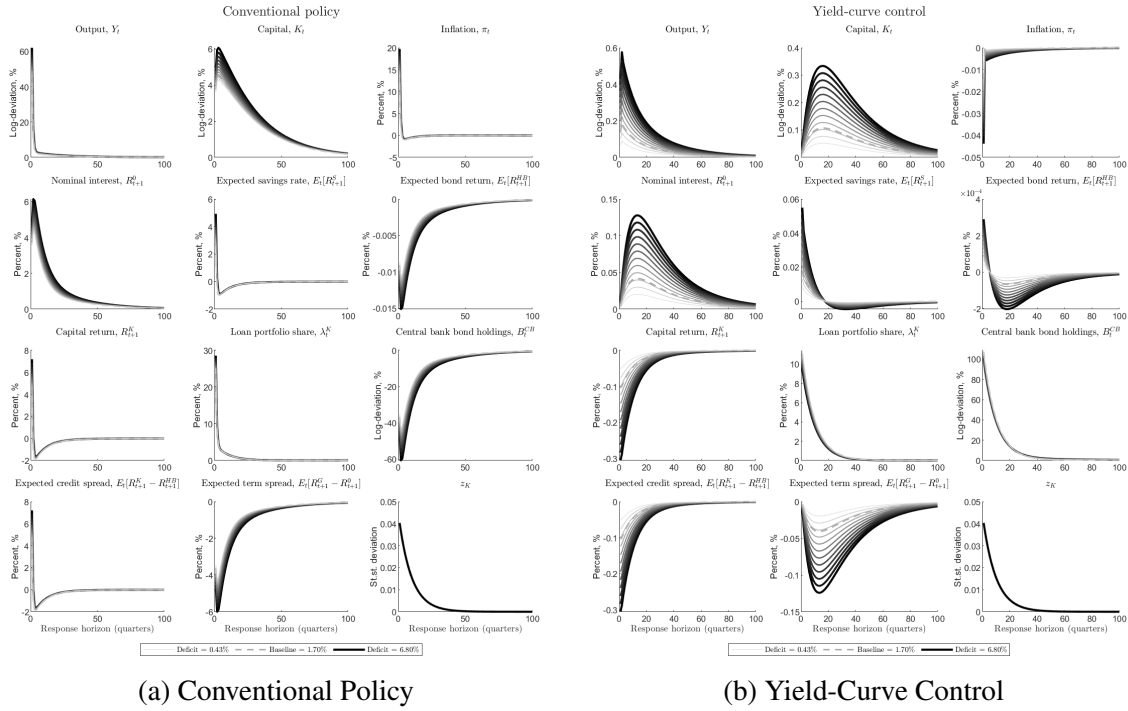


Figure H.11: z_t^K shock under different κ_B

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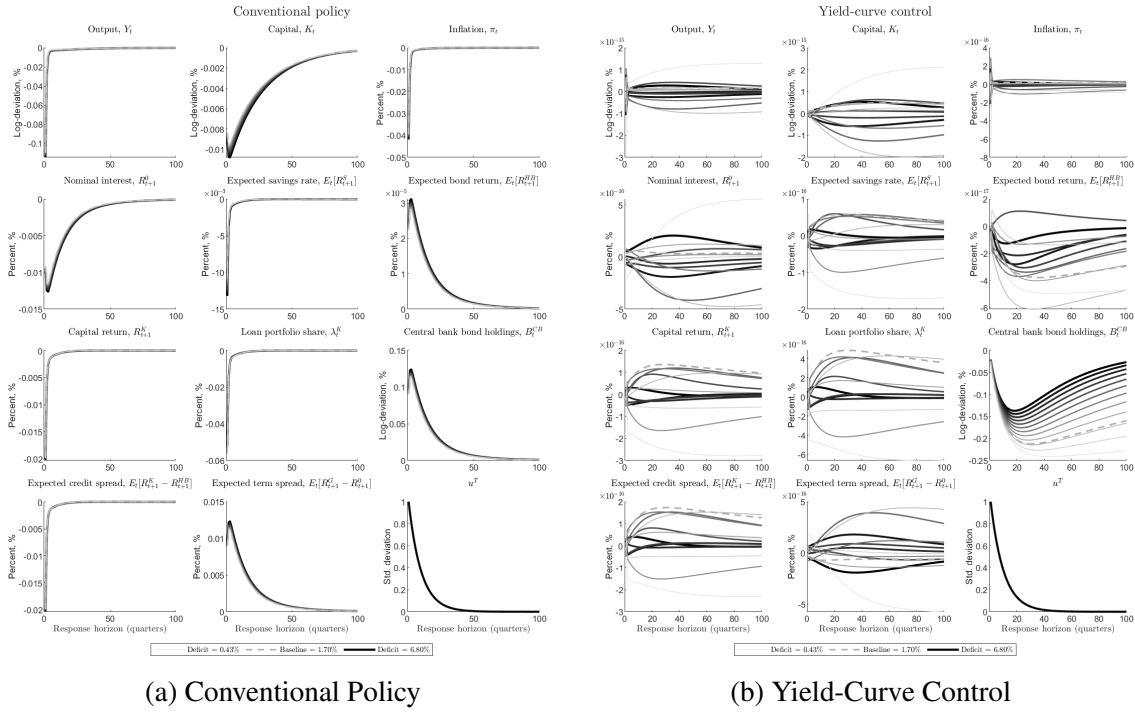


Figure H.12: Tax shock under different κ_B

I Additional Results: Operation Twist Shock and Pure Quantity Shock

Operation Twist Shock In this exercise we change the composition of the central bank portfolio while leaving the total balance sheet size unaffected (except for general equilibrium effects). The formula for the log-deviation in the bond holdings of the central bank is given by:

$$\hat{b}_t^{CB} = \sum_{f=1}^F \lambda^{CB,f} \hat{b}_t^{CB,f}.$$

Under conventional policy, maturity $f = 1$ is endogenous and will be capturing the general equilibrium effects, while maturities $f \geq 2$ are exogenous and depend on portfolio shocks.

We assume two quantity shocks: $\hat{u}_t^{CB,1}$ that affects maturities $f = 80, \dots, F$, and $\hat{u}_t^{CB,2}$ that affects maturities $f = 2, \dots, 79$. In order to have a shock that affects the composition of the central bank balance sheet but not its size, we require:

$$\sum_{f=2}^F \lambda^{CB,f} \hat{b}_t^{CB,f} = 0.$$

Using the mapping of the bond shocks to $\hat{b}_t^{CB,f}$ and rearranging, this requires:

$$\left[\sum_{f=80}^F \lambda^{CB,f} \right] \cdot \hat{u}_t^{CB,1} + \left[\sum_{f=2}^{79} \lambda^{CB,f} \right] \cdot \hat{u}_t^{CB,2} = 0.$$

Rearranging the above expression in order to obtain the necessary $\hat{u}_t^{CB,2}$ shock to counter-balance a $\hat{u}_t^{CB,1}$ shock, we obtain:

$$\hat{u}_t^{CB,2} = - \left(\frac{\sum_{f=80}^F \lambda^{CB,f}}{\sum_{f=2}^{79} \lambda^{CB,f}} \right) \cdot \hat{u}_t^{CB,1} \quad (\text{I.1})$$

The exercise will be as follows. We simulate impulse responses in response to a shock to $\hat{u}_t^{CB,1}$ of the same magnitude as the currently simulated long-duration purchase shock in Section 5.2.3. At the same time, we simultaneously shock the model with a $\hat{u}_t^{CB,2}$ shock of the magnitude given by (I.1).

The left panel of Figure I.13 shows that the impulse responses under this operation twist shock is expansionary.

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Portfolio Size In this exercise, we are going to implement a proportional shift in the yield curve by assuming a monetary policy shock of the same exact size and direction across all maturities under YCC, i.e., shocks ε^{YDf} , $\forall f$, so that we create a perfect parallel shift in the yield curve (before accounting for any potential general equilibrium effects).

The right panel of Figure I.13 shows that the impulse responses under this pure quantity shock is contractionary, which is easy to understand given that the central bank reduces its bond holdings so that the yield curve shifts upwards in a parallel manner.

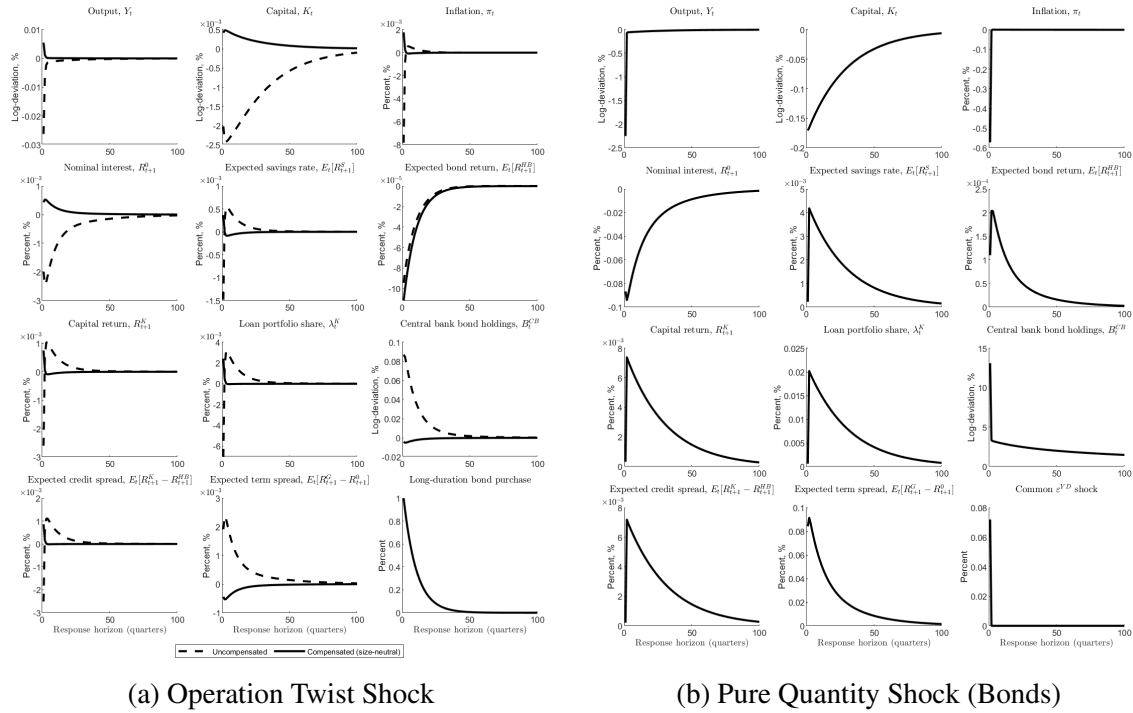


Figure I.13: Operation Twist Shock and Pure Quantity Shock (Bonds)

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